GRADE 8 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 9) • Forward 2 (Grade 10)

58 question sets • 1450 questions

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SET 1 • 8.NS.A.1 • THE NUMBER SYSTEM

Classify numbers as rational or irrational

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Write a fraction as a terminating or repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Decide whether sums and products are rational or irrational	**HSN-RN.B.3**
FORWARD 2	Extend the number system to complex numbers using i	**HSN-CN.A.1**

SET 1 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

Which number is irrational?

A. $\sqrt{49}$

B. $\frac{-7}{3}$

C. 0.1212... (the digits 12 repeat)

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

Is 0.375 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

Is $\sqrt{50}$ rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

True or false?

Every number whose decimal expansion repeats is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

True or false?

π is rational because it is often written as 3.14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 1

Write a fraction as a terminating or repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

Write $\frac{2}{3}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

Which fraction has a repeating decimal?

A. $\frac{3}{5}$

B. $\frac{1}{6}$

C. $\frac{1}{4}$

D. $\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{11}$ = 0.3636... (the digits 36 repeat)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{3} = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-2\frac{1}{4}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

Write 0.7 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

Which number is equal to $\frac{-5}{6}$?

A. $\frac{5}{6}$

B. $\frac{-5}{-6}$

C. $\frac{6}{-5}$

D. $\frac{5}{-6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

True or false?

-12 can be written as $\frac{-12}{1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{-3}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 1

Decide whether sums and products are rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is 4 + √3 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

Is $\sqrt{8} \times \sqrt{2}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

Which product is irrational?

A. $\sqrt{3} \times \sqrt{12}$

B. $3\sqrt{5}$

C. $0 \times \sqrt{7}$

D. $\frac{2}{3} \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 2

Extend the number system to complex numbers using i • High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

The number i has $i^2 = -1$.

Write $\sqrt{-16}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

The number i has $i^2 = -1$.

What is i^4?

$i^4 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

The number i has $i^2 = -1$.

Which number is a solution of $x^2 = -9$?

A. 3

B. -3

C. 9i

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q3**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number a is also a complex number, a + 0i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\sqrt{-25}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q5**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

SET 2 • 8.NS.A.1 • THE NUMBER SYSTEM

Convert a repeating decimal into a fraction

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Use long division to write a fraction as a repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Multiply decimals by powers of 10	**5.NBT.A.2**
FORWARD 1	Solve linear equations whose solutions are fractions	**HSA-REI.B.3**
FORWARD 2	Write a geometric sequence from a repeating decimal	**HSF-BF.A.2**

SET 2 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

Write 0.444... (the 4 repeats) as a fraction. Show your work.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

Write 0.2727... (the digits 27 repeat) as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

Which fraction is equal to 0.1666... (the 6 repeats)?

A. $\frac{1}{16}$

B. $\frac{1}{6}$

C. $\frac{16}{100}$

D. $\frac{6}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

True or false?

2.333... (the 3 repeats) is equal to $\frac{7}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

True or false?

0.0909... (the digits 09 repeat) is equal to $\frac{9}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 1

Use long division to write a fraction as a repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{7}{9}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

Write $\frac{5}{12}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{3}{11}$?

A. 0.27 (the decimal ends)

B. 0.3

C. 0.2727... (the digits 27 repeat)

D. 0.311

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{9} = 0.111...$ (the 1 repeats)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

True or false?

$\frac{5}{6} = 0.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 2

Multiply decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

0.454×100

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

Multiply.

3.27×10

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

Multiply.

$0.0081 \times 1{,}000$

A. 81

B. 8.1

C. 0.81

D. 0.081

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 \times 100 = 250$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$0.36 \times 10 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 1

Solve linear equations whose solutions are fractions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Write the answer as a fraction in simplest form.

$100x - x = 45$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

Solve for x.

$10x = x + 6.3$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

Solve for x.

$1{,}000x - 10x = 270$

A. $x = \frac{3}{11}$

B. $x = \frac{1}{37}$

C. $x = \frac{27}{100}$

D. $x = 27$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

True or false?

$x = \frac{4}{9}$ is the solution of $10x - x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = 0.12$ is the solution of $100x - x = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 2

Write a geometric sequence from a repeating decimal • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

0.333... = 0.3 + 0.03 + 0.003 + ... The terms 0.3, 0.03, 0.003, ... form a geometric sequence.

What is the common ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

Write an explicit formula for the nth term of the sequence 0.3, 0.03, 0.003, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

Which recursive formula gives the sequence 0.27, 0.0027, 0.000027, ...?

A. $a(1) = 0.27;\ a(n) = 100 \cdot a(n-1)$

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

C. $a(1) = 0.27;\ a(n) = a(n-1) - 0.01$

D. $a(1) = 0.0027;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

The sequence 0.5, 0.05, 0.005, ... is geometric with a common ratio of $\frac{1}{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

The sequence 0.4, 0.04, 0.004, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • 8.NS.A.2 • THE NUMBER SYSTEM

Estimate the value of an irrational number

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Round decimals to the nearest tenth	**5.NBT.A.4**
FORWARD 1	Solve quadratic equations by taking square roots, exactly and ap…	**HSA-REI.B.4.b**
FORWARD 2	Find arc lengths and sector areas exactly in terms of π and appro…	**HSG-C.B.5**

SET 3 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is $\sqrt{40}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

Estimate $\sqrt{20}$ to the nearest tenth. Explain how you know.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

Which is the best estimate of $\sqrt{75}$ to the nearest tenth?

A. 8.7

B. 8.6

C. 37.5

D. 9.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

True or false?

$\sqrt{2}$ is between 1.4 and 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

True or false?

$\sqrt{90}$ is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

Find the value.

12^2

A. 24

B. 121

C. 122

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$8^2 = 64$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

$3^2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 2

Round decimals to the nearest tenth • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 4.472 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

Round 8.660 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

Round 3.162 to the nearest tenth.

A. 3.1

B. 3.0

C. 3.2

D. 3.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

True or false?

6.325 rounded to the nearest tenth is 6.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

True or false?

2.449 rounded to the nearest tenth is 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 1

Solve quadratic equations by taking square roots, exactly and approximately • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 = 30$. Give the exact solutions, then round each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

Solve $x^2 - 7 = 0$. Give the exact solutions.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

Solve $(x - 1)^2 = 5$.

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

B. $x = 1 + \sqrt{5}$ only

C. $x = 6$ or $x = -4$

D. $x = \sqrt{6}$ or $x = -\sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

True or false?

The solutions of $x^2 = 12$ are $x = 2\sqrt{3}$ and $x = -2\sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

True or false?

$x^2 = -9$ has the real solutions 3 and -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 2

Find arc lengths and sector areas exactly in terms of π and approximately • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 6 cm.

Find the length of a 90° arc exactly in terms of π, then to the nearest tenth.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5

A circle has a radius of 4 inches.

Find the area of a 45° sector exactly in terms of π, then to the nearest tenth.

Sector area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5

A circle has a radius of 3 m.

To the nearest tenth, how long is a 60° arc?

A. 6.3 m

B. 9.4 m

C. 3.1 m

D. 18.8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5

True or false?

The arc of a semicircle with a radius of 1 is π units long, about 3.14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5

True or false?

A 180° sector of a circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 8.NS.A.2 • THE NUMBER SYSTEM

Locate and compare irrational numbers on a number line

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Read inequalities as positions on a number line	**6.NS.C.7.a**
BACKWARD 2	Position decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Report irrational values to a level of accuracy that fits the situati…	**HSN-Q.A.3**
FORWARD 2	Use coordinates to find exact lengths and perimeters with radicals	**HSG-GPE.B.7**

SET 4 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Plot $\sqrt{5}$ on the number line. Use an estimate to the nearest tenth.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

Which list is in order from least to greatest?

A. 2, √5, √3, 1.8

B. √3, 1.8, 2, √5

C. √3, 2, 1.8, √5

D. 1.8, √3, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

Write <, >, or =.

π ___ $\sqrt{10}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

Which number is the best name for point P: √2, √3, or √6?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

True or false?

$\sqrt{30}$ is to the left of 5.5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 1

Read inequalities as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

2.4 is to the right of 2.25 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

Which number is farthest to the left on a number line?

A. 0.5

B. −1.9

C. −2.05

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

What does 1.7 < 1.75 tell you about the positions of the two numbers on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

True or false?

$-1.4 > -1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

True or false?

$3.09 > 3.1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 2

Position decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot 1.4 and 2.75 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

What decimal is at point A?



Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

Which decimal is between 1.7 and 1.8?

A. 1.81

B. 1.73

C. 1.69

D. 1.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

True or false?

0.9 is to the right of 1.1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 1

Report irrational values to a level of accuracy that fits the situation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A square garden has an area of 50 square meters, so its side is √50 m. A tape measure reads to the nearest centimeter.

Report the side length to that accuracy, in meters.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

A coin has a diameter of 4 cm. Its circumference is 4π cm.

Report the circumference to the nearest tenth of a centimeter.

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

Which value of $\sqrt{2}$ is given to the nearest thousandth?

A. 1.41

B. 1.414

C. 1.4142

D. 1.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

True or false?

Reporting $\sqrt{3}$ as 1.7 is accurate to the nearest tenth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

True or false?

Writing more digits of $\sqrt{5}$ always makes a measured length more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 2

Use coordinates to find exact lengths and perimeters with radicals • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (2, 0), and (0, 2).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

Find the exact length of the segment from (1, 1) to (4, 3).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

A square has vertices (0, 0), (1, 2), (−1, 3), and (−2, 1).

What is its perimeter?

A. 2√5 units

B. 20 units

C. 4√5 units

D. 4√3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

True or false?

The distance from $(-1, -1)$ to $(2, 3)$ is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

True or false?

A rectangle with vertices (0, 0), (3, 0), (3, 1), and (0, 1) has a diagonal 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

SET 5 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the product and quotient properties of integer exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Evaluate whole-number exponents	**6.EE.A.1**
BACKWARD 2	Write powers of 10 with whole-number exponents	**5.NBT.A.2**
FORWARD 1	Multiply and divide monomials using exponent properties	**HSA-SSE.A.2**
FORWARD 2	Use exponent properties to rewrite and interpret exponential funct	**HSF-IF.C.8.b**

SET 5 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$3^4 \times 3^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

Write as a single power.

$7^{10} \div 7^{4}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

Which expression is equal to 2^6?

A. $2^{12} \div 2^2$

B. 2×6

C. $2^2 \times 2^3$

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

True or false?

$5^3 \times 5^2 = 5^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

True or false?

$4^8 \div 4^2 = 4^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 1

Evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

2^5

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

Find the value.

10^4

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

Find the value.

3^4

A. 81

B. 64

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$6^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

True or false?

$2^3 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 2

Write powers of 10 with whole-number exponents • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

Write 1,000,000 as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

What is the value of 10^5?

A. 10,000

B. 50

C. 100,000

D. 1,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

True or false?

$10^3 = 1{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

True or false?

$10^2 \times 10 = 10^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 1

Multiply and divide monomials using exponent properties • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Simplify.

$(3x^4)(5x^2)$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

Simplify. Assume $x \neq 0$.

$$\frac{12x^7}{4x^3}$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

Simplify.

$x^5 \cdot x \cdot x^2$

A. x^7

B. $3x^8$

C. x^{10}

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

True or false?

$(2a^3)(4a^2) = 8a^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

True or false?

Assume $y \neq 0$. $\frac{y^9}{y^3} = y^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 2

Use exponent properties to rewrite and interpret exponential functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b

$f(t) = 2^{(t + 3)}$

Use a property of exponents to write f(t) in the form $a \cdot 2^t$.

$f(t) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

A population is P(t) = 200 · 2^t after t years.

By what factor does the population grow each year, and what is P(5)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

$g(t) = 5^{(t - 1)}$

Which expression is equal to g(t)?

A. $5^t - 5$

B. $5 \div 5^t$

C. $5^t \div 5$

D. $5^t - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

True or false?

If f(x) = 5 · 2^x, then f(x + 1) = 2 · f(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

True or false?

If f(x) = 2^x, then f(3) · f(4) = f(12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

SET 6 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the power-of-a-power property and zero and negative exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Find powers of fractions and decimals	**6.EE.A.1**
BACKWARD 2	Multiply signed numbers	**7.NS.A.2.a**
FORWARD 1	Explain rational exponents with the properties of exponents	**HSN-RN.A.1**
FORWARD 2	Rewrite exponential expressions with the properties of exponents	**HSA-SSE.B.3.c**

SET 6 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$(5^2)^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

Find the value.

4^{-2}

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

Which expression is equal to 1?

A. 0^9

B. 9^0

C. $1^{-9} \times 9$

D. 9^{-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

True or false?

$3^{-2} \times 3^{5} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

True or false?

$(2^3)^2 = 2^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 1

Find powers of fractions and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

$(\frac{1}{2})^3$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

Find the value.

0.3^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

Find the value.

$(\frac{2}{3})^2$

A. $\frac{4}{3}$

B. $\frac{4}{9}$

C. $\frac{2}{9}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$0.1^3 = 0.001$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

True or false?

$(\frac{3}{4})^2 = \frac{6}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 2

Multiply signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$(-2)(-2)(-2)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

Multiply.

$(-3)(-3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

Multiply.

$(-1)(-1)(-1)(-1)$

A. -1

B. -4

C. 1

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

True or false?

$(-5)(-5) = 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

True or false?

$$(-2)(-2)(-2)(-2) = -16$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 1

Explain rational exponents with the properties of exponents • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1

$(5^{1/2})^2 = 5^{(1/2) \cdot 2} = 5$.

Which radical is equal to $5^{1/2}$?

Radical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

Write 7^(1/3) as a radical.

Radical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

Which number is equal to 9^(3/2)?

A. 6

B. 27

C. 13.5

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

True or false?

Because $(8^{(1/3)})^3 = 8$, $8^{(1/3)} = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

True or false?

x^(1/2) means x ÷ 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 2

Rewrite exponential expressions with the properties of exponents • High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

Rewrite 1.21^t as a power of 1.1.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

A value grows by a factor of 1.04 each month, so after t years it has grown by (1.04^12)^t.

Write the yearly growth factor, rounded to the nearest hundredth.

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

Which expression is equivalent to 8^x?

A. 2^x ÷ 3

B. 3^(2x)

C. 2^(3x)

D. 2^(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

True or false?

$(0.5)^{(-t)} = 2^t$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

True or false?

$4^x \times 4^x = 16^{(2x)}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

SET 7 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Solve equations of the form $x^2 = p$ and $x^3 = p$

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Solve one-step multiplication equations	**6.EE.B.7**
BACKWARD 2	Find squares and cubes of whole numbers	**6.EE.A.1**
FORWARD 1	Solve quadratic equations by taking square roots	**HSA-REI.B.4.b**
FORWARD 2	Find a dimension from a volume with square and cube roots	**HSG-GMD.A.3**

SET 7 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x. Give all solutions.

$x^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

Solve for x.

$x^3 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

Solve for x.

$x^2 = \frac{9}{16}$

A. $x = \frac{3}{4}$ only

B. $x = \frac{3}{4}$ or $x = -\frac{3}{4}$

C. $x = \frac{9}{32}$

D. $x = \frac{81}{256}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

Solve for x.

$x^3 = -27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

True or false?

$x^2 = 10$ has the two solutions $x = \sqrt{10}$ and $x = -\sqrt{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 1

Solve one-step multiplication equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$4x = 64$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

Solve for x.

$\frac{x}{3} = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

Solve for x.

$9x = 45$

A. $x = 36$

B. $x = 5$

C. $x = 405$

D. $x = 54$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

True or false?

$x = 12$ is the solution of $3x = 36$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

True or false?

$x = 5$ is the solution of $7x = 42$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 2

Find squares and cubes of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

9^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

Find the value.

5^3

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which whole number cubed is 8?

A. 3

B. 4

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$10^4 = 10{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$4^2 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 1

Solve quadratic equations by taking square roots • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve for x. Give all solutions.

$2x^2 = 50$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

Solve for x. Give all solutions.

$(x + 2)^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$x^2 - 9 = 0$

A. $x = 9$ or $x = -9$

B. $x = 3$ only

C. $x = 3$ or $x = -3$

D. $x = 4.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

True or false?

The solutions of $3x^2 = 27$ are 3 and -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

True or false?

The solutions of $(x - 1)^2 = 16$ are 5 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 2

Find a dimension from a volume with square and cube roots • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cube has a volume of 343 cubic centimeters.

How long is each edge?

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

A sphere has a volume of 36π cubic centimeters.

$(V = \frac{4}{3}\pi r^3)$

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

A cylinder has a volume of 200π cubic units and a height of 8 units.

What is its radius?

A. 10 units

B. 25 units

C. 5 units

D. $\sqrt{200}$ units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

True or false?

A cone with a volume of 12π cubic units and a height of 4 units has a radius of 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

True or false?

A cube with a volume of 1,000 cubic centimeters has edges 100 cm long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

SET 8 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Evaluate square roots and cube roots and recognize irrational roots

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Find factor pairs and square numbers	**4.OA.B.4**
BACKWARD 2	Find the volume of a cube	**5.MD.C.5.b**
FORWARD 1	Simplify square roots by factoring out perfect squares	**HSN-RN.A.2**
FORWARD 2	Solve quadratic equations by completing the square	**HSA-REI.B.4.a**

SET 8 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

$\sqrt{144}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

Find the value.

$\sqrt[3]{125}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

Which square root is irrational?

A. $\sqrt{\frac{1}{4}}$

B. $\sqrt{0.09}$

C. $\sqrt{20}$

D. $\sqrt{36}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

True or false?

$\sqrt{2}$ can be written as a fraction of two whole numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 1

Find factor pairs and square numbers • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 16.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

What whole number times itself is 49?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

Which number has a factor pair made of two equal factors?

A. 30

B. 24

C. 18

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

True or false?

81 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

True or false?

12 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 2

Find the volume of a cube • Grade 5 • Measurement & Data • 5.MD.C.5.b

A cube has edges 3 units long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

A cube has edges 5 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

Which cube has a volume of 64 cubic units?

A. A cube with 32-unit edges

B. A cube with 4-unit edges

C. A cube with 16-unit edges

D. A cube with 8-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

True or false?

A cube with 2-unit edges has a volume of 8 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

True or false?

A cube with 10-unit edges has a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 1

Simplify square roots by factoring out perfect squares • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify.

$\sqrt{48}$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

Simplify.

$\sqrt{200}$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

Which expression is equal to $\sqrt{18}$?

A. $2\sqrt{3}$

B. $9\sqrt{2}$

C. $3\sqrt{2}$

D. $6\sqrt{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q3**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

True or false?

$\sqrt{50} = 5\sqrt{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

True or false?

$\sqrt{27} = 9\sqrt{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 2

Solve quadratic equations by completing the square • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

What number goes in the blank?

$x^2 + 6x + 9 = (x + ___)^2$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

Solve by completing the square. Give exact solutions.

$x^2 + 4x = 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

Which number completes the square?

$x^2 - 10x + ___$

A. 5

B. −25

C. 100

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q3**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

True or false?

$$x^2 + 8x + 16 = (x + 4)^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

True or false?

Completing the square on $x^2 + 2x = 3$ gives $(x + 1)^2 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

SET 9 • 8.EE.A.3 • EXPRESSIONS & EQUATIONS

Estimate very large and very small quantities as a single digit times a power of 10

MAIN	Grade 8 main question	**8.EE.A.3**
BACKWARD 1	Multiply and divide by powers of 10	**5.NBT.A.2**
BACKWARD 2	Round whole numbers to any place	**4.NBT.A.3**
FORWARD 1	Choose a level of accuracy for very large and very small quantities	**HSN-Q.A.3**
FORWARD 2	Use base-10 logarithms to describe orders of magnitude	**HSF-LE.A.4**

SET 9 • 8.EE.A.3

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.3

Estimate 412,000,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

Estimate 0.00079 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

A country has about 3×10^8 people. A city has about 6×10^5 people.

About how many times as many people live in the country?

A. 2

B. 50

C. 500

D. 5,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

A whale has a mass of about 2×10^{5} kg. A mouse has a mass of about 2×10^{-2} kg.

About how many times as large is the whale's mass?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

True or false?

6×10^{-3} is greater than 9×10^{-4}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 1

Multiply and divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

4×10^3

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

Divide.

$7 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

What is the value of 3×10^5?

A. 15

B. 3,000,000

C. 300,000

D. 30,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$6 \div 10^3 = 0.006$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$8 \times 10^2 = 8{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 2

Round whole numbers to any place • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 412,000,000 to the nearest hundred million.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

Round 6,750 to the nearest thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

Round 58,300 to the nearest ten thousand.

A. 50,000

B. 59,000

C. 60,000

D. 58,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

245,000 rounded to the nearest hundred thousand is 200,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

8,500 rounded to the nearest thousand is 8,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 1

Choose a level of accuracy for very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Earth's mean radius is about 6,371 km.

Write it to one significant digit as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

A bacterium is 0.0000023 m long.

Write its length in micrometers. (1 μm = 10^{-6} m)

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

A sheet of paper is about 0.0001 m thick.

Which unit best expresses this thickness?

A. Millimeters (0.1 mm)

B. Megameters

C. Meters

D. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reporting 7,900,000,000 people as 8×10^9 is accurate to one significant digit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

True or false?

A distance known to be about 3×10^8 m is reported more accurately as 300,000,000.0 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 2

Use base-10 logarithms to describe orders of magnitude • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Find the value.

$\log_{10}(1{,}000{,}000)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

Solve for x.

$10^x = 0.001$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

Between which two whole numbers is $\log_{10}(4{,}500)$?

A. 2 and 3

B. 45 and 46

C. 4 and 5

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

True or false?

$\log_{10}(10^{-5}) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

True or false?

$\log_{10}(50)$ is between 5 and 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

SET 10 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Multiply and divide numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Multiply decimals	**5.NBT.B.7**
BACKWARD 2	Multiply by powers of 10	**5.NBT.A.2**
FORWARD 1	Use units to guide multistep calculations in scientific notation	**HSN-Q.A.1**
FORWARD 2	Model exponential growth and decay with scientific notation	**HSF-LE.A.1.c**

SET 10 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Multiply. Write the answer in scientific notation.

$(4 \times 10^{6})(2.5 \times 10^{3})$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

Divide. Write the answer in scientific notation.

$(6.3 \times 10^{8}) \div (9 \times 10^{2})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

Light travels about 3×10^5 km per second.

How far does it travel in 5×10^2 seconds?

A. 1.5×10^7 km

B. 6×10^2 km

C. 1.5×10^8 km

D. 8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

True or false?

$$(2 \times 10^{-3})(3 \times 10^{-4}) = 6 \times 10^{-7}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

True or false?

$(8 \times 10^9) \div (4 \times 10^3) = 2 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 1

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.5×4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

Multiply.

1.2×3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

Multiply.

6×0.5

A. 6.5

B. 3

C. 0.3

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$2.4 \times 0.5 = 1.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$3.5 \times 2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 2

Multiply by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

2.5×10^4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

Multiply.

3.1×10^2

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

Multiply.

0.6×10^3

A. 60

B. 0.0006

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$7 \times 10^3 = 7{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 1

Use units to guide multistep calculations in scientific notation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A drive transfers 2×10^{8} bytes per second.

How many seconds does it take to transfer 6×10^{10} bytes?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

Light travels 3×10^8 meters per second.

How many meters does it travel in 1 hour (3.6×10^3 seconds)? Use scientific notation.

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

The Sun is about 1.5×10^{11} m from Earth. Light travels 3×10^{8} m per second.

About how long does sunlight take to reach Earth?

A. 500 seconds

B. 4.5×10^{19} seconds

C. 5×10^{19} seconds

D. 50 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

True or false?

A flow of 5×10^3 liters per minute is 3×10^5 liters per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} km is equal to 6×10^{-6} m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 2

Model exponential growth and decay with scientific notation • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A colony of 2×10^3 bacteria triples every hour.

How many bacteria are there after 4 hours? Use scientific notation.

Bacteria: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

A quantity of 5×10^9 is cut in half each year.

What is it after 3 years? Use scientific notation.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

A population of 4×10^6 grows by 10% each year.

What is the population after 2 years?

A. 4.84×10^6

B. 8×10^6

C. 4.4×10^6

D. 4.8×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

True or false?

Doubling 3×10^5 ten times gives about 3.07×10^8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

True or false?

A quantity that grows by a constant percent rate increases by the same amount each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

SET 11 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Add and subtract numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Add decimals	**5.NBT.B.7**
BACKWARD 2	Write a number times a power of 10 in standard form	**5.NBT.A.2**
FORWARD 1	Add and subtract polynomials	**HSA-APR.A.1**
FORWARD 2	Add and subtract functions	**HSF-BF.A.1.b**

SET 11 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Add. Write the answer in scientific notation.

$(3.2 \times 10^5) + (4.5 \times 10^4)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

Subtract. Write the answer in scientific notation.

$(9 \times 10^{7}) - (6 \times 10^{6})$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

Add. Write the answer in scientific notation.

$(1.5 \times 10^{-3}) + (2 \times 10^{-4})$

A. 1.7×10^{-4}

B. 3.5×10^{-3}

C. 1.7×10^{-3}

D. 3.5×10^{-7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

True or false?

$(5 \times 10^8) + (5 \times 10^8) = 1 \times 10^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

True or false?

$(4 \times 10^6) - (3 \times 10^5) = 1 \times 10^1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 1

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.2 + 0.45

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

Add.

8.4 + 1.6

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

Add.

1.5 + 0.2

A. 0.17

B. 1.7

C. 1.52

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

True or false?

$2.75 + 0.25 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

True or false?

$4.6 + 0.04 = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 2

Write a number times a power of 10 in standard form • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write 6×10^6 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

Write 0.6×10^{7} in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

Which number is equal to 45×10^4?

A. 450,000

B. 4,500

C. 45,000

D. 4,500,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$3.2 \times 10^2 = 320$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$0.5 \times 10^3 = 50$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 1

Add and subtract polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add.

$(3x^2 + 5x) + (4x^2 - 2x)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

Subtract.

$(6a^3 - 2a + 1) - (2a^3 + 5a - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

Add.

$(x^2 - 4) + (3x^2 + x)$

A. $4x^2 + x - 4$

B. $3x^4 + x - 4$

C. $4x^2 - 3$

D. $4x^2 + x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

True or false?

$$(2y^2 + 3) - (y^2 - 3) = y^2 + 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

True or false?

$$(5n - 2) - (3n - 4) = 2n - 6$$

A. True

B. False

SET 11 • 8.EE.A.4

FORWARD 2

Add and subtract functions • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b

Revenue is $R(x) = 25x$ and cost is $C(x) = 4{,}000 + 9x$ for x items.

Write the profit function $P(x) = R(x) - C(x)$.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5

The profit function is $P(x) = 16x - 4{,}000$.

Find $P(1{,}000)$.

$P(1{,}000) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5

$f(x) = 3x^2$ and $g(x) = x^2 - 5$.

Which expression is $(f + g)(x)$?

A. $2x^2 + 5$

B. $4x^2 - 5$

C. $3x^4 - 5$

D. $4x^2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q3**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5

True or false?

If $f(x) = 2x$ and $g(x) = x + 1$, then $(f - g)(3) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x + 4$ and $g(x) = 2x$, then $(f + g)(x) = 2x^2 + 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5

SET 12 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Choose units for scientific notation and read calculator notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Convert metric units	**5.MD.A.1**
BACKWARD 2	Divide by powers of 10	**5.NBT.A.2**
FORWARD 1	Convert units of very large and very small quantities	**HSN-Q.A.1**
FORWARD 2	Define the quantities and units a model needs	**HSN-Q.A.2**

SET 12 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

A calculator shows 4.7E8.

Write the number in scientific notation and in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

A calculator shows 2.05E-6.

Write the number in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

The distance from Earth to the Moon is about 3.84×10^8 m.

Which unit is the most sensible for this distance?

A. Kilometers

B. Centimeters

C. Millimeters

D. Micrometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

A dust particle has a mass of 7.53×10^{-10} kg.

Write the mass in micrograms. (1 μg = 10^{-9} kg)

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

True or false?

A calculator display of 3E-4 means −30,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 1

Convert metric units • Grade 5 • Measurement & Data • 5.MD.A.1

How many grams are in 3.5 kilograms?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

How many meters are in 4,200 millimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

How many centimeters are in 6 meters?

A. 60

B. 0.06

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

True or false?

5 km = 5,000 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

True or false?

250 cm = 25 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 2

Divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$3{,}840 \div 10^3$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

Divide.

$75 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

Divide.

$6 \div 10^4$

A. 0.006

B. 0.00006

C. 60,000

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

True or false?

$520 \div 10 = 52$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$9 \div 10^3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 1

Convert units of very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

Convert 3.6×10^6 mm to kilometers.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

A server sends 4.5×10^9 bytes per minute.

Write this rate in gigabytes per second. (1 GB = 10^9 bytes)

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

A virus is 1.2×10^{-7} m wide.

What is its width in nanometers? ($1 \text{ nm} = 10^{-9}$ m)

A. 1,200 nm

B. 120 nm

C. 12 nm

D. 0.12 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

True or false?

7×10^5 g = 700 kg

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

True or false?

2.5×10^{-3} km = 2.5 cm

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 2

Define the quantities and units a model needs • High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2

You want to estimate how much water a city of 2×10^6 people uses each day.

Which quantities do you need, and with what units?

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

2×10^6 people each use about 3×10^2 liters of water per day.

About how many liters does the city use per day? Use scientific notation.

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

Which unit is the best choice for the mass of a large ship?

A. Metric tons

B. Grams

C. Ounces

D. Milligrams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q3**

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

True or false?

Choosing units that keep numbers between about 1 and 1,000 can make a model easier to read.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

True or false?

Kilometers are the best unit for the width of a cell.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Graph a proportional relationship and interpret the unit rate as the slope

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Find the constant of proportionality from a graph	**7.RP.A.2.b**
FORWARD 1	Calculate and interpret an average rate of change	**HSF-IF.B.6**
FORWARD 2	Compare properties of functions represented in different ways	**HSF-IF.C.9**

SET 13 • 8.EE.B.5

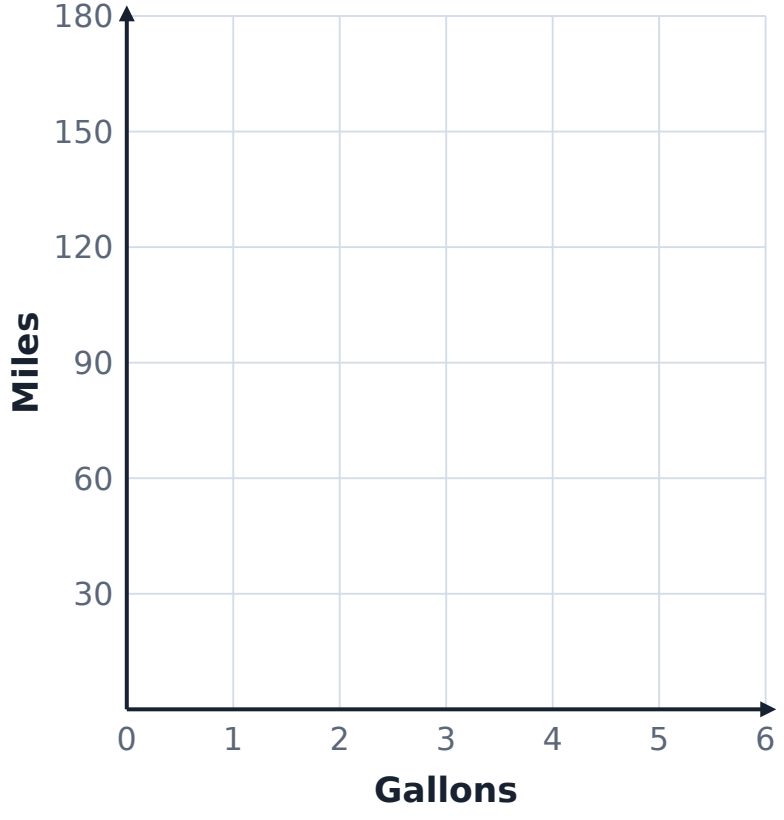
MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

A car uses gas at a constant rate: 3 gallons for every 90 miles.

Graph the relationship between gallons (x) and miles (y). What is the slope, and what does it mean?

Slope and meaning: _____

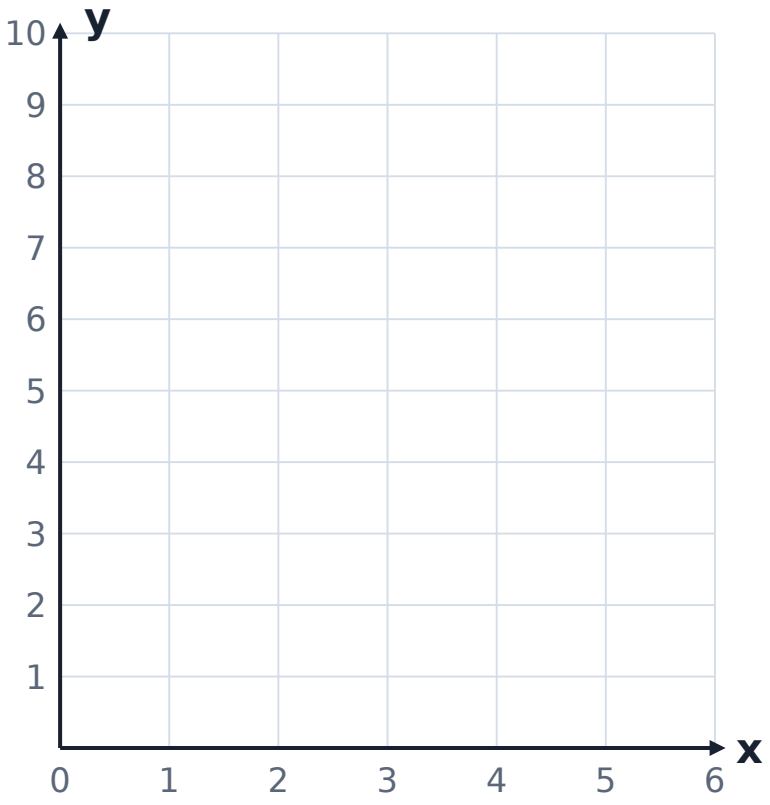


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

Graph y = 1.5x on the grid for x from 0 to 6.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

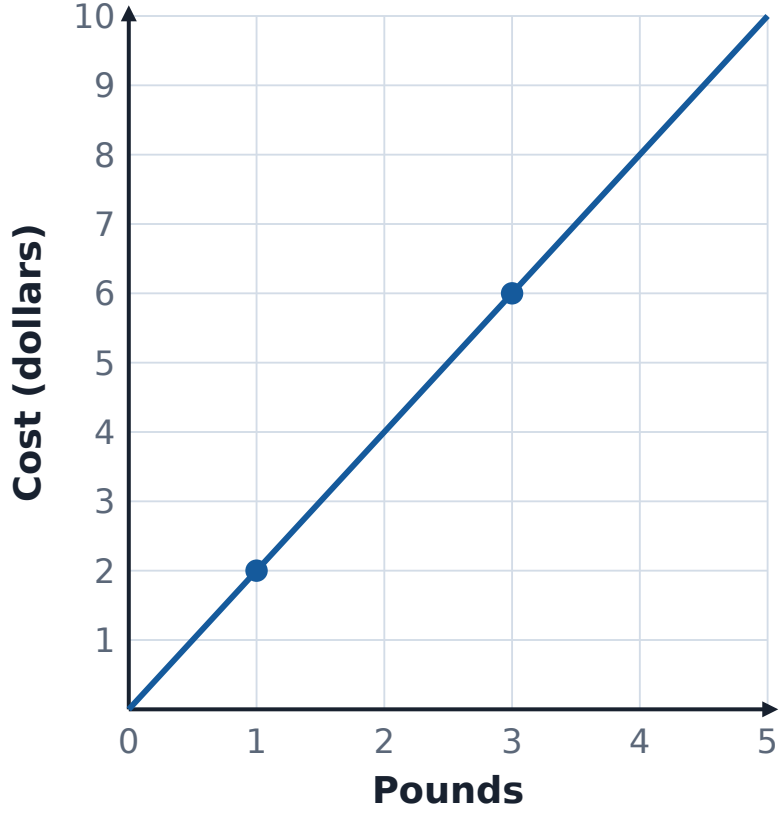
ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

The graph shows the cost of apples.

What is the slope, and what does it represent?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

The graph of a proportional relationship passes through (4, 10).

What is the unit rate?

A. 14

B. 2.5

C. 0.4

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

True or false?

On graphs of distance (y) against time (x) that pass through the origin, a steeper line means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of hours (x) and pay in dollars (y) passes through (1, 15).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

A proportional graph shows cups of flour (x) and cookies (y).

What does the point (0, 0) mean?

A. Each cup makes 0 cookies.

B. 0 cups of flour make 0 cookies.

C. The recipe starts with some cookies.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and pages read (y) passes through (5, 20).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) tells you the unit rate is r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

True or false?

On a graph of hours (x) and miles (y), the point (3, 12) means 3 miles in 12 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 2

Find the constant of proportionality from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A proportional graph passes through (0, 0) and (5, 20).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

Which point shows k = 3 on the graph of a proportional relationship?

A. (3, 6)

B. (6, 2)

C. (3, 1)

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

A proportional graph passes through (8, 2).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

True or false?

A line through (0, 0) and (10, 25) has a constant of proportionality of 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

True or false?

A line through (0, 0) and (6, 3) has a constant of proportionality of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 1

Calculate and interpret an average rate of change • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 3x + 2$

What is the average rate of change of f from $x = 1$ to $x = 5$?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

The table shows a trip.

What is the average rate of change from 2 hours to 5 hours?

Hours	0	2	5
Miles	0	110	260

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

$h(t) = t^2$

What is the average rate of change of h from t = 1 to t = 4?

A. 5

B. 3

C. 16

D. 15

True or false?

For a linear function, the average rate of change is the same over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 2

Compare properties of functions represented in different ways • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 2.5x. The graph of g is a line through (0, 0) and (4, 12).

Which function has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

f(x) = 40 + 5x. The table shows g.

Which function has the greater initial value, and which has the greater rate of change?

x	0	1	2
g(x)	30	37	44

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

f(x) = 2^x and g(x) = 10x.

For which x-value is f(x) > g(x)?

A. x = 3

B. x = 4

C. x = 6

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q3**

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

True or false?

A linear function with a slope of 4 eventually grows more slowly than f(x) = 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

True or false?

Two functions with the same y-intercept must be the same function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Compare two proportional relationships represented in different ways

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Write an equation for a proportional relationship	**7.RP.A.2.c**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Create and compare linear equations in two variables	**HSA-CED.A.2**
FORWARD 2	Compare linear and exponential growth	**HSF-LE.A.3**

SET 14 • 8.EE.B.5

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

Runner A's distance is d = 6.2t (miles, hours). Runner B's graph is a line through (0, 0) and (2, 13).

Which runner is faster?

A. Runner A

B. Runner B

C. They run at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

Plan A costs c = 0.12m dollars for m minutes. The table shows Plan B.

Which plan costs less per minute, and by how much?

Minutes	100	250	400
Cost ($)	15	37.5	60

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

Pump A fills 45 gallons in 3 minutes at a constant rate. Pump B fills according to g = 14m.

Which pump is faster, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

True or false?

The graph of $y = 0.8x$ is less steep than the graph of a proportional relationship through (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

True or false?

A proportional relationship through (3, 7) has a greater unit rate than $y = 2.5x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost $0.45 each.

Write an equation for the cost c of n pencils.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	2	4	6
y	7	14	21

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

A car travels 55 miles per hour.

Which equation gives the distance d after t hours?

A. $d = \frac{t}{55}$

B. $d = 55t$

C. $t = 55d$

D. $d = t + 55$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

True or false?

A recipe uses 6 cups of water for every cup of rice. The equation $y = 6x$ gives the cups of water y for x cups of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

True or false?

A printer prints 30 pages in 2 minutes. The equation $p = 30m$ gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ty walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

$\frac{2}{5}$ pound of nuts costs $2.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

A hose fills $\frac{1}{2}$ gallon in $\frac{1}{6}$ minute.

How many gallons per minute is this?

A. $\frac{1}{12}$

B. 6

C. $\frac{1}{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{3}$ mile in $\frac{1}{6}$ hour is a speed of 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

True or false?

A drink uses $\frac{3}{4}$ cup of juice for every $\frac{1}{2}$ cup of water. That is $\frac{3}{8}$ cup of juice per cup of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 1

Create and compare linear equations in two variables • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

Tank A has 20 gallons and gains 5 gallons per minute. Tank B starts empty and gains 7 gallons per minute.

Write an equation for each tank, with y gallons after x minutes.

Equations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

Tank A holds $y = 5x + 20$ gallons and Tank B holds $y = 7x$ gallons after x minutes.

When do they hold the same amount?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

A car starts 30 miles from home and drives away at 50 miles per hour.

Which equation gives its distance d from home after t hours?

A. $d = 50t - 30$

B. $d = 50t + 30$

C. $d = 30t + 50$

D. $d = 80t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

True or false?

The graphs of $y = 5x + 20$ and $y = 7x$ intersect at $(10, 70)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

True or false?

$y = 3x + 4$ and $y = 3x - 2$ have different rates of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 2

Compare linear and exponential growth • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

f(x) = 100 + 50x and g(x) = 100(1.5)^x.

Which is greater at x = 5? Give both values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

Which function eventually has the greater values: y = 1,000x or y = 2^x?

A. It depends only on the y-intercepts

B. y = 1,000x

C. y = 2^x

D. Neither; they are always equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q2**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

Which function in the table is exponential? Explain.

x	0	1	2	3
f(x)	2	4	8	16
g(x)	2	4	6	8

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

A quantity that increases exponentially eventually exceeds a quantity that increases linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

At x = 2, 3^x is greater than 10x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to explain slope and derive y = mx and y = mx + b

MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Use a scale factor to find a length in a scaled copy	**7.G.A.1**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Use similarity transformations to decide whether triangles are sim…	**HSG-SRT.A.2**
FORWARD 2	Use slopes of parallel and perpendicular lines	**HSG-GPE.B.5**

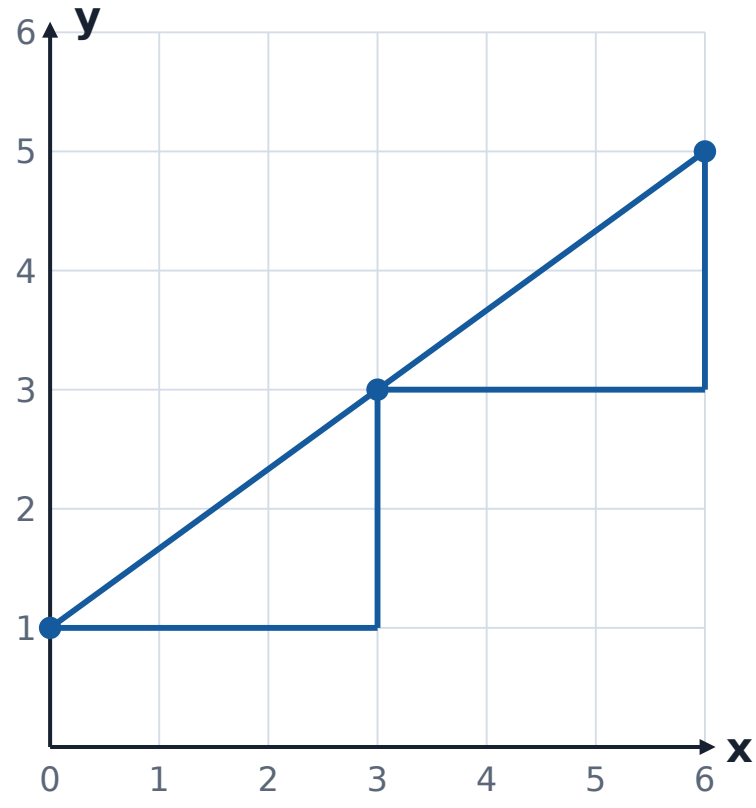
SET 15 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

Two slope triangles are drawn under the line. Explain why they give the same slope.

Explanation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

A line passes through the origin and (4, 6). By similar triangles, any point (x, y) on the line has y ÷ x = 6 ÷ 4.

Write the equation of the line.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

A line has a slope of 2 and crosses the y-axis at (0, 3).

Which equation describes it?

A. $y = 2x$

B. $y = 2x + 3$

C. $y = 3x + 2$

D. $y = x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

A line crosses the y-axis at (0, −2) and passes through (5, 8).

Write its equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

True or false?

Any two slope triangles drawn on the same non-vertical line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 1

Use a scale factor to find a length in a scaled copy • Grade 7 • Geometry • 7.G.A.1

A right triangle has legs of 2 and 3. In a scaled copy, the leg of 2 becomes 8.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

A right triangle has legs of 4 and 6. A scaled copy has a leg of 6 matching the leg of 4.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

Which pair of right triangles are scaled copies of each other (matching the shorter legs)?

A. Legs 2 and 5; legs 4 and 7

B. Legs 2 and 5; legs 6 and 12

C. Legs 2 and 5; legs 3 and 6

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

True or false?

A scaled copy with a scale factor of 3 has every length 3 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

True or false?

Right triangles with legs 3 and 4 and legs 6 and 9 are scaled copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	6	7	8

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	1	2	3
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

Which equation matches the table?

x	0	1	2
y	1	3	5

A. $y = 2x$

B. $y = x + 1$

C. $y = 3x$

D. $y = 2x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

True or false?

For $y = 2x + 4$, $y = 10$ when $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

True or false?

For $y = x + 7$, $y = 14$ when $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 3, 4, and 5. Triangle DEF has sides 6, 8, and 10.

Are the triangles similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

Explain how a dilation shows that two slope triangles on the same line are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

Which information is enough to show that two triangles are similar?

A. One pair of equal sides

B. Two pairs of equal angles

C. Equal perimeters

D. One pair of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

True or false?

If one triangle is a dilation of another, their corresponding sides are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 2

Use slopes of parallel and perpendicular lines • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line p has a slope of $\frac{3}{4}$.

What is the slope of a line perpendicular to p?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

Write an equation of the line through (0, 2) that is parallel to $y = -3x + 7$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

Which line is perpendicular to $y = 2x - 5$?

A. $y = -2x - 5$

B. $y = 2x + 1$

C. $y = -\frac{1}{2}x + 1$

D. $y = \frac{1}{2}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

Parallel lines that are not vertical have equal slopes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

True or false?

Lines with slopes 3 and −3 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a • EXPRESSIONS & EQUATIONS

Decide whether a linear equation has one solution, no solution, or infinitely many

MAIN	Grade 8 main question	**8.EE.C.7.a**
BACKWARD 1	Expand linear expressions	**7.EE.A.1**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Solve systems and recognize when they have no solution or infinit…	**HSA-REI.C.6**
FORWARD 2	Use the form $(x - p)^2 = q$ to find how many real solutions a quadra…	**HSA-REI.B.4.a**

SET 16 • 8.EE.C.7.a

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x + 3) = 4x + 12$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

How many solutions does this equation have?

$5x - 2 = 5x + 7$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

Which equation has exactly one solution?

A. $3(x + 1) = 3x + 3$

B. $2x - 2x = 0$

C. $3x + 1 = 2x + 9$

D. $3x + 1 = 3x + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

What number in the blank makes the equation have no solution?

$6x + 5 = ___x + 1$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

True or false?

$2(x - 4) = 2x - 4$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 1

Expand linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$4(x + 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

Expand.

$-2(3x - 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

Which expression is equal to $5(2x - 1)$?

A. $10x + 5$

B. $10x - 1$

C. $10x - 5$

D. $7x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$$3(x + 2) = 3x + 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$$-(x - 4) = -x - 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$3x + 7 = 22$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

Solve for x.

$-2x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

Solve for x.

$5x - 6 = 19$

A. $x = \frac{13}{5}$

B. $x = -5$

C. $x = 5$

D. $x = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x = 2$ is the solution of $4x + 1 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 1

Solve systems and recognize when they have no solution or infinitely many • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

How many solutions does the system have?

$y = 2x + 3$

$y = 2x - 1$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

Solve the system.

$x + y = 7$

$x - y = 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

How many solutions does the system have?

$2x + 4y = 8$

$x + 2y = 4$

A. Two

B. Infinitely many

C. One

D. None

True or false?

A system of two different parallel lines has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

True or false?

The system $y = x + 1$ and $y = -x + 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 2

Use the form $(x - p)^2 = q$ to find how many real solutions a quadratic equation has • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

How many real solutions does $(x - 3)^2 = 0$ have? Give any solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

How many real solutions does $(x + 1)^2 = -4$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

Completing the square on $x^2 + 6x + 13 = 0$ gives $(x + 3)^2 = q$.

What is q, and how many real solutions are there?

A. q = 4; two real solutions

B. q = −4; one real solution

C. q = 13; two real solutions

D. q = −4; no real solutions

True or false?

$(x - 2)^2 = 9$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

True or false?

$x^2 - 6x + 9 = 0$ has two different real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Solve multistep linear equations by expanding and collecting like terms

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Solve two-step equations with rational numbers	**7.EE.B.4.a**
FORWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
FORWARD 2	Justify each step of a solution and find an invalid step	**HSA-REI.A.1**

SET 17 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$3(2x - 4) = 4x + 6$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

Solve for x.

$5 - 2(x + 1) = 3x - 7$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

Solve for x.

$4(x - 1) - (x + 2) = 2x + 3$

A. $x = 1$

B. $x = 5$

C. $x = 9$

D. $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

Solve. Show each step.

$-3(x + 5) + 2x = 4(x - 10)$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

True or false?

$x = -1$ is the solution of $2(3x + 1) = 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$6x - 4 + 2x + 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

Simplify.

$-3a + 7 - a$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

Simplify.

$2(x + 3) + 5x$

A. $10x + 6$

B. $2x + 11$

C. $7x + 3$

D. $7x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$$4y - y = 3y$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$$5n + 2 - 2n = 3n$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 2

Solve two-step equations with rational numbers • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$6x - 5 = 31$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

Solve for x.

$\frac{x}{4} + 3 = 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

Solve for x.

$-3x + 8 = -10$

A. $x = -6$

B. $x = -\frac{2}{3}$

C. $x = \frac{2}{3}$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $5x + 12 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ is the solution of $3x - 2 = 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 1

Solve linear equations with letter coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Assume $a \neq 3$.

$ax + 2 = 3x + 8$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

Solve for x. Assume $k \neq 0$.

$k(x - 4) = 2k$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

Solve for x. Assume $b \neq 0$.

$bx - 5b = 2b$

A. $x = 7b$

B. $x = \frac{7}{b}$

C. $x = 7$

D. $x = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

True or false?

If $c \neq 1$, the solution of $cx + 1 = x + 1$ is $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

True or false?

The solution of $mx = m + 4$ is $x = 5$ for every nonzero value of m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 2

Justify each step of a solution and find an invalid step • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve $2(x - 3) + 4 = 5x - 11$. Give a reason for each step.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

Dee solved $4 - 2(x - 1) = 8$ like this:

Step 1: $4 - 2x - 2 = 8$

Step 2: $2 - 2x = 8$

Step 3: $x = -3$

Which step is wrong? Correct it and give the solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

Which reason justifies rewriting $3x - 5 = 7$ as $3x = 12$?

A. Addition property of equality

B. Multiplication property of equality

C. Commutative property of addition

D. Distributive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

True or false?

Each step of a valid solution gives an equation with the same solutions as the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Write and solve linear equations with rational coefficients for real-world problems

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Solve word problems that lead to px + q = r	**7.EE.B.4.a**
BACKWARD 2	Multiply rational numbers in context	**7.NS.A.3**
FORWARD 1	Solve systems of linear equations from real-world problems	**HSA-REI.C.6**
FORWARD 2	Create equations and inequalities in one variable to solve proble…	**HSA-CED.A.1**

SET 18 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Gym A charges $30 to join plus $12.50 per month. Gym B charges $55 to join plus $10 per month.

Write and solve an equation to find when the total costs are equal.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

Plant A is $6\frac{1}{2}$ inches tall and grows $\frac{3}{4}$ inch per week.

Plant B is 10 inches tall and grows $\frac{1}{4}$ inch per week.

Write and solve an equation to find when they are the same height.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

Solve for x.

$\frac{2}{3}x - 4 = \frac{1}{2}x + 1$

A. $x = 18$

B. $x = \frac{30}{7}$

C. $x = 30$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

Solve for x.

$0.4(x + 15) = 0.25x + 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x + 2 = \frac{1}{2}x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 1

Solve word problems that lead to px + q = r • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Ana has $40. She buys some $6.50 tickets and has $1 left.

How many tickets did she buy?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

A rental costs $20 plus $8 per hour. The total is $68.

For how many hours was it rented?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

Solve for x.

$0.5x + 3 = 11$

A. $x = 7$

B. $x = 16$

C. $x = 28$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

True or false?

$x = 9$ is the solution of $\frac{1}{3}x + 2 = 5$.

A. True

B. False

True or false?

$x = 4$ is the solution of $2.5x - 1 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 2

Multiply rational numbers in context • Grade 7 • The Number System • 7.NS.A.3

A plant grows $\frac{3}{4}$ inch per week.

How much does it grow in 8 weeks?

Growth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

A gym charges $12.50 per month.

How much do 10 months cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

A cyclist rides 0.4 mile per minute for 15 minutes.

How far does she ride?

A. 6 miles

B. 60 miles

C. 4 miles

D. 15.4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3}$ of 9 feet is 6 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

0.25 of a 12-hour day is 4 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 1

Solve systems of linear equations from real-world problems • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve the system.

$y = 12.5x + 30$

$y = 10x + 55$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

Adult tickets cost $8 and student tickets cost $5. 120 tickets were sold for $750.

How many adult tickets were sold?

Adult tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

Solve the system.

$2x + 3y = 12$

$x - y = 1$

A. $x = 1, y = 0$

B. $x = 3, y = 2$

C. $x = 4, y = 3$

D. $x = 2, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

True or false?

$x = 2, y = 5$ is a solution of both $y = 3x - 1$ and $y = x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

True or false?

The system $x + y = 10$ and $2x + 2y = 15$ has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 2

Create equations and inequalities in one variable to solve problems • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

Plan A costs $25 plus $0.15 per text. Plan B costs $40 with unlimited texts.

Write and solve an inequality to find when Plan A costs less.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

Two trains leave a station at the same time in opposite directions, at 55 and 65 miles per hour.

Write and solve an equation to find when they are 300 miles apart.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

A rectangle's length is 3 more than twice its width w. Its perimeter is 54.

Which equation can find w?

A. $2(2w) + 3 = 54$

B. $w + 2w + 3 = 54$

C. $2w + 2(2w + 3) = 54$

D. $w(2w + 3) = 54$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

True or false?

A rectangle has a length of 2w + 3 and a perimeter of 54. Its width w is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

True or false?

The inequality $25 + 0.15t < 40$ is true when $t = 120$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations by graphing

MAIN	Grade 8 main question	**8.EE.C.8.a**
BACKWARD 1	Make a table and graph an equation	**6.EE.C.9**
BACKWARD 2	Graph points in all four quadrants	**6.NS.C.8**
FORWARD 1	Solve f(x) = g(x) from the intersection of two graphs	**HSA-REI.D.11**
FORWARD 2	Graph the solution set of a system of linear inequalities	**HSA-REI.D.12**

SET 19 • 8.EE.C.8.a

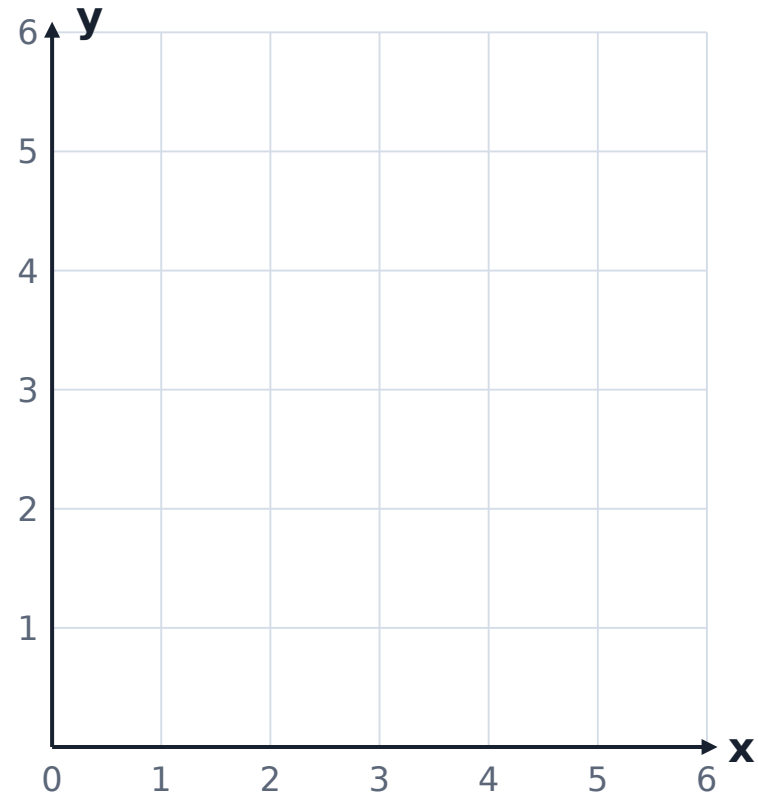
MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Graph $y = x + 1$ and $y = -x + 5$ on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

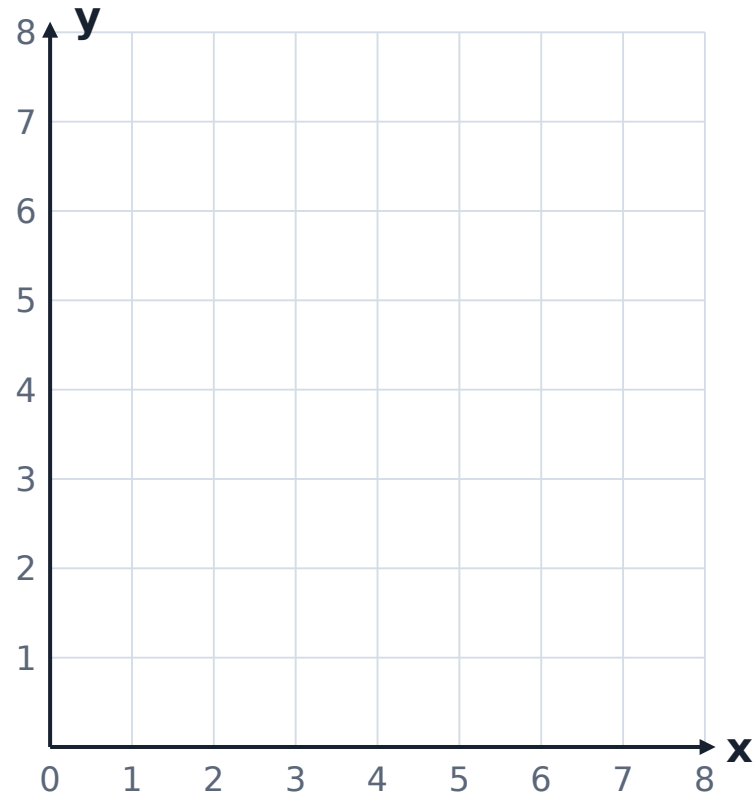
ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

Graph $y = 0.5x + 1$ and $y = -x + 7$ on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

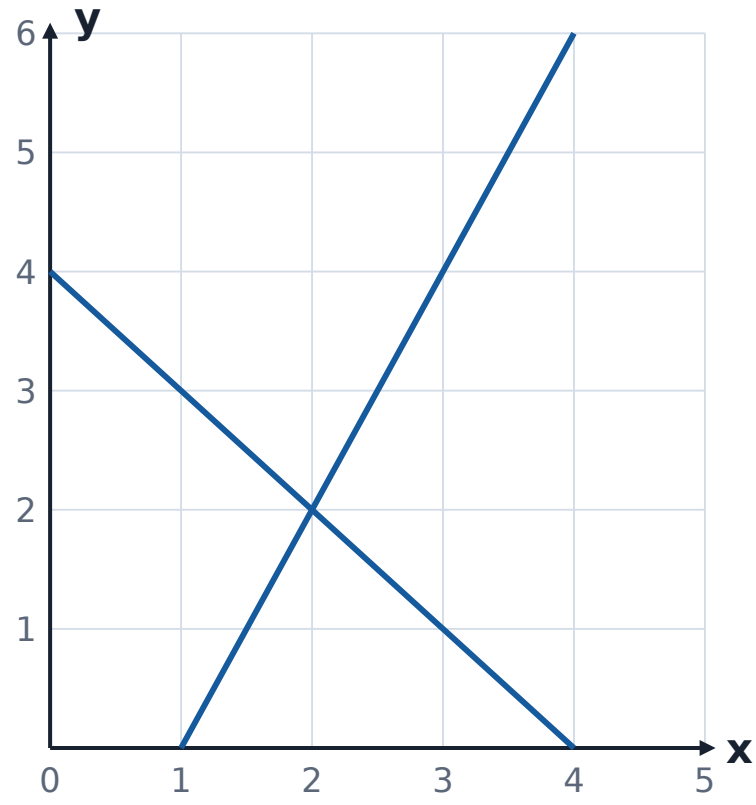
ID S19-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

The graph shows the lines $y = -x + 4$ and $y = 2x - 2$.

What is the solution of the system? Check it in both equations.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

The graphs of the two equations in a system are parallel lines.

What does this tell you about the system?

A. It has no solution.

B. It has infinitely many solutions.

C. It has one solution.

D. Its solution is the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

True or false?

When the graphs of two linear equations are two different lines that cross, the crossing point is the only solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

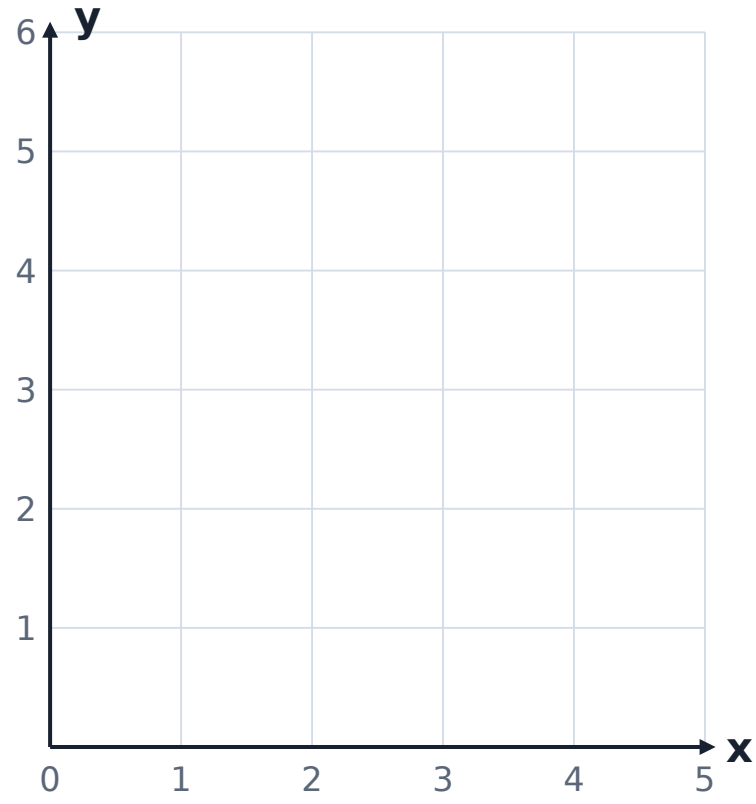
Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 1

Make a table and graph an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

Make a table for y = x + 2 with x = 0, 1, 2, and 3. Graph the points on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

What is the value of y when x = 4 in y = 3x?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 3?

A. (1, 3)

B. (3, 0)

C. (5, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

True or false?

(4, 8) is on the graph of $y = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of $y = x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

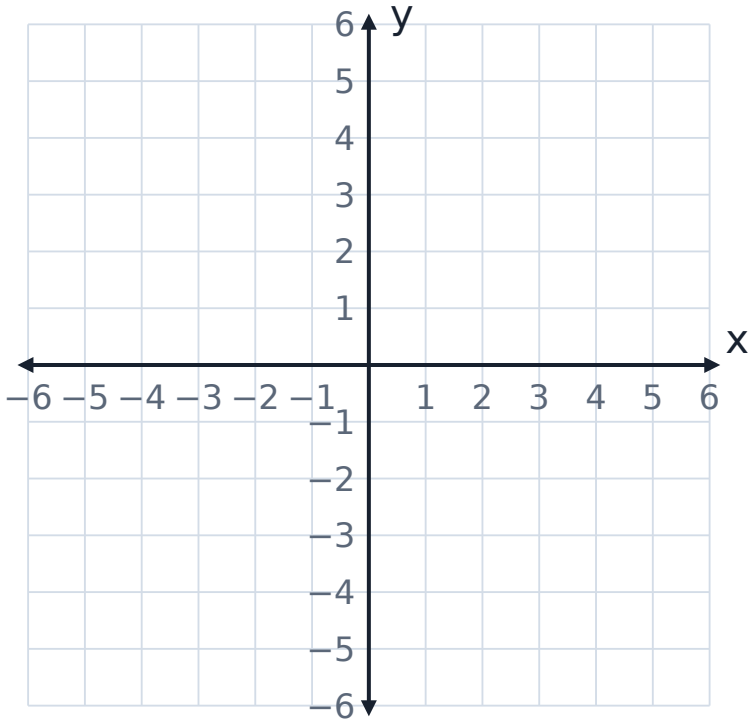
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 2

Graph points in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Plot A(2, −3) and B(−4, 1) on the coordinate plane.

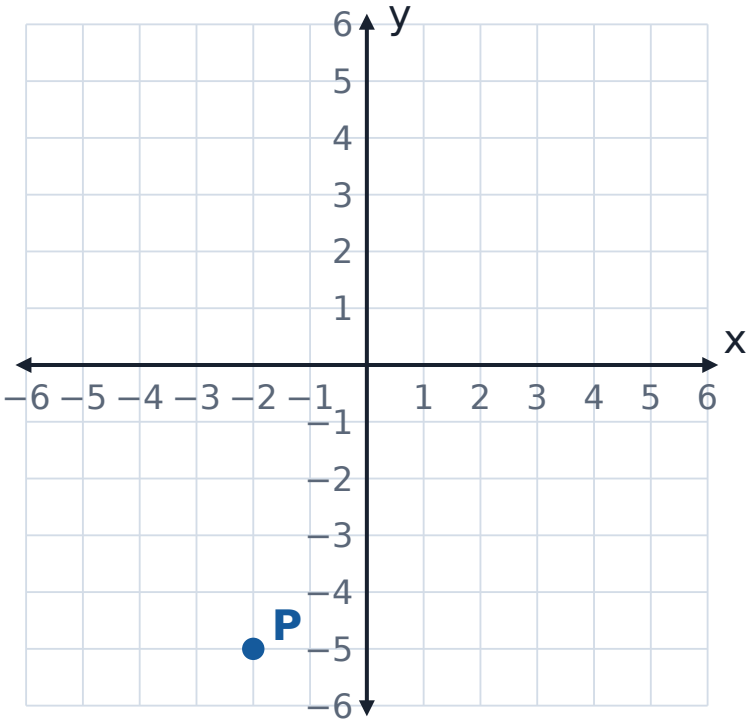


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

What are the coordinates of point P?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

Which point is in Quadrant II?

A. (3, −4)

B. (−3, −4)

C. (3, 4)

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

(0, −2) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

(5, −1) is in Quadrant III.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a

FORWARD 1

Solve f(x) = g(x) from the intersection of two graphs • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $f(x) = x + 1$ and $g(x) = 7 - x$ intersect at $(3, 4)$.

What is the solution of $x + 1 = 7 - x$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

Use a table of values for $x = 0, 1, 2$, and 3 to solve $2x + 1 = 7 - x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ intersect at $(2, 4)$ and $(-1, 1)$.

What are the solutions of $x^2 = x + 2$?

A. $x = 2$ only

B. $x = -1$ only

C. $x = 4$ and $x = 1$

D. $x = 2$ and $x = -1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

True or false?

If the graphs of $y = f(x)$ and $y = g(x)$ intersect at $(5, 9)$, then $f(5) = g(5) = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

True or false?

If the graphs of f and g never meet, then f(x) = g(x) has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

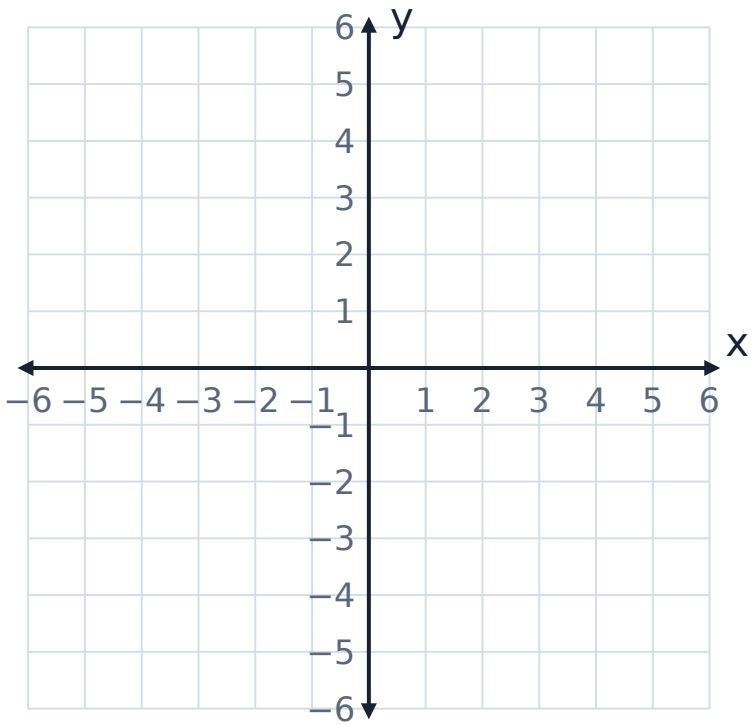
FORWARD 2

Graph the solution set of a system of linear inequalities • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph the system on the coordinate plane.

$y \geq x$

$y < 4$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

Is (1, 3) a solution of the system $y > 2x$ and $y < x + 3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

Which point is a solution of the system $y \leq -x + 4$ and $y > 1$?

A. (1, 2)

B. (4, 3)

C. (3, 2)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

True or false?

The boundary line of $y < 2x + 1$ is drawn dashed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

True or false?

The solution set of a system of two inequalities is every point that satisfies at least one of them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations algebraically

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve equations with the distributive property	**7.EE.B.4.a**
FORWARD 1	Explain why adding equations (elimination) keeps the same solut…	**HSA-REI.C.5**
FORWARD 2	Solve a system of a linear equation and a quadratic equation	**HSA-REI.C.7**

SET 20 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve the system. Show your work.

$y = 3x - 2$

$2x + y = 13$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

Solve the system. Show your work.

$x + 2y = 10$

$3x - 2y = 6$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

Solve the system.

$y = x + 4$

$y = -2x + 1$

A. $x = 1, y = 5$

B. $x = 3, y = 7$

C. $x = -1, y = 5$

D. $x = -1, y = 3$

Solve the system.

$4x + 3y = 18$

$2x - y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

True or false?

$x = 2, y = -1$ is a solution of both $3x + y = 5$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 2x + y when x = 3 and y = 7.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

Evaluate $3x - 2$ when $x = 4$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

Evaluate x + 2y when x = 4 and y = 3.

A. 9

B. 10

C. 11

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

True or false?

When $x = 2$, the value of $5x - 1$ is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

True or false?

When y = 3, the value of 2y + 4 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 2

Solve equations with the distributive property • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$5x - 2 = 13$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

Solve for x.

$2x + 3(x - 2) = 9$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

Solve for x.

$4x + 2 = -10$

A. $x = 3$

B. $x = -2$

C. $x = -3$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

True or false?

$x = 4$ is the solution of $3x - 5 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

True or false?

$x = -1$ is the solution of $2x + 9 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 1

Explain why adding equations (elimination) keeps the same solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

Add the equations $x + y = 9$ and $x - y = 3$.

What new equation do you get, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

Multiply $x + 2y = 7$ by 3, then add the result to $-3x + y = 0$.

What equation results?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

In a system, one equation is replaced by its sum with a multiple of the other equation.

What happens to the solutions?

A. The solutions always change.

B. The number of solutions doubles.

C. All solutions are removed.

D. The solutions stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

The system $x + y = 5$ and $x - y = 1$ has the same solution as the system $x + y = 5$ and $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

Multiplying one equation of a system by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 2

Solve a system of a linear equation and a quadratic equation • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

Solve the system. Give all solutions.

$y = x^2$

$y = 4$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

Solve the system. Give all solutions.

$y = x^2 - 1$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

How many solutions does the system have?

$y = x^2 + 3$

$y = 1$

A. Infinitely many

B. Two

C. One

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q3**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

True or false?

The line $y = 2x$ and the parabola $y = x^2$ meet at $(0, 0)$ and $(2, 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

True or false?

A line and a parabola always meet at exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve simple systems by inspection

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Recognize equivalent expressions	**7.EE.A.1**
BACKWARD 2	Decide whether a value makes two equations true	**6.EE.B.5**
FORWARD 1	Solve or classify systems of linear equations	**HSA-REI.C.6**
FORWARD 2	Classify lines as parallel, perpendicular, or neither from their equ..	**HSG-GPE.B.5**

SET 21 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Without solving, explain why this system has no solution.

$3x + 2y = 5$

$3x + 2y = 6$

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

Without solving, how many solutions does this system have? Explain.

$y = 4x - 1$

$y = 4x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

Which system has infinitely many solutions?

A. $y = x$ and $y = -x$

B. $x + y = 4$ and $x + y = 8$

C. $x + y = 4$ and $x - y = 4$

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

Without solving, how many solutions does this system have? Explain.

$y = -2x + 5$

$y = 3x + 5$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

True or false?

The system $2x - y = 7$ and $6x - 3y = 21$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 1

Recognize equivalent expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Is 2x + 2y equivalent to 2(x + y)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

Write 3(2x + y) without parentheses.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

Which expression is equivalent to 4x + 6y?

A. 6(4x + y)

B. 4(x + 6y)

C. 2(4x + 6y)

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

True or false?

$$5x - 5y = 5(x - y)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

True or false?

$$2(x + 3) = 2x + 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 2

Decide whether a value makes two equations true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 2 a solution of both x + 3 = 5 and 2x = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 makes both $2x = 6$ and $x + 1 = 4$ true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

Which value makes both x + 2 = 7 and 3x = 15 true?

A. 2

B. 7

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

True or false?

No value of x makes both $x + 1 = 3$ and $x + 1 = 4$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ makes both $x - 1 = 3$ and $2x = 6$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 1

Solve or classify systems of linear equations • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve the system, or say that it has no solution or infinitely many.

$2x + y = 3$

$4x + 2y = 6$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

Solve the system.

$3x - y = 5$

$x + y = 7$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

Which value of k makes the system have infinitely many solutions?

$x + ky = 4$

$2x + 6y = 8$

A. 4

B. 2

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

True or false?

If the two equations of a system describe the same line, every point on the line is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

True or false?

The system $y = 3x + 1$ and $y = 3x + 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 2

Classify lines as parallel, perpendicular, or neither from their equations • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are the lines $y = 2x + 1$ and $4x - 2y = 6$ parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

Are the lines $3x + y = 2$ and $x - 3y = 9$ parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

One line passes through (0, 0) and (2, 4). Another passes through (1, 1) and (3, 5).

How are the lines related?

A. Perpendicular

B. The same line

C. Parallel

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

True or false?

Lines with slopes $\frac{2}{3}$ and $-\frac{3}{2}$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

True or false?

The lines $y = x$ and $y = -x$ are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c • EXPRESSIONS & EQUATIONS

Write and solve systems of linear equations for real-world problems

MAIN	Grade 8 main question	**8.EE.C.8.c**
BACKWARD 1	Solve word problems with one unknown	**7.EE.B.4.a**
BACKWARD 2	Use variables to represent quantities in a situation	**6.EE.B.6**
FORWARD 1	Create equations in two variables to model and solve problems	**HSA-CED.A.2**
FORWARD 2	Represent constraints with a system of inequalities	**HSA-CED.A.3**

SET 22 • 8.EE.C.8.c

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.c

A school sells 200 tickets. Adult tickets cost $6 and student tickets cost $4. The sales total $940.

Write a system and solve it to find how many of each ticket were sold.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

Plan A costs $15 plus $0.05 per minute. Plan B costs $0.10 per minute.

Write a system and solve it to find when the plans cost the same.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all.

Which system models this?

A. $d - q = 25$ and $0.10d + 0.25q = 4$

B. $d + q = 25$ and $0.10d + 0.25q = 4$

C. $d + q = 4$ and $0.10d + 0.25q = 25$

D. $d + q = 25$ and $0.25d + 0.10q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all: d + q = 25 and 0.10d + 0.25q = 4.

How many of each coin does Jo have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

True or false?

Two numbers have a sum of 40 and a difference of 12. The larger number is 26.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 1

Solve word problems with one unknown • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride costs $19.

How many miles long was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

Concert tickets cost $9 each, plus a $5 fee for the whole order. An order costs $50.

How many tickets were bought?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

The sum of a number and 3 times the number is 48.

What is the number?

A. 16

B. 45

C. 12

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

True or false?

If 2x + 7 = 25, then x = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

True or false?

Items cost $4 each, and shipping is $6. A $30 order has 7 items.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 2

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

Write an expression for the value, in dollars, of d dimes.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Adult tickets cost $6.

Write an expression for the cost of a adult tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

There are 25 coins: d dimes and the rest quarters.

Which expression gives the number of quarters?

A. $d - 25$

B. $25d$

C. $25 - d$

D. $d + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

True or false?

q quarters are worth 0.25q dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

There are s students and 3 more teachers than students. There are 3s teachers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 1

Create equations in two variables to model and solve problems • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

A company's cost is C = 2,000 + 15x and its revenue is R = 40x for x items.

Write and solve an equation to find the break-even number of items.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

A boat travels 24 miles downstream in 2 hours and 24 miles back upstream in 3 hours. Let b be the boat's speed in still water and c the speed of the current.

Write a system of two equations.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

A boat's trips give the system $2(b + c) = 24$ and $3(b - c) = 24$.

What is the boat's speed b in still water?

A. 10 miles per hour

B. 2 miles per hour

C. 12 miles per hour

D. 8 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

True or false?

For the system $2(b + c) = 24$ and $3(b - c) = 24$, the current's speed c is 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

True or false?

The break-even point is where revenue is twice the cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 2

Represent constraints with a system of inequalities • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

A bakery has 30 pounds of flour. Each batch of bread uses 2 pounds and each batch of muffins uses 0.5 pound.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

A bakery has at most 20 hours of oven time. Each batch of bread takes 1 hour and each batch of muffins takes 2 hours.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

A bakery's limits are $2b + 0.5m \leq 30$ (flour) and $b + 2m \leq 20$ (oven hours).

Can it make 10 batches of bread and 5 batches of muffins?

A. No; both limits are broken.

B. No; the flour runs out.

C. Yes; both limits are met.

D. No; the oven time runs out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

In a model of batches of bread b and muffins m, the constraints $b \geq 0$ and $m \geq 0$ make sense because batches cannot be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The point (12, 6) meets both limits $2b + 0.5m \leq 30$ and $b + 2m \leq 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 8.F.A.1 • FUNCTIONS

Decide whether a relation is a function

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Find the output of a rule	**6.EE.C.9**
BACKWARD 2	Read points in the first quadrant	**5.G.A.2**
FORWARD 1	Use function notation and find the domain and range	**HSF-IF.A.1**
FORWARD 2	Find the inverse of a linear function	**HSF-BF.B.4.a**

SET 23 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

True or false?

The table shows y as a function of x.

x	1	2	3	4
y	5	5	7	9

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

Is the relation in the table a function? Explain.

x	2	3	3	5
y	4	6	8	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

Which set of ordered pairs is a function?

A. (5, 1), (6, 2), (5, 3)

B. (0, 0), (0, 1), (1, 1)

C. (1, 2), (1, 3), (2, 4)

D. (1, 4), (2, 4), (3, 4)

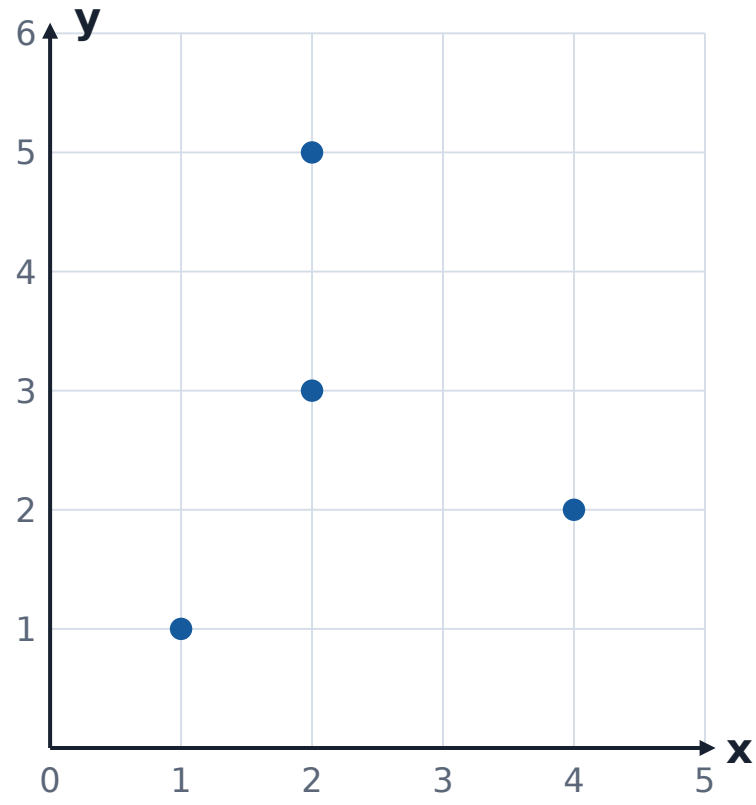
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

Do the points on the graph show y as a function of x? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

True or false?

A rule assigns each month of the year to the students in a class born in that month. This rule is a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

SET 23 • 8.F.A.1

BACKWARD 1

Find the output of a rule • Grade 6 • Expressions & Equations • 6.EE.C.9

The rule is y = 2x + 1.

What is y when x = 4?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

The table follows a pattern.

What is y when x = 4?

x	1	2	3
y	4	7	10

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

The rule is: output = input × 3.

What is the output for an input of 7?

A. 4

B. 73

C. 21

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

True or false?

For $y = x - 5$, an input of $x = 9$ gives $y = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

True or false?

For $y = 4x$, an input of $x = 3$ gives $y = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

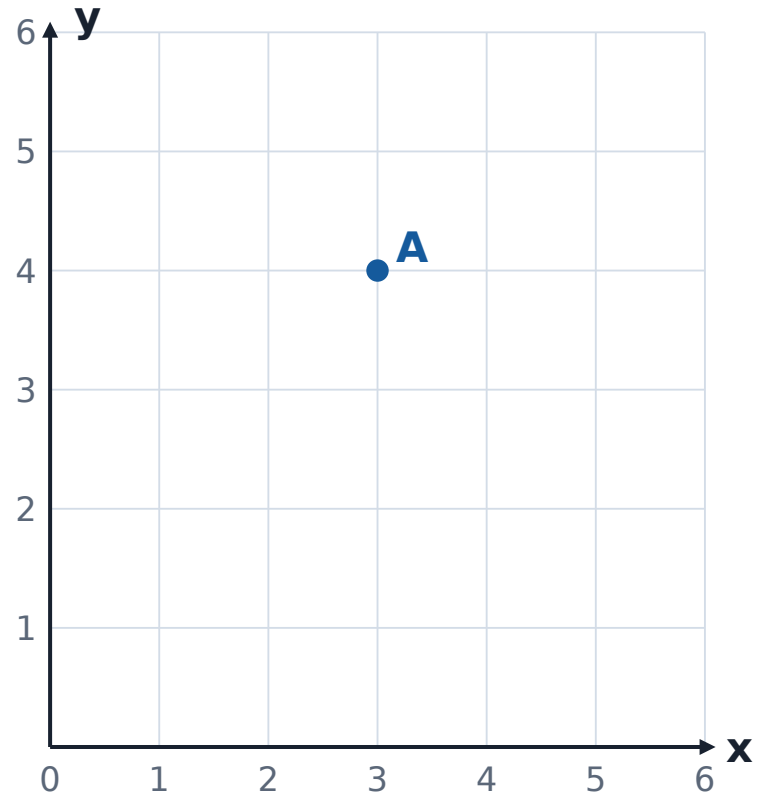
SET 23 • 8.F.A.1

BACKWARD 2

Read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point A?

Coordinates: _____



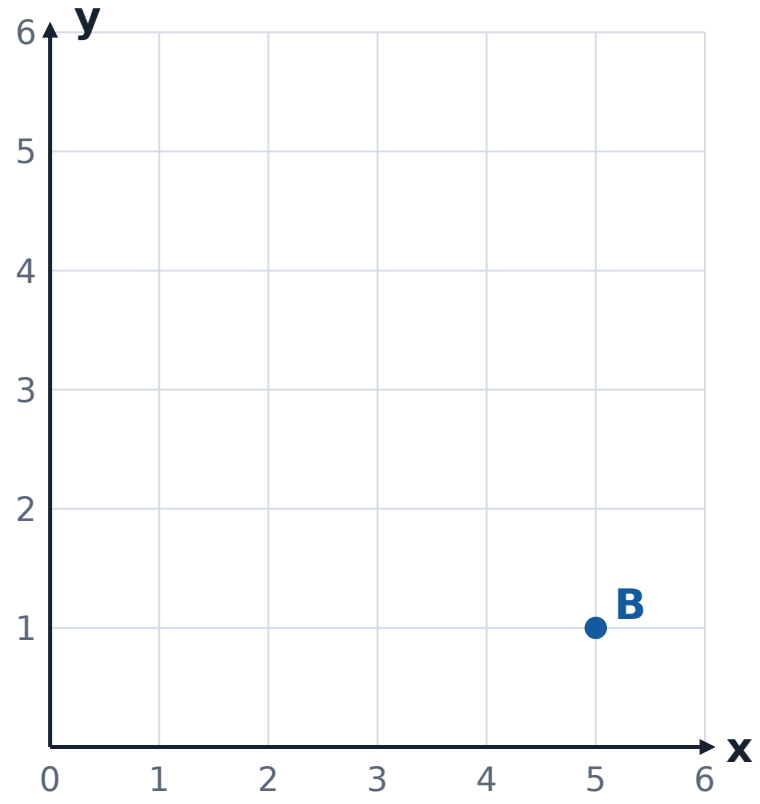
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

What are the coordinates of point B?

Coordinates: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

Which point is 2 units to the right of the origin and 5 units up?

A. (5, 5)

B. (5, 2)

C. (2, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

True or false?

The point (0, 3) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

True or false?

The point (4, 0) is 4 units up from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 8.F.A.1

FORWARD 1

Use function notation and find the domain and range • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its range?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

The output of f is 7 when the input is 2.

Which statement says this in function notation?

A. $f = 2 \times 7$

B. $f(7) = 2$

C. $2f = 7$

D. $f(2) = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x^2$, then $f(-3) = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

True or false?

A function can assign two different outputs to the same input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 8.F.A.1

FORWARD 2

Find the inverse of a linear function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = 2x + 3$

Write the inverse function.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

$f(x) = 2x + 3$

Solve $f(x) = 11$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

$g(x) = \frac{x}{5}$

What is the inverse function?

A. $g^{-1}(x) = x + 5$

B. $g^{-1}(x) = x - 5$

C. $g^{-1}(x) = 5x$

D. $g^{-1}(x) = \frac{5}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q3**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

True or false?

If $f(2) = 7$ and f has an inverse, then $f^{-1}(7) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

SET 24 • 8.F.A.1 • FUNCTIONS

Graph a function as the set of ordered pairs of inputs and outputs

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Make ordered pairs from an equation	**6.EE.C.9**
BACKWARD 2	Graph points from a pattern	**5.G.A.2**
FORWARD 1	Choose a domain that fits a function's situation	**HSF-IF.B.5**
FORWARD 2	Graph quadratic functions and show their key features	**HSF-IF.C.7.a**

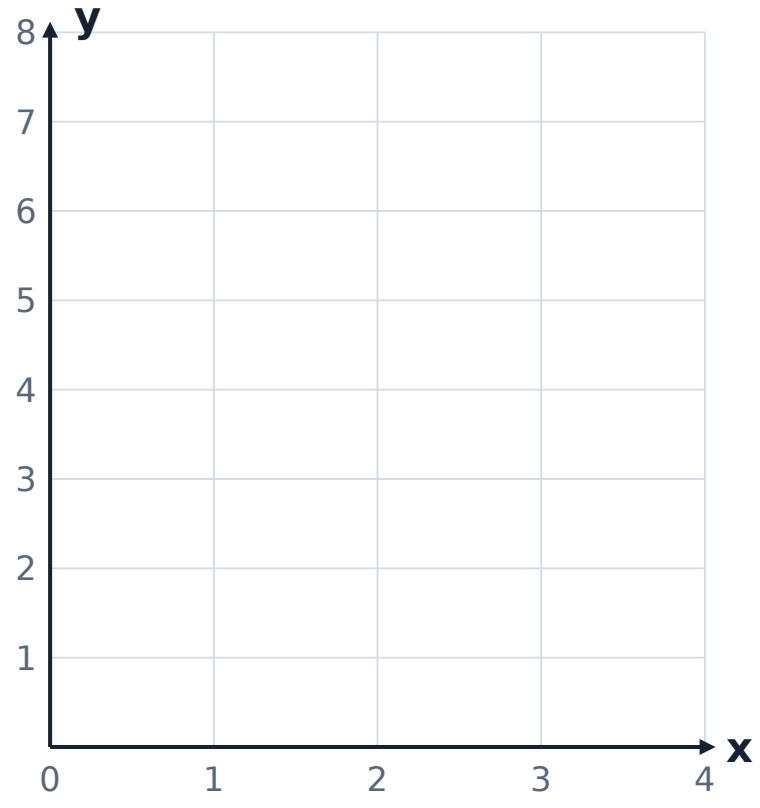
SET 24 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

The function $y = 2x + 1$ has the inputs 0, 1, 2, and 3.

Plot the ordered pairs (input, output) on the grid.



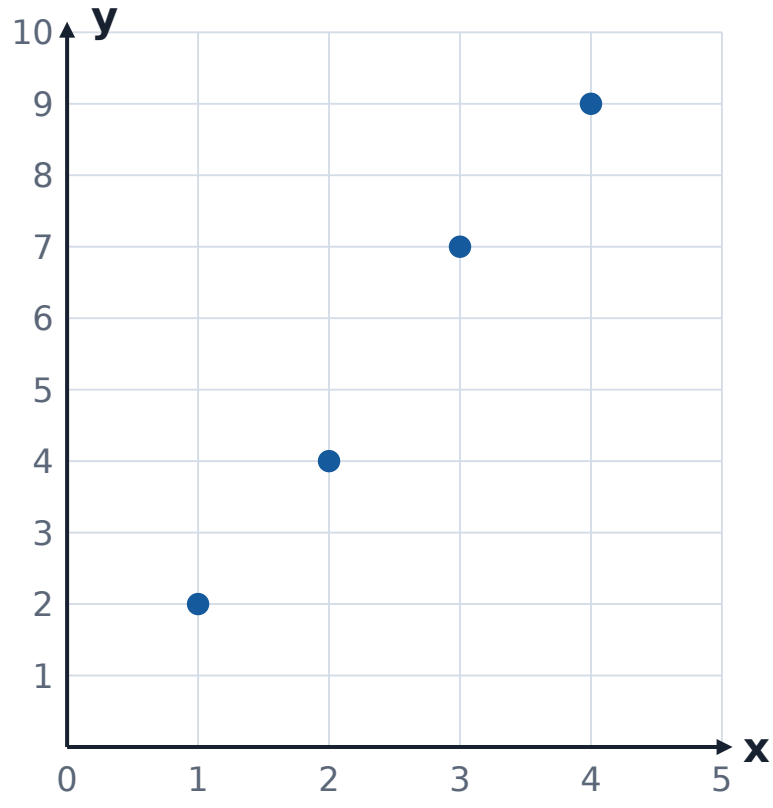
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

The points show a function.

What is the output when the input is 3?



Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

The graph of a function contains the point (4, 10).

Which statement is true?

A. When the input is 4, the output is 10.

B. When the input is 10, the output is 4.

C. The graph has only one point.

D. The function adds 4 to every input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

The rule is "output = input squared."

Write the ordered pairs (input, output) for the inputs −2, −1, 0, 1, and 2.

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

True or false?

The graph of a function can contain both (2, 5) and (2, 8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

SET 24 • 8.F.A.1

BACKWARD 1

Make ordered pairs from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 3x, what are the values of y when x = 0, 1, and 2?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

For y = x + 4, what is y when x = 5?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

Which pair (x, y) fits y = 5x?

A. (2, 7)

B. (10, 2)

C. (2, 10)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

True or false?

The pair (3, 9) fits $y = 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

The pair (4, 6) fits $y = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

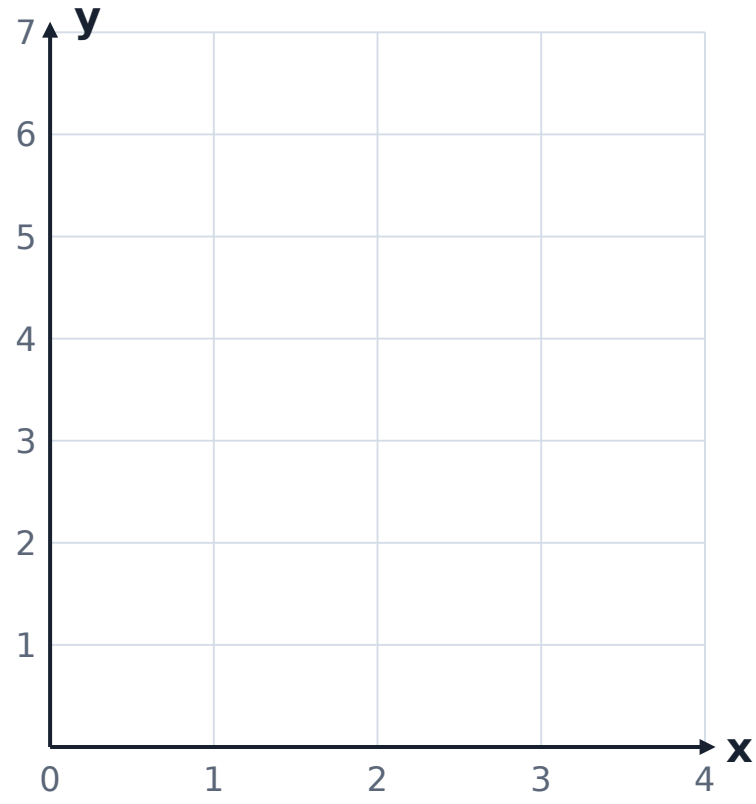
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 8.F.A.1

BACKWARD 2

Graph points from a pattern • Grade 5 • Geometry • 5.G.A.2

Plot the points (1, 2), (2, 4), and (3, 6) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

The y-values start at 1 and increase by 2: (1, 1), (2, 3), (3, 5), ...

What is the point for x = 4?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which point is 3 units to the right of the origin and 1 unit up?

A. (3, 1)

B. (3, 3)

C. (1, 3)

D. (1, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

The point (2, 6) is 2 units to the right of the origin and 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

To plot (5, 2), you move 5 units up and 2 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 8.F.A.1

FORWARD 1

Choose a domain that fits a function's situation • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5

A function gives the cost of n concert tickets at $12 each. A buyer may buy at most 8 tickets.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

h(t) gives the height of a ball t seconds after it is thrown, until it lands at t = 3.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

p is the number of people in a car that seats 5.

Which domain makes sense for p?

A. $-5 \leq p \leq 5$

B. 0, 1, 2, 3, 4, 5

C. All real numbers

D. $p \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

True or false?

For the area of a square as a function of its side length, the domain contains only positive numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

True or false?

A function that gives the number of buses needed for n students should have all real numbers as its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

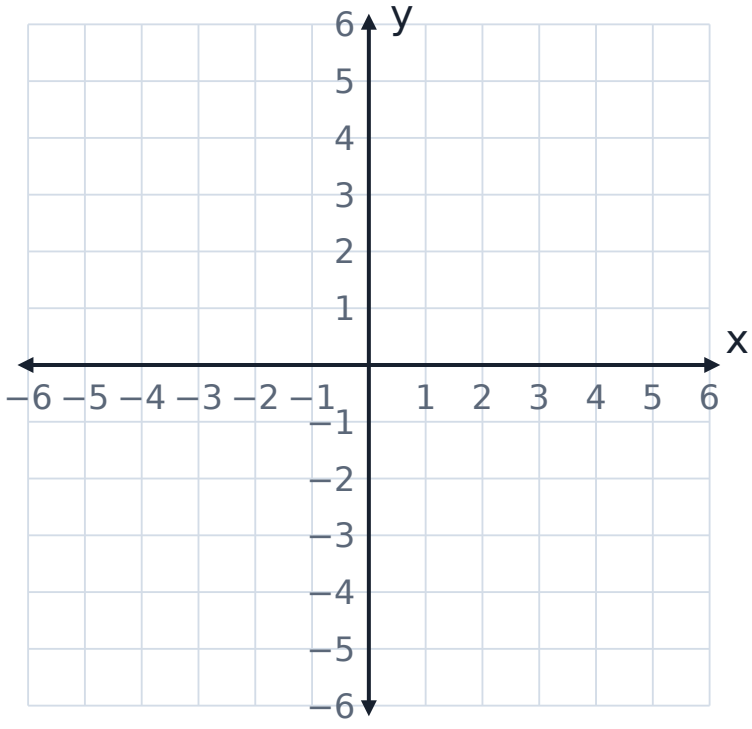
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 8.F.A.1

FORWARD 2

Graph quadratic functions and show their key features • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $y = x^2 - 4$ on the coordinate plane. Mark the vertex and the x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

What is the vertex of the graph of $y = (x - 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

Where does the graph of $y = x^2 - 9$ cross the x-axis?

A. At (3, 0) and (−3, 0)

B. At (9, 0) and (−9, 0)

C. At (0, −9)

D. At (3, 0) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -x^2$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 2x + 3$ has a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

SET 25 • 8.F.A.2 • FUNCTIONS

Compare two functions represented in different ways

MAIN	Grade 8 main question	**8.F.A.2**
BACKWARD 1	Find a unit rate from a table, graph, or equation	**7.RP.A.2.b**
BACKWARD 2	Find the starting value of a relationship	**6.EE.C.9**
FORWARD 1	Compare average rates of change of linear and nonlinear function…	**HSF-IF.B.6**
FORWARD 2	Describe the effect of transformations on the graph of a function	**HSF-BF.B.3**

SET 25 • 8.F.A.2

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.2

$f(x) = 3x + 2$. The table shows g.

Which function has the greater rate of change?

x	0	2	4
g(x)	1	9	17

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

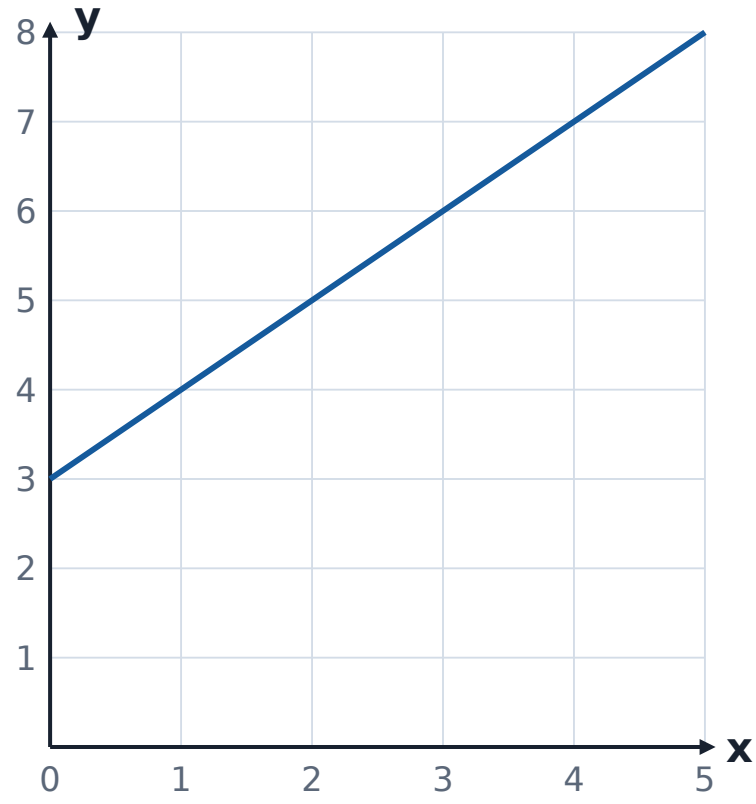
ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

The graph shows f.

$g(x) = -2x + 5$.

Which function has the greater initial value?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

Function A is $y = 0.5x + 10$. Function B is a line through $(0, 4)$ and $(2, 8)$.

Which statement is true?

A. B has the greater rate of change; A has the greater initial value.

B. B has the greater rate of change and the greater initial value.

C. A has the greater rate of change and the greater initial value.

D. They have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

Car A travels d = 45t miles in t hours. The table shows Car B.

Which car is faster, and by how much?

Hours	1	2	3
Miles	50	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

True or false?

A function with x-values 0, 1, 2 and y-values 6, 4, 2 decreases faster than $y = -x + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

SET 25 • 8.F.A.2

BACKWARD 1

Find a unit rate from a table, graph, or equation • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the unit rate?

x	2	4	6
y	5	10	15

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

What is the unit rate in y = 7x?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

A proportional graph passes through (0, 0) and (3, 12).

What is the unit rate?

A. 36

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (1, 9) and (2, 18) has a unit rate of 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The unit rate in $y = 0.5x$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 8.F.A.2

BACKWARD 2

Find the starting value of a relationship • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 4x + 7, what is y when x = 0?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

For the table, what is y when x = 0?

x	0	1	2
y	9	11	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

Which equation gives y = 5 when x = 0?

A. y = 5x + 2

B. y = 2x + 5

C. y = 5x

D. y = x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

True or false?

For $y = 3x + 1$, $y = 1$ when $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

For $y = 6x$, $y = 6$ when $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 8.F.A.2

FORWARD 1

Compare average rates of change of linear and nonlinear functions over intervals • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. The table shows g.

Find the average rate of change of each function from $x = 0$ to $x = 3$.

x	0	1	2	3
g(x)	0	3	6	9

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

$f(x) = 2^x$ and $g(x) = 3x + 5$.

Find the average rate of change of each function from $x = 0$ to $x = 4$. Which is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

$f(x) = x^2$ and $g(x) = 4x$.

Which statement about the average rates of change from $x = 1$ to $x = 3$ is true?

A. f's rate is greater.

B. f's rate is 9 and g's rate is 12.

C. g's rate is greater.

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

True or false?

The average rate of change of $f(x) = -x^2 + 4$ from $x = 0$ to $x = 2$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

True or false?

The average rate of change of a linear function depends on which interval you use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 8.F.A.2

FORWARD 2

Describe the effect of transformations on the graph of a function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3

How does the graph of g(x) = f(x) + 3 compare with the graph of f?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

How does the graph of $g(x) = (x - 2)^2$ compare with the graph of $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

Which function shifts the graph of $y = |x|$ down 4 units?

A. $y = |x| - 4$

B. $y = |x + 4|$

C. $y = 4|x|$

D. $y = |x - 4|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

The graph of $y = f(x + 5)$ is the graph of $y = f(x)$ shifted 5 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 8.F.A.3 • FUNCTIONS

Decide whether a function is linear or nonlinear

MAIN	Grade 8 main question	**8.F.A.3**
BACKWARD 1	Decide whether a table shows a proportional relationship	**7.RP.A.2.a**
BACKWARD 2	Find how much y changes for each change in x	**6.EE.C.9**
FORWARD 1	Show that linear functions grow by equal differences and expone..	**HSF-LE.A.1.a**
FORWARD 2	Use factoring to find the zeros and maximum of a quadratic functi..	**HSF-IF.C.8.a**

SET 26 • 8.F.A.3

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.3

Which equation is a linear function?

A. $y = 4x - 7$

B. $y = x^2 + 1$

C. $y = 2^x$

D. $y = \frac{3}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

Is the function in the table linear? Explain.

x	0	1	2	3
y	1	2	5	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

$A = s^2$ gives the area of a square with side length s.

Is A a linear function of s? Use three points to support your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

Is the function in the table linear? Explain.

x	0	2	4	6
y	7	4	1	−2

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

True or false?

$y = 3(x + 2)$ is a linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

SET 26 • 8.F.A.3

BACKWARD 1

Decide whether a table shows a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	1	2	3
y	4	8	12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	3
y	3	5	7

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

Which table shows a proportional relationship?

A. x: 0, 1, 2 and y: 1, 2, 3

B. x: 1, 2, 3 and y: 1, 4, 9

C. x: 1, 2, 3 and y: 2, 3, 4

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

The pairs (5, 2) and (10, 4) are from a proportional relationship.

What is the constant of proportionality (y per x)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

True or false?

$y = x + 1$ is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 8.F.A.3

BACKWARD 2

Find how much y changes for each change in x • Grade 6 • Expressions & Equations • 6.EE.C.9

How much does y increase each time x increases by 1?

x	0	1	2	3
y	5	8	11	14

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

How much does y increase each time x increases by 2?

x	0	2	4
y	1	5	9

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

In y = 6x, how much does y increase when x increases by 1?

A. 1

B. 7

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

True or false?

In the table with x-values 1, 2, 3 and y-values 10, 20, 30, y increases by 10 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

True or false?

In $y = x + 9$, y increases by 9 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 8.F.A.3

FORWARD 1

Show that linear functions grow by equal differences and exponential functions by equal factors • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

Show that f(x) = 4x + 1 grows by the same amount over every interval of length 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

Show that g(x) = 3^x grows by the same factor over every interval of length 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

Which function grows by equal factors over equal intervals?

A. $y = 2x^2$

B. $y = x + 5$

C. $y = 5x + 2$

D. $y = 5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

True or false?

The linear function $f(x) = mx + b$ changes by $m(x_2 - x_1)$ from x_1 to x_2.

A. True

B. False

True or false?

$y = x^2$ grows by equal differences over equal intervals.

A. True

B. False

SET 26 • 8.F.A.3

FORWARD 2

Use factoring to find the zeros and maximum of a quadratic function • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a

A ball's height is $h(t) = -16t^2 + 32t = -16t(t - 2)$ feet after t seconds.

Use the factored form to find when the ball is at height 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

A ball's height is $h(t) = -16t(t - 2)$ feet after t seconds. Its greatest height is halfway between the zeros.

When does it reach its greatest height, and what is that height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

$f(x) = x^2 - 6x + 8 = (x - 2)(x - 4)$

What are the zeros of f?

A. 2 and 4

B. -2 and -4

C. 8 only

D. 6 and 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

True or false?

The graph of a quadratic function is symmetric about a vertical line through its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

True or false?

The zeros of $f(x) = (x - 3)(x + 5)$ are 3 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 8.F.B.4 • FUNCTIONS

Find the rate of change and initial value from two points, a table, or a graph

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Find a constant rate from a table	**7.RP.A.2.b**
BACKWARD 2	Subtract integers	**7.NS.A.1.c**
FORWARD 1	Construct linear and exponential functions from two points	**HSF-LE.A.2**
FORWARD 2	Write arithmetic and geometric sequences recursively and explicit..	**HSF-BF.A.2**

SET 27 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A linear function passes through (2, 7) and (5, 16).

What are its rate of change and its initial value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

The table shows a linear function.

What are its rate of change and its initial value?

x	1	3	5
y	10	16	22

Answer: _____

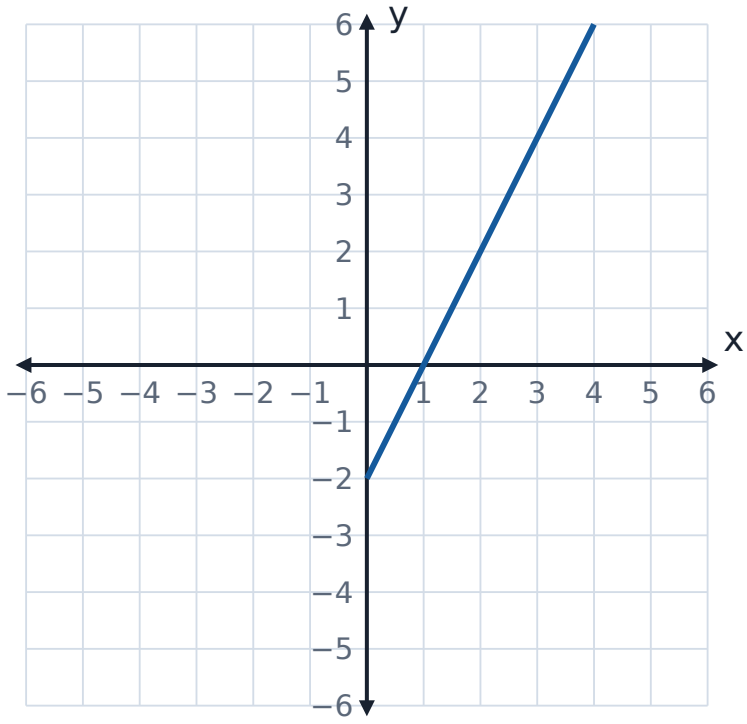
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

The graph shows a linear function.

What are its rate of change and its initial value?



A. Rate 4, initial value −2

B. Rate −2, initial value 2

C. Rate 2, initial value −2

D. Rate 2, initial value 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

A line passes through $(-1, 8)$ and $(3, 0)$.

Write its equation in the form $y = mx + b$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

True or false?

A linear function with f(0) = 5 and f(2) = 1 has a rate of change of −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

SET 27 • 8.F.B.4

BACKWARD 1

Find a constant rate from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant rate (y per x)?

x	3	6	9
y	12	24	36

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

What is the constant rate (y per x)?

Hours	2	5	8
Miles	7	17.5	28

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

A proportional table contains the pair (4, 10).

What is the constant rate (y per x)?

A. 0.4

B. 40

C. 2.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (2, 1) and (6, 3) has a constant rate of 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

True or false?

A proportional table with the pairs (5, 15) and (8, 24) has a constant rate of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 8.F.B.4

BACKWARD 2

Subtract integers • Grade 7 • The Number System • 7.NS.A.1.c

Subtract.

$6 - (-2)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

Subtract.

$-3 - 5$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

Subtract.

$-4 - (-9)$

A. 5

B. 13

C. -13

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

True or false?

$2 - 7 = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

True or false?

$-6 - (-6) = -12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 8.F.B.4

FORWARD 1

Construct linear and exponential functions from two points • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the linear function whose graph passes through (1, 4) and (3, 10).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

Write the exponential function f(x) = a · b^x whose graph passes through (0, 5) and (1, 15).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

Which exponential function passes through (0, 2) and (2, 18)?

A. f(x) = 8x + 2

B. f(x) = 2 · 3^x

C. f(x) = 18 · 2^x

D. f(x) = 2 · 9^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q3**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

True or false?

The linear function through $(0, -1)$ and $(4, 7)$ is $f(x) = 2x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

True or false?

The exponential function through (0, 4) and (1, 2) is f(x) = 4 · 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

SET 27 • 8.F.B.4

FORWARD 2

Write arithmetic and geometric sequences recursively and explicitly • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula for the arithmetic sequence 7, 11, 15, ..., starting with n = 1.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

Write a recursive formula for the sequence 7, 11, 15, ...

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

Which explicit formula gives 3, 6, 12, 24, ..., starting with n = 1?

A. a(n) = 2 · 3^(n − 1)

B. a(n) = 3 · 2^(n − 1)

C. a(n) = 3 + 2n

D. a(n) = 3n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function whose domain is the counting numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

The sequence 2, 5, 8, 11, ... has a common ratio of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 8.F.B.4 • FUNCTIONS

Construct a linear function to model a situation

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Write an equation for a situation	**6.EE.C.9**
BACKWARD 2	Solve a two-step equation from a model	**7.EE.B.4.a**
FORWARD 1	Write a function for a situation and use it to answer a question	**HSF-BF.A.1.a**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 28 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A plumber charges $65 for a visit plus $40 per hour.

Write a function for the cost C after h hours.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

A candle is 30 cm tall and burns down 2.5 cm each hour.

Write a function for its height h after t hours.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

A phone plan costs $20 per month after a $50 activation fee.

Which function gives the total cost C after m months?

A. $C = 20m - 50$

B. $C = 20m + 50$

C. $C = 50m + 20$

D. $C = 70m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

The cost of a taxi ride is a linear function of the miles m. A 3-mile ride costs $11 and a 7-mile ride costs $21.

Write the function.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

True or false?

A tank holds 500 liters and drains 20 liters each minute. The function $V = 500 - 20t$ gives the liters V after t minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

SET 28 • 8.F.B.4

BACKWARD 1

Write an equation for a situation • Grade 6 • Expressions & Equations • 6.EE.C.9

Tickets cost $9 each.

Write an equation for the cost c of t tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

Mia has $15 and saves $5 each week.

Write an equation for her savings s after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

A pool holds 100 gallons and fills at 10 gallons per minute.

Which equation gives the gallons g after m minutes?

A. g = 110m

B. g = 100m + 10

C. g = 10m + 100

D. g = 10m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

True or false?

Apples cost $3 per pound. The equation c = 3p gives the cost c of p pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A game costs a $10 fee plus $2 per game. The equation $c = 10g + 2$ gives the cost c of g games.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 8.F.B.4

BACKWARD 2

Solve a two-step equation from a model • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for h.

$40h + 65 = 185$

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

Solve for t.

$30 - 2.5t = 10$

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

Solve for m.

$20m + 50 = 170$

A. $m = 3$

B. $m = 11$

C. $m = 8.5$

D. $m = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

True or false?

$t = 4$ is the solution of $500 - 20t = 420$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

True or false?

$h = 5$ is the solution of $12h + 8 = 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 8.F.B.4

FORWARD 1

Write a function for a situation and use it to answer a question • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a

A gym charges $25 per month, but the first month is free.

Write a function for the total cost C after m months, where m ≥ 1.

C(m) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

A savings account starts with $300, and $40 is added each week.

Write a function S(w), then find when the balance reaches $700.

Function and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

A pool is 4 feet deep and drains 0.25 foot each hour.

Which function gives the depth after t hours, and when is the pool empty?

A. d(t) = 4 − 0.25t; after 16 hours

B. d(t) = 0.25t + 4; after 16 hours

C. d(t) = 4t − 0.25; after 1 hour

D. d(t) = 4 − 0.25t; after 4 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q3**

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

True or false?

A rental costs $30 plus $0.20 per mile. The function C(m) = 0.2m + 30 gives the cost for m miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

True or false?

A balance of $200 decreases by $15 each month. The function B(m) = 15m + 200 gives the balance after m months.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

SET 28 • 8.F.B.4

FORWARD 2

Rearrange a formula to solve for a chosen quantity • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve $C = 40h + 65$ for h.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

Solve $F = \frac{9}{5}C + 32$ for C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

Solve $y = mx + b$ for b.

A. $b = \frac{y}{mx}$

B. $b = mx - y$

C. $b = y + mx$

D. $b = y - mx$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

True or false?

For $t \neq 0$, solving $d = rt$ for r gives $r = \frac{d}{t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

True or false?

For $l \neq 0$, solving $A = lw$ for w gives $w = A - l$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 8.F.B.4 • FUNCTIONS

Interpret the rate of change and initial value of a linear function in context

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Identify the independent and dependent variables	**6.EE.C.9**
FORWARD 1	Interpret the slope and intercept of a linear model of data	**HSS-ID.C.7**
FORWARD 2	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**

SET 29 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

Renting a bike costs $C = 4h + 10$ dollars for h hours.

What do the 4 and the 10 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

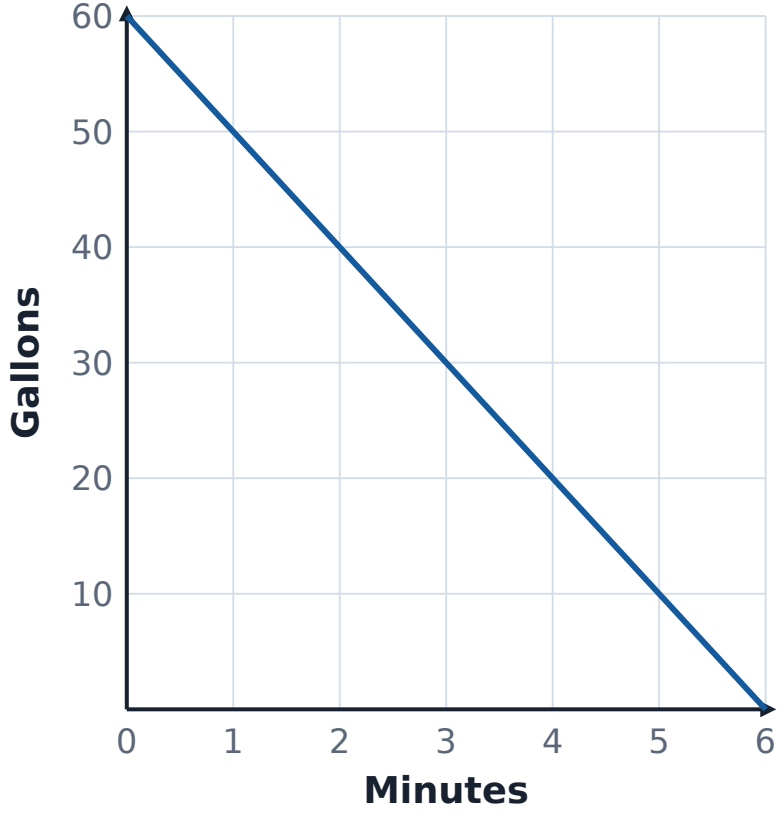
ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

The graph shows the water in a tank.

What do the rate of change and the initial value mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

A plant's height is $h = 0.5w + 12$ centimeters after w weeks.

What does 0.5 mean?

A. The plant grows for 0.5 week.

B. The plant grows 0.5 cm each week.

C. The plant was 0.5 cm tall at the start.

D. The plant grows 12 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

The table shows the money in a savings account.

What does the rate of change mean?

Weeks	0	4	8
Dollars	150	210	270

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

True or false?

For C = 4h + 10, the initial value means the bike costs $4 before any hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

SET 29 • 8.F.B.4

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of gallons (x) and miles (y) passes through (1, 28).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

A proportional graph shows hours worked (x) and pay (y).

What does the point (0, 0) mean?

A. 0 hours of work earn $0.

B. The worker starts with $0 per hour.

C. The pay is $0 per hour.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and laps (y) passes through (6, 3).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

True or false?

On a graph of tickets (x) and cost (y), the point (2, 18) means each ticket costs $18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 8.F.B.4

BACKWARD 2

Identify the independent and dependent variables • Grade 6 • Expressions & Equations • 6.EE.C.9

In C = 4h + 10, which variable is the independent variable?

Variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

"Pay depends on the hours worked." Which quantity is the dependent variable?

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

In d = 60t, what is d?

A. The rate

B. The independent variable

C. A constant

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

True or false?

In a table of hours worked and pay, the hours are usually the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

True or false?

In $y = 3x + 2$, x depends on y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 8.F.B.4

FORWARD 1

Interpret the slope and intercept of a linear model of data • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A line of fit predicts a test score $y = 4.5x + 60$ for x hours of study.

What does 4.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

A line of fit predicts a test score $y = 4.5x + 60$ for x hours of study.

What does 60 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

A model gives a used car's value as $y = 18{,}000 - 1{,}500x$ dollars after x years.

What does $-1{,}500$ mean?

A. The car is 1,500 years old.

B. The value drops about $1,500 each year.

C. The car costs $1,500.

D. The value rises $1,500 each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

True or false?

A model gives a child's height as $y = 2.1x + 30$ cm at age x months. The intercept 30 is the predicted height at 0 months.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

True or false?

A slope of −3 in a linear model means y increases by 3 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

SET 29 • 8.F.B.4

FORWARD 2

Interpret the parameters of linear and exponential functions • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

P(t) = 2,000(1.05)^t gives an account balance after t years.

What do 2,000 and 1.05 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

V(t) = 25,000(0.85)^t gives a car's value after t years.

What is the yearly percent change?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

A phone plan costs C(m) = 0.12m + 9.99 dollars for m minutes.

What does 9.99 represent?

A. A fixed monthly charge

B. The number of minutes

C. The total cost

D. The cost per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

True or false?

In f(t) = 100(2)^t, the quantity doubles each time t increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

True or false?

In f(t) = 50(0.9)^t, the quantity grows by 90% each period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 8.F.B.5 • FUNCTIONS

Describe the relationship shown by a graph

MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Interpret points on a graph of a situation	**5.G.A.2**
BACKWARD 2	Compare speeds as unit rates	**6.RP.A.3.b**
FORWARD 1	Interpret the key features of a graph in context	**HSF-IF.B.4**
FORWARD 2	Graph and evaluate piecewise-defined functions	**HSF-IF.C.7.b**

SET 30 • 8.F.B.5

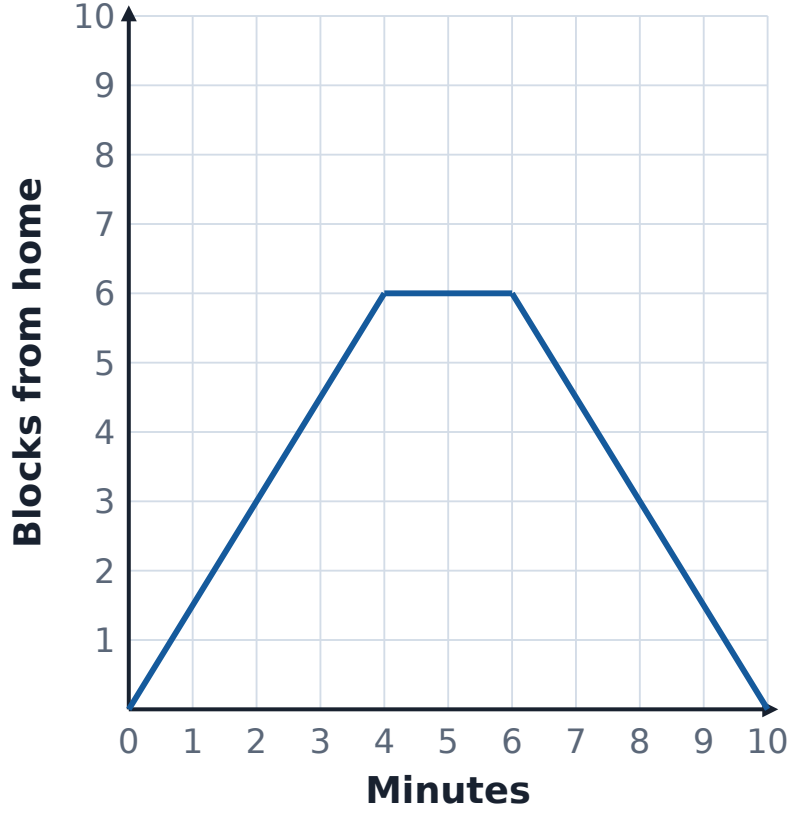
MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

The graph shows Ana's walk.

Describe what happens from 4 to 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

The graph shows Ana's walk.

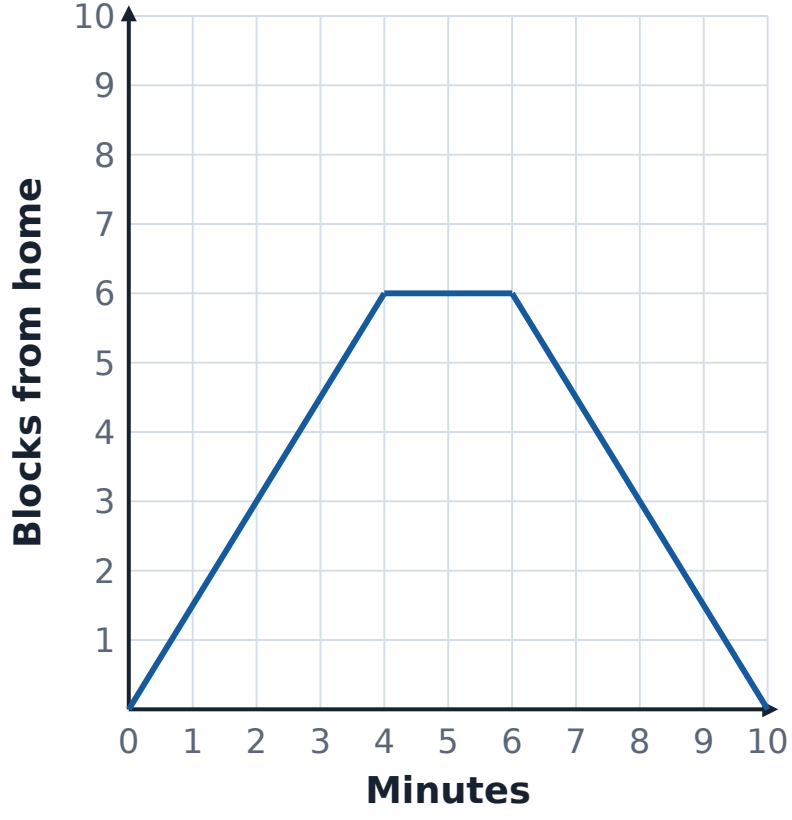
During which interval is Ana moving toward home?

A. 0 to 4 minutes

B. 6 to 10 minutes

C. 4 to 6 minutes

D. 0 to 6 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

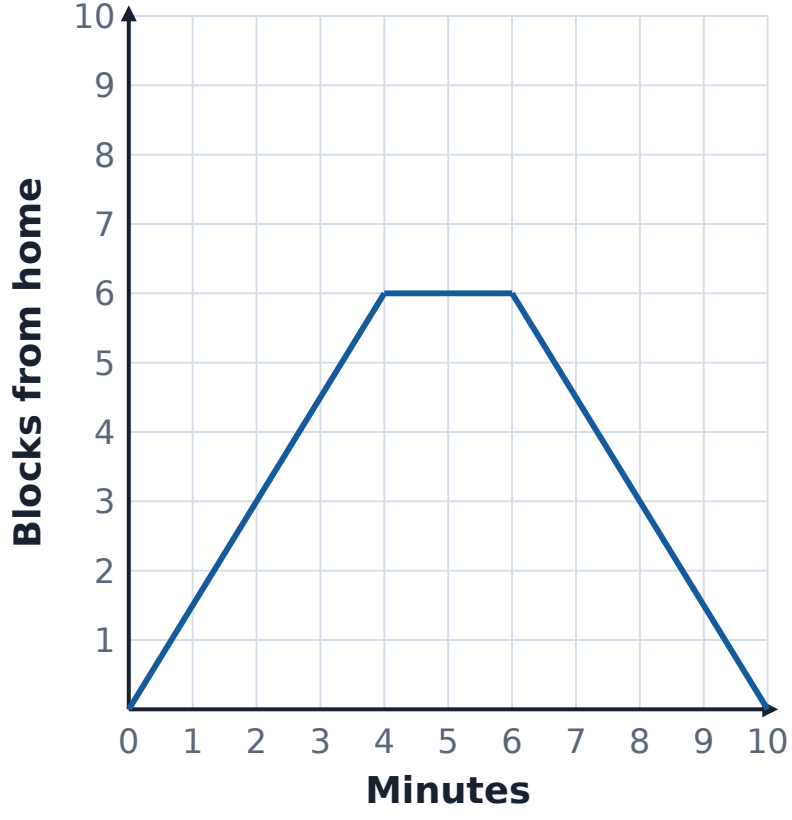
ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

The graph shows Ana's walk.

Is the relationship linear or nonlinear over the whole walk? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

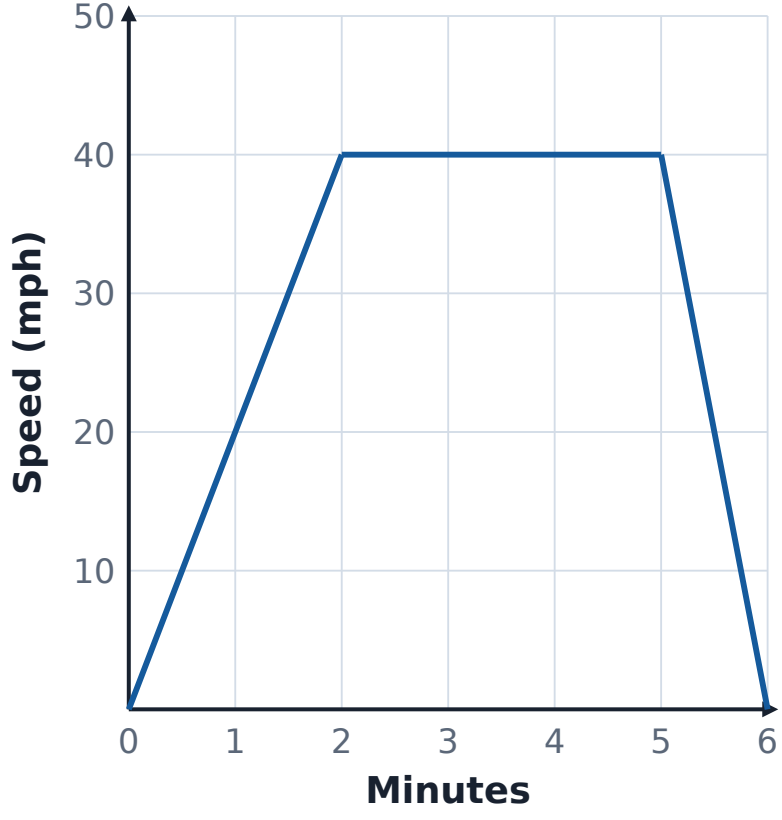
ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

The graph shows a car's speed.

Describe how the speed changes over the 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

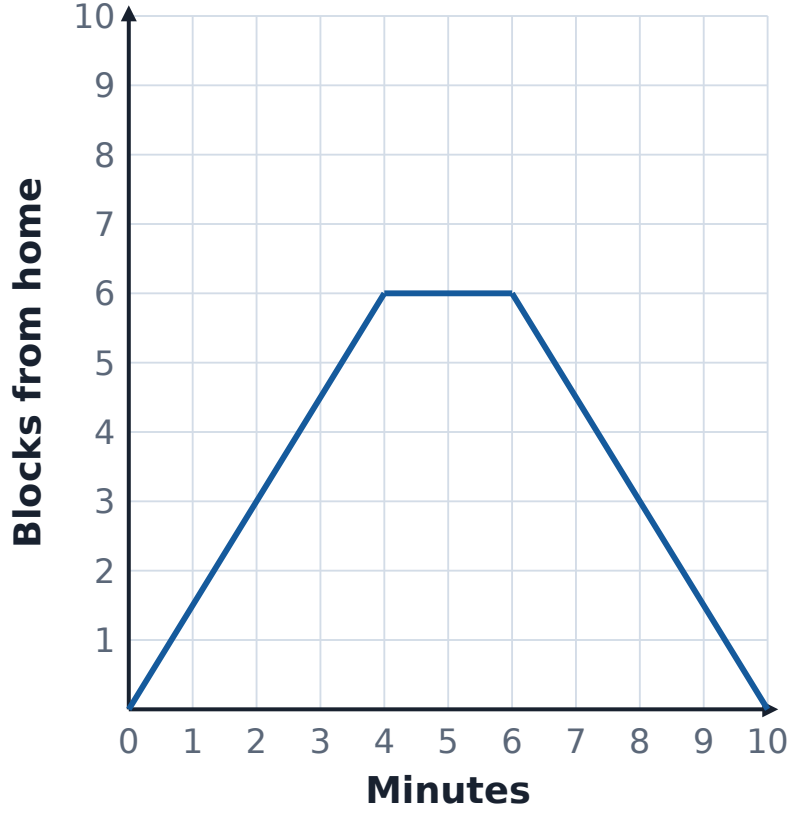
Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

True or false?

In the graph of Ana's walk, she walks faster going home than going away from home.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

SET 30 • 8.F.B.5

BACKWARD 1

Interpret points on a graph of a situation • Grade 5 • Geometry • 5.G.A.2

A graph shows minutes (x) and blocks from home (y). What does the point (3, 5) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

A graph shows days (x) and plant height in centimeters (y). What does the point (10, 4) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

A graph shows hours (x) and miles driven (y).

Which point shows 120 miles after 2 hours?

A. (2, 120)

B. (60, 2)

C. (2, 60)

D. (120, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

True or false?

On a graph of time (x) and distance (y), the point (0, 0) can mean the start of a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (4, 1) on a graph of hours (x) and miles (y) means 4 miles in 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 8.F.B.5

BACKWARD 2

Compare speeds as unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana walks 6 blocks in 4 minutes at a steady pace.

What is her speed in blocks per minute?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

Which is faster: 12 miles in 3 hours, or 10 miles in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

Which rate is the greatest?

A. 60 miles in 1 hour

B. 100 miles in 2 hours

C. 20 miles in 0.5 hour

D. 150 miles in 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

True or false?

Walking 3 miles in 1 hour is faster than walking 5 miles in 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

True or false?

Running 8 km in 40 minutes is the same rate as running 2 km in 20 minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 8.F.B.5

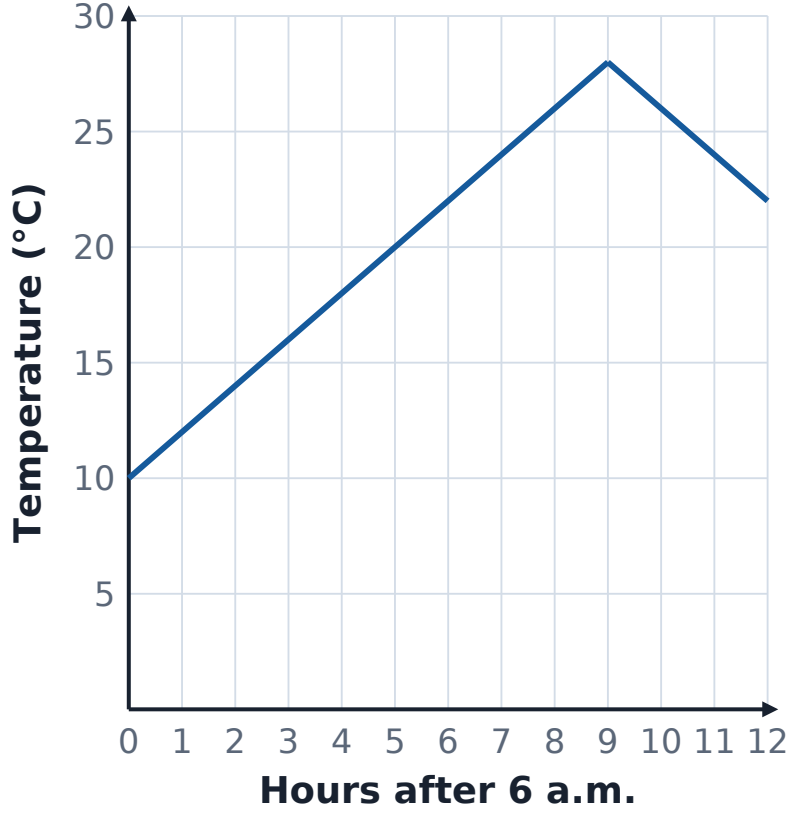
FORWARD 1

Interpret the key features of a graph in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows the temperature during a day.

When is the temperature highest, and what is it?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

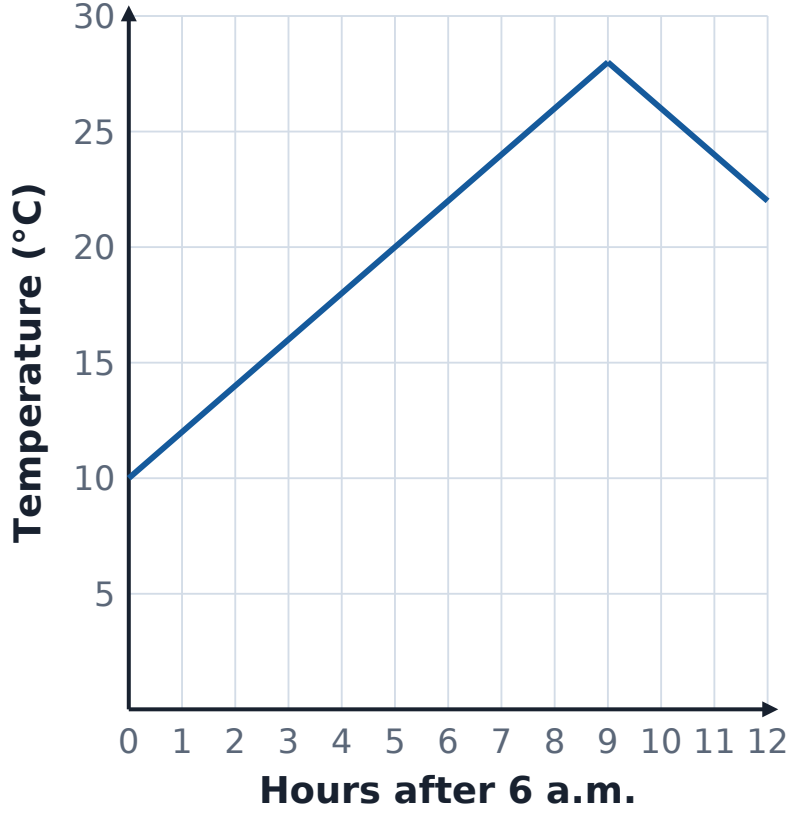
ID S30-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

The graph shows the temperature during a day.

On what interval is the temperature increasing?

Interval: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

The graph shows the temperature during a day.

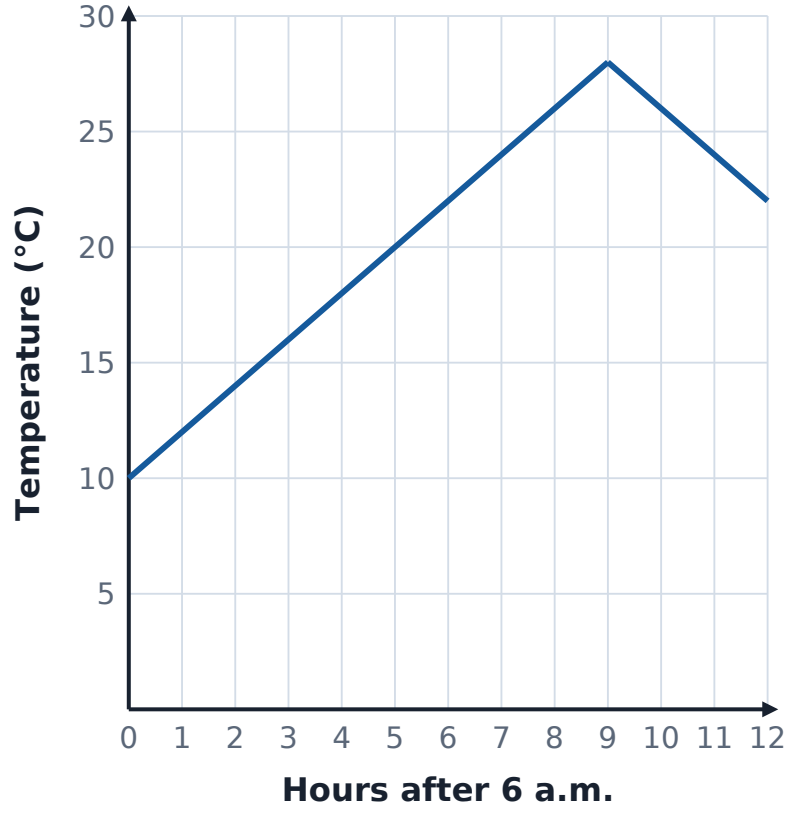
What does the y-intercept mean?

A. It was 28°C at 3 p.m.

B. It was 10°C at 6 a.m.

C. It was 22°C at 6 p.m.

D. The temperature rose for 9 hours.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

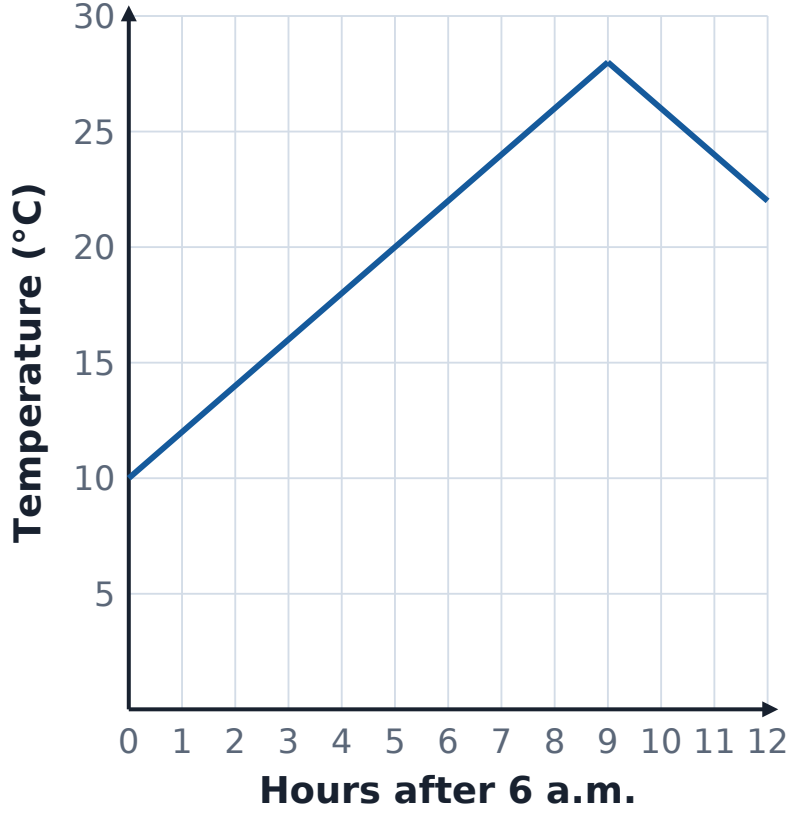
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

True or false?

In the graph, the temperature is decreasing from 9 to 12 hours after 6 a.m.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

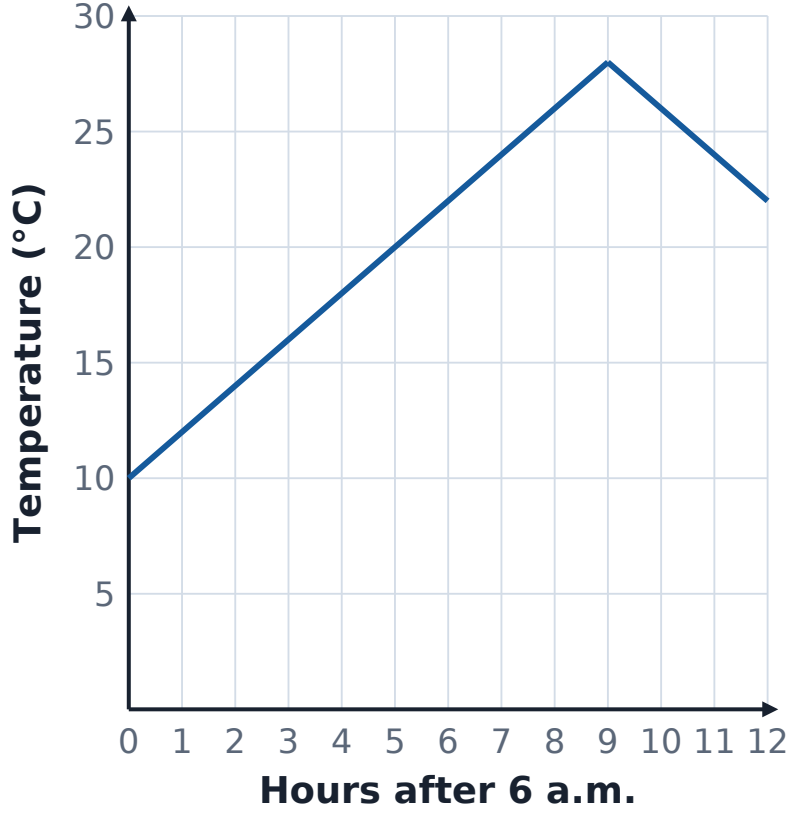
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

True or false?

The graph shows that the temperature never dropped below 22°C.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 8.F.B.5

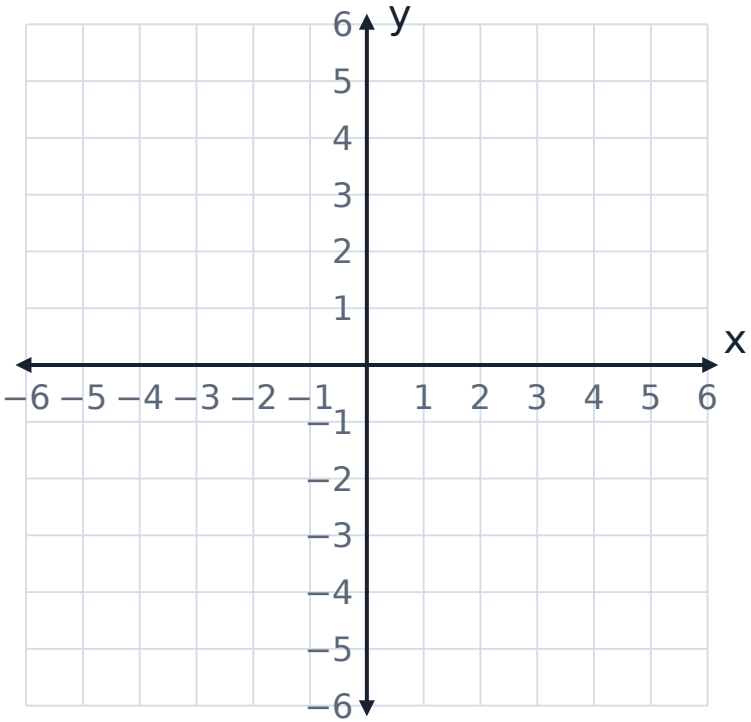
FORWARD 2

Graph and evaluate piecewise-defined functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the coordinate plane.

$f(x) = x + 2$ for $x < 0$

$f(x) = 2$ for $x \geq 0$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

$f(x) = 3$ for $x \leq 2$, and $f(x) = x + 1$ for $x > 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

A taxi charges $3 for the first mile and $2 for each mile after that.

Which function gives the cost for m miles?

A. $C(m) = 2m + 3$ for all m

B. $C(m) = 3m$ for all m

C. $C(m) = 3$ for $0 < m \leq 1$; $C(m) = 3 + 2(m - 1)$ for $m > 1$

D. $C(m) = 3 + 2m$ for $m \leq 1$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

True or false?

For $f(x) = |x|$, $f(-4) = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

True or false?

A piecewise-defined function can never have an unbroken graph.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

SET 31 • 8.F.B.5 • FUNCTIONS

Sketch a graph from a description of a relationship

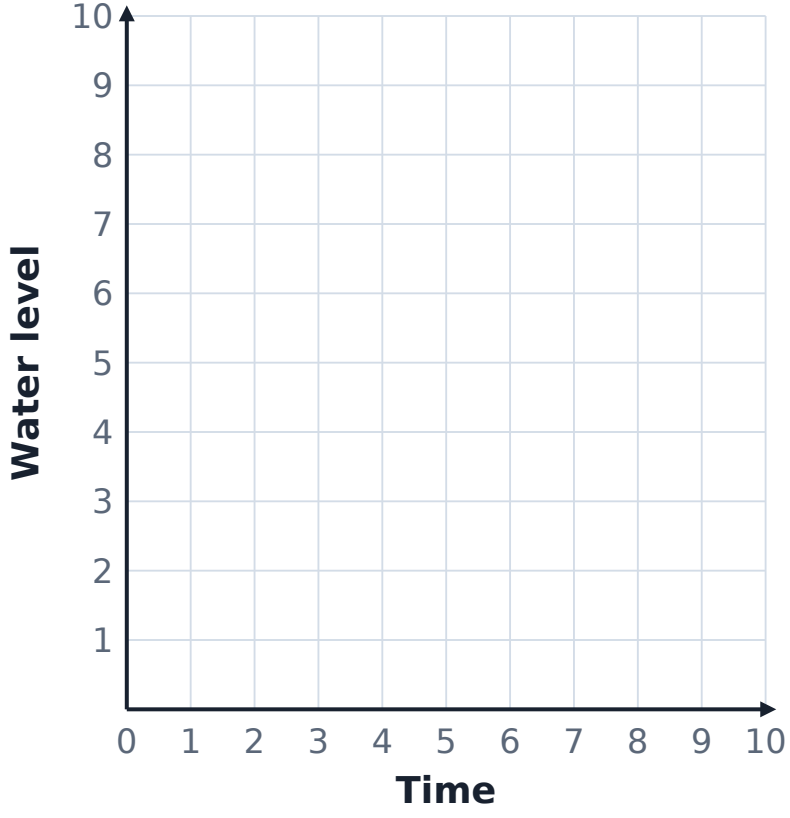
MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Graph a relationship from a table	**6.EE.C.9**
BACKWARD 2	Plot points that represent a situation	**5.G.A.2**
FORWARD 1	Sketch graphs that show given key features	**HSF-IF.B.4**
FORWARD 2	Graph exponential functions and describe their end behavior	**HSF-IF.C.7.e**

SET 31 • 8.F.B.5

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

Sketch a graph of the water level in a bathtub over time: it fills at a steady rate, stays the same while someone bathes, and then drains quickly.

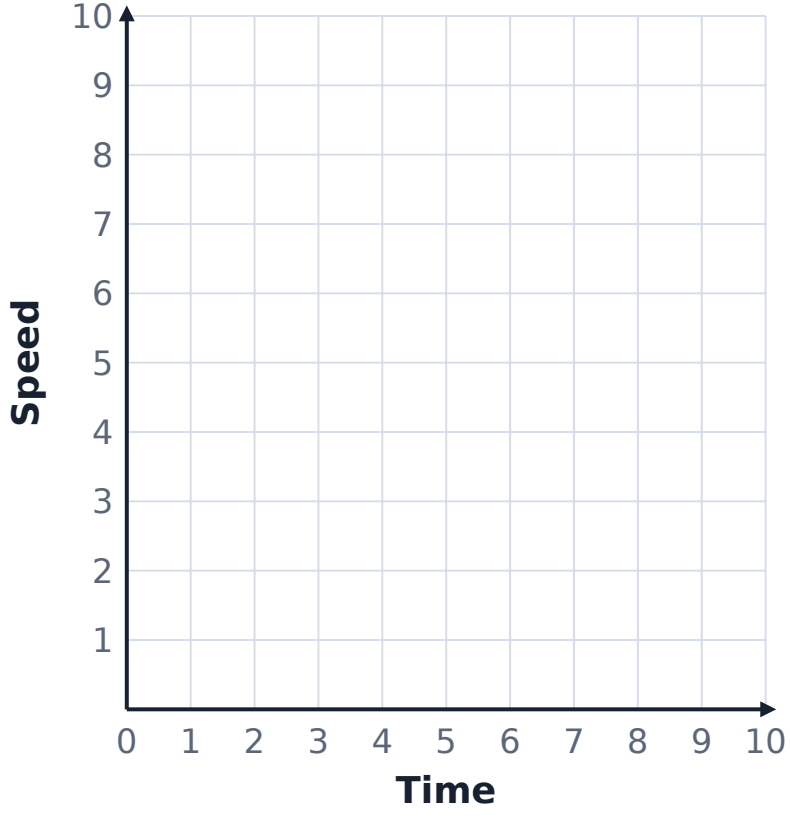


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

Sketch a graph of a car's speed over time: it speeds up, travels at a steady speed, and then slows down to a stop.

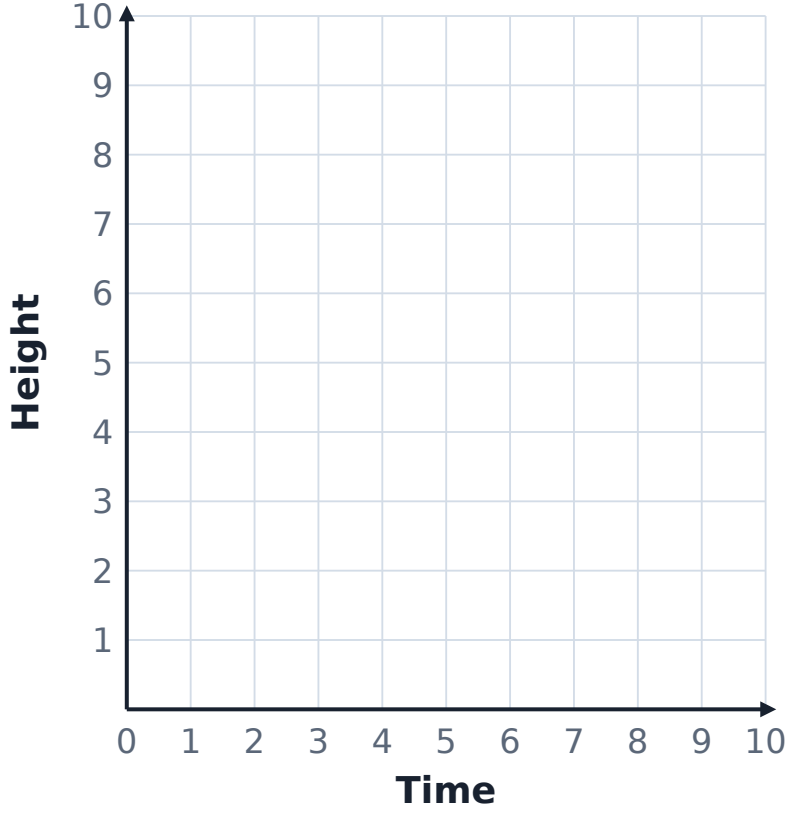


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

Sketch the height of a ball thrown straight up over time: it rises, slows down, reaches a highest point, and falls back to the ground.

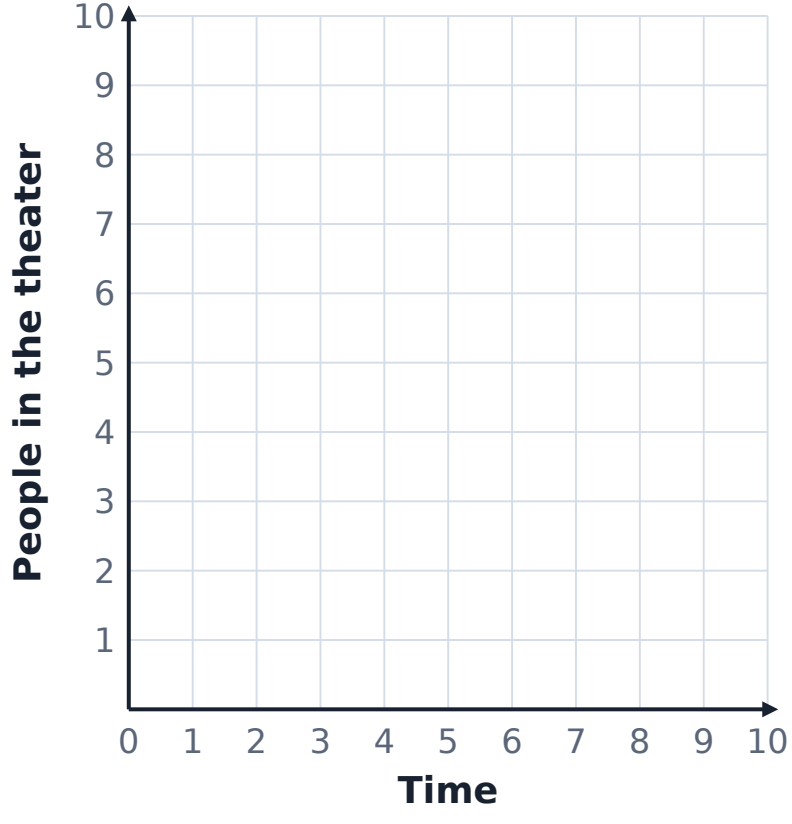


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

Sketch the number of people in a theater from 6:00 to 10:00. It fills quickly before a 7:00 show, stays full, and empties after the show ends at 9:30.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

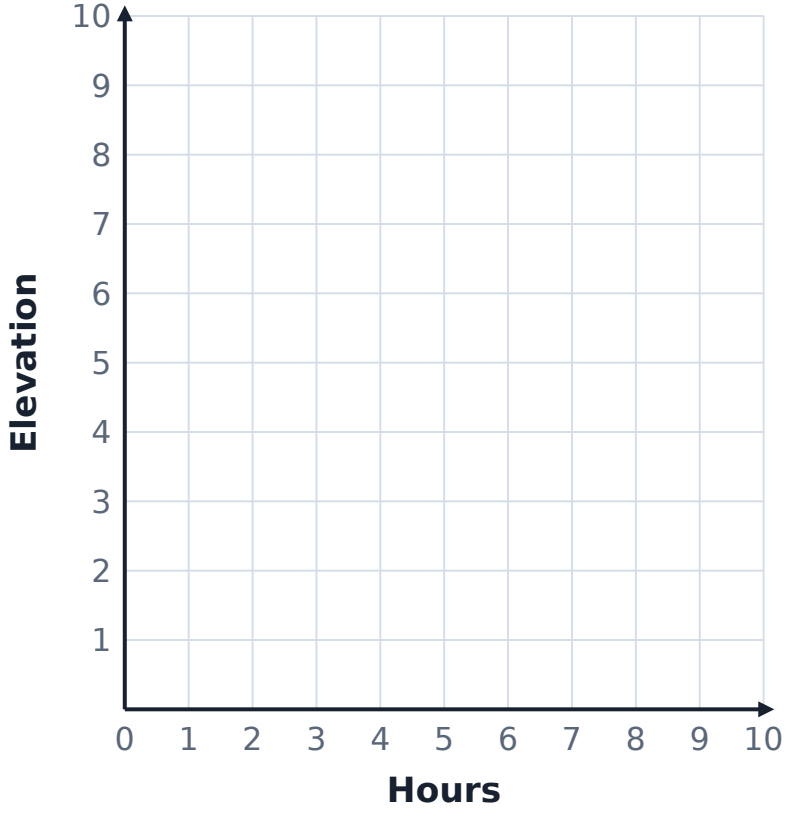
ID S31-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

Sketch a graph of a hike: the hiker climbs a steep trail for 2 hours, then a gentle slope for 1 hour, then rests for 1 hour.

Is elevation a linear function of time for the whole hike? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

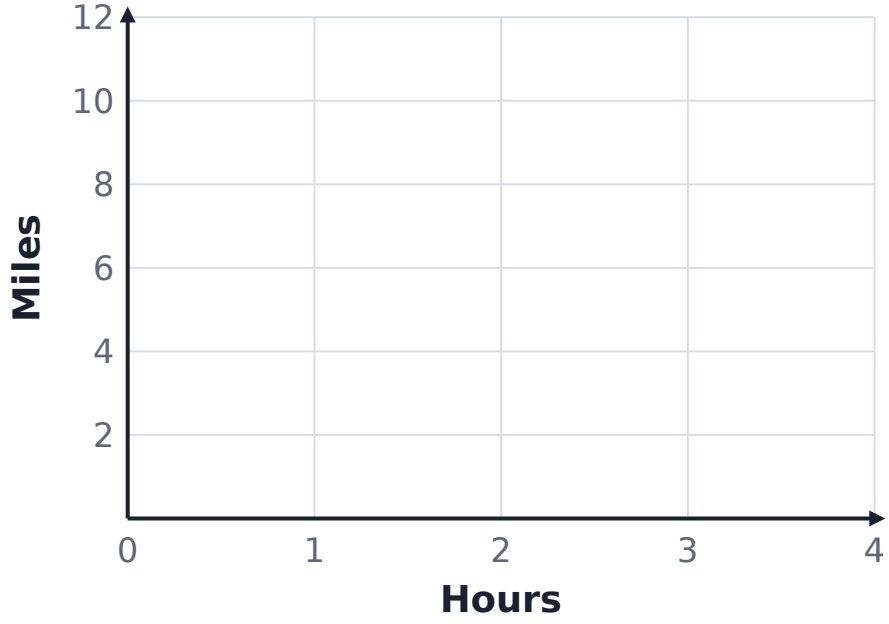
SET 31 • 8.F.B.5

BACKWARD 1

Graph a relationship from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

Graph the pairs in the table.

Hours	0	1	2	3
Miles	0	4	8	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

A hiker walks 4 miles each hour, so miles = 4 × hours.

How many miles does she walk in 2.5 hours?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

Which table shows y increasing by the same amount each time?

A. x: 0, 1, 2 and y: 0, 3, 4

B. x: 0, 1, 2 and y: 5, 5, 6

C. x: 0, 1, 2 and y: 6, 3, 9

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A table where y is 5 for every x shows y not changing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

True or false?

A table with y-values 2, 4, 3 shows y always increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

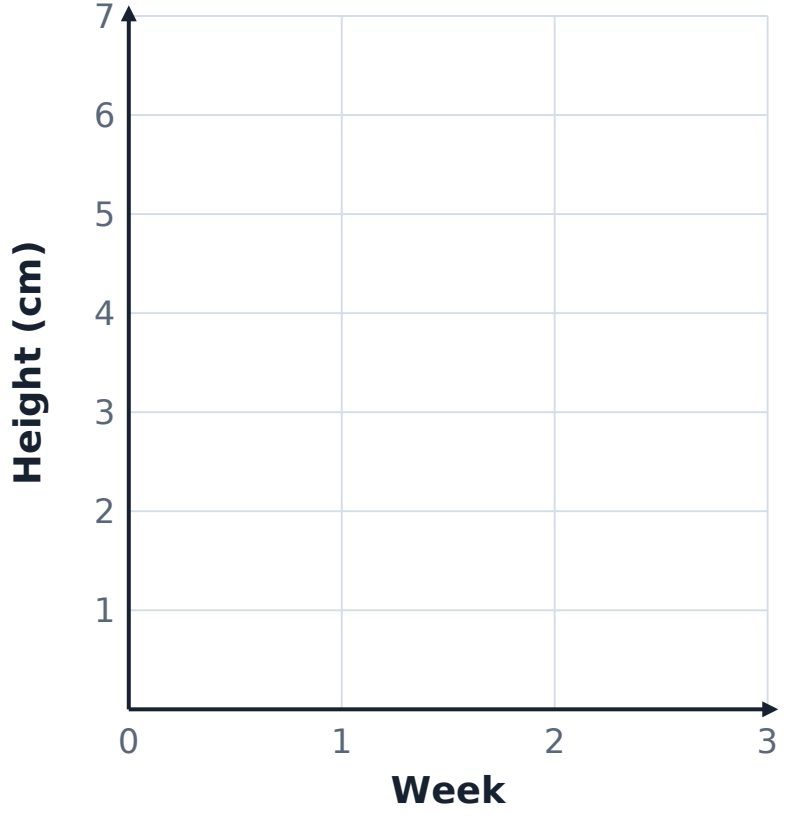
SET 31 • 8.F.B.5

BACKWARD 2

Plot points that represent a situation • Grade 5 • Geometry • 5.G.A.2

A plant is 2 cm tall at week 0, 4 cm at week 1, and 6 cm at week 2.

Plot the points (week, height).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

A plant is 2 cm tall at week 0 and grows 2 cm each week.

Which point shows its height at week 3?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

On a graph of weeks (x) and plant height in cm (y), what does the point (2, 6) mean?

A. The plant grows 2 cm.

B. At week 6 the plant is 2 cm tall.

C. The plant grows 6 cm each week.

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

True or false?

On a graph of weeks (x) and height (y), the point (0, 2) shows the height at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

True or false?

On a graph of weeks (x) and height in cm (y), the point (1, 4) means the plant is 1 cm tall at week 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

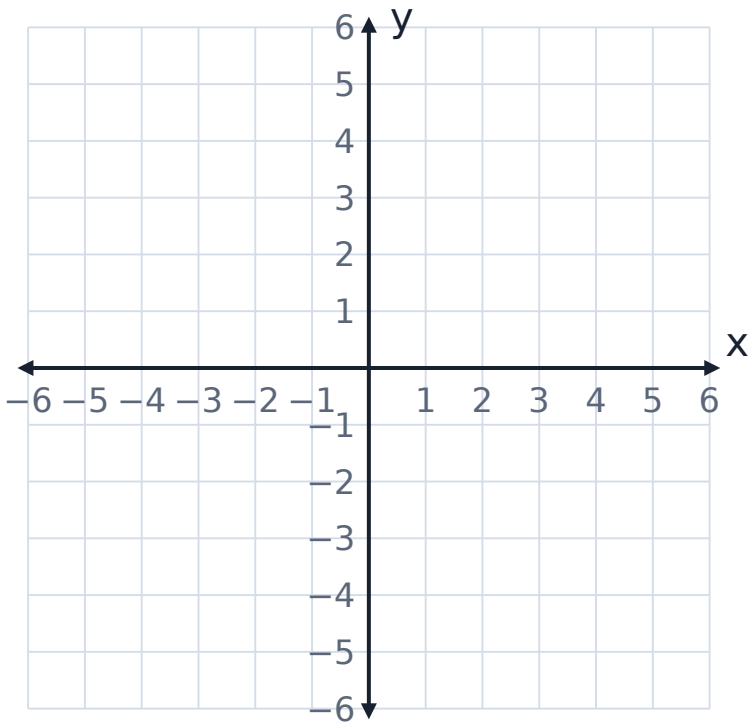
Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 8.F.B.5

FORWARD 1

Sketch graphs that show given key features • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

Sketch a function that increases for $0 < x < 3$, has a maximum value of 5 at $x = 3$, and decreases for $x > 3$.

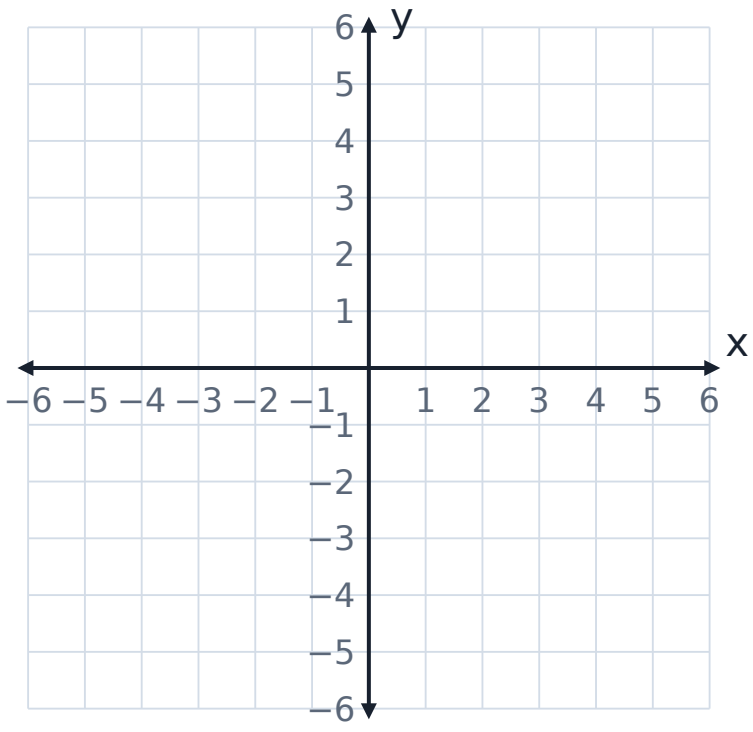


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

Sketch a function whose graph crosses the y-axis at 2 and that increases for all x shown.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

Which description matches $y = -(x - 2)^2 + 4$?

A. A maximum of 4 at $x = 2$; increasing for $x < 2$

B. A minimum of 4 at $x = 2$

C. A maximum of 2 at $x = 4$

D. Increasing for every x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

A graph with a maximum at x = 3 decreases for some x-values greater than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

A function that is always decreasing can have its maximum in the middle of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

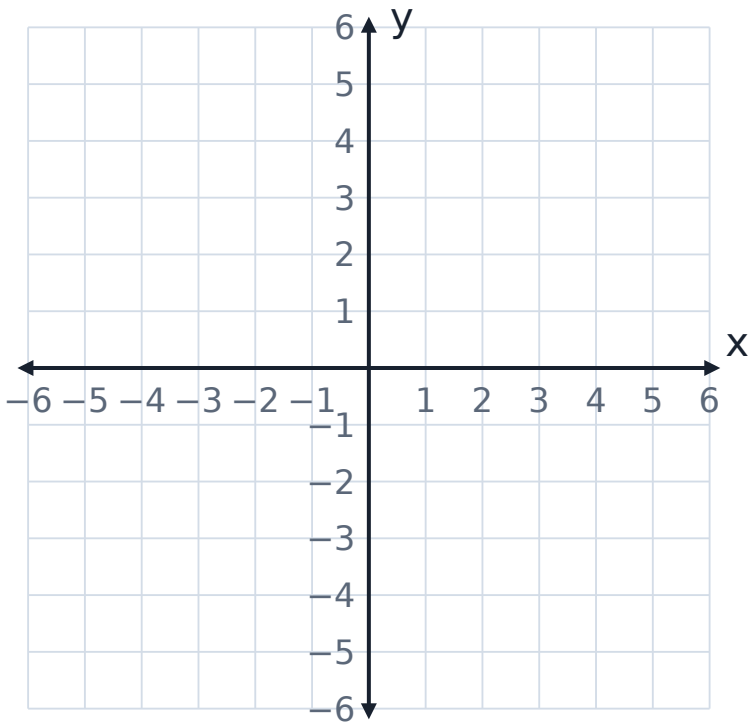
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • 8.F.B.5

FORWARD 2

Graph exponential functions and describe their end behavior • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph y = 2^x for x from −2 to 2 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

What is the y-intercept of the graph of y = 5(0.5)^x?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

For y = 3^x, what happens to y as x decreases without bound?

A. y gets closer to 3.

B. y decreases without bound.

C. y increases.

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

True or false?

The graph of y = 2^x never touches the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

True or false?

The function y = (0.5)^x is increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

SET 32 • 8.G.A.1.a • GEOMETRY

Verify that rigid motions take lines to lines and segments to segments of the same length

MAIN	Grade 8 main question	**8.G.A.1.a**
BACKWARD 1	Reflect points across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
FORWARD 1	Define rotations, reflections, and translations precisely	**HSG-CO.A.4**
FORWARD 2	Use rigid motions to decide whether figures are congruent	**HSG-CO.B.6**

SET 32 • 8.G.A.1.a

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.a

Segment AB has endpoints A(1, 2) and B(5, 2). It is translated 3 units down.

What are the endpoints of the image, and how long is the image?

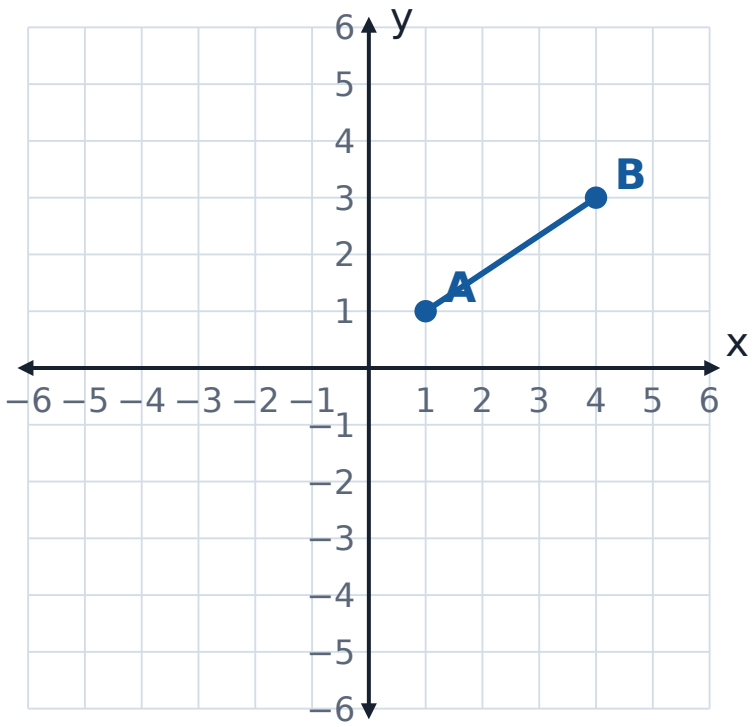
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

Reflect segment AB across the y-axis. Draw the image A'B'.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

A segment 7 cm long is rotated 90° about a point.

How long is its image?

A. 3.5 cm

B. 7 cm

C. 14 cm

D. It depends on the center of rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

True or false?

The image of a straight line under a translation is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

True or false?

A reflection can map a segment 5 units long to a segment 10 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point $(3, -4)$ is reflected across the x-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

The point $(-2, 5)$ is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

Which point is the reflection of (6, 1) across the y-axis?

A. (1, 6)

B. (6, −1)

C. (−6, −1)

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The reflection of (−3, −3) across the x-axis is (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

True or false?

The reflection of (4, 2) across the x-axis is (−4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 2

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

Find the distance between (1, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

Find the distance between $(1, -1)$ and $(5, -1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which two points are 6 units apart?

A. (−2, 3) and (2, 3)

B. (0, 1) and (0, 5)

C. (3, −2) and (3, 2)

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

$(0, -3)$ and $(0, 4)$ are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

$(-5, 1)$ and $(2, 1)$ are 3 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 1

Define rotations, reflections, and translations precisely • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A reflection across line m maps point P to point P'.

How is line m related to segment PP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

A rotation of 90° about point O maps P to P'.

What is true about OP, OP', and angle POP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

In a translation, the segments that join each point to its image are...

A. perpendicular to each other

B. all through one point

C. different lengths

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

True or false?

A rotation keeps every point the same distance from the center of rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

True or false?

In a reflection, a point (not on the line) and its image are on the same side of the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 2

Use rigid motions to decide whether figures are congruent • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is mapped onto triangle DEF by a rotation followed by a translation.

What can you conclude about the triangles?

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

Explain why two figures are congruent when a sequence of rigid motions maps one onto the other.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

Which transformation does NOT always map a figure onto a congruent figure?

A. A translation

B. A rotation

C. A dilation with a scale factor of 2

D. A reflection

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

True or false?

If a translation maps segment AB onto segment CD, then AB = CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

True or false?

Two triangles with the same area must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

SET 33 • 8.G.A.1.b • GEOMETRY

Verify that rigid motions take angles to angles of the same measure

MAIN	Grade 8 main question	**8.G.A.1.b**
BACKWARD 1	Find unknown angles with supplementary, complementary, and v…	**7.G.B.5**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Specify a sequence of transformations that maps a figure onto a…	**HSG-CO.A.5**
FORWARD 2	Prove theorems about parallelograms	**HSG-CO.C.11**

SET 33 • 8.G.A.1.b

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.b

Angle ABC measures 65°. It is reflected across a line.

What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

Triangle PQR has angles of 40°, 60°, and 80°. It is rotated 180° about the origin.

What are the angle measures of the image?

Measures: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

A right angle is translated 5 units up.

What is its image?

A. A right angle

B. A straight angle

C. An obtuse angle

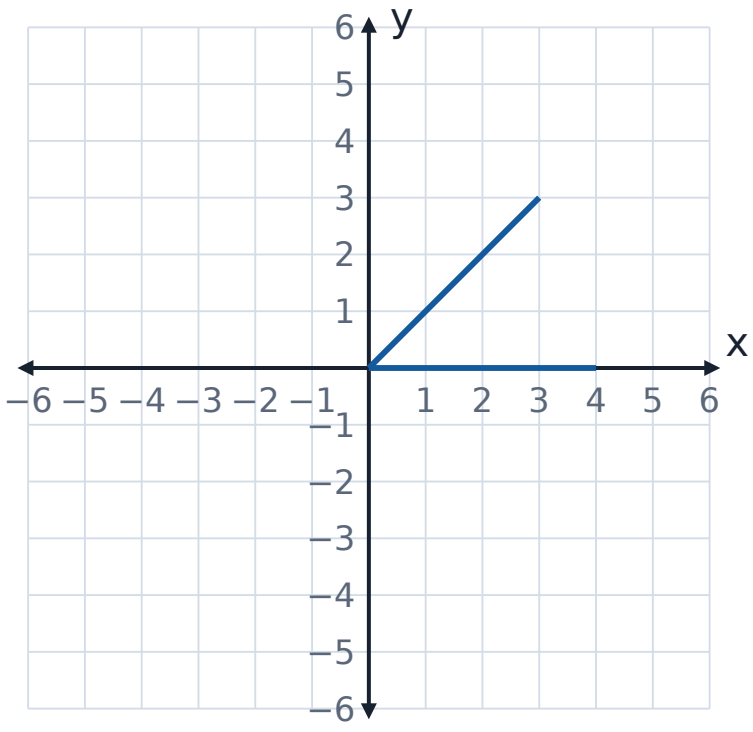
D. An acute angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

The angle has its vertex at the origin. Rotate it 90° counterclockwise about the origin and draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

True or false?

A rotation can map a 30° angle onto a 60° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 1

Find unknown angles with supplementary, complementary, and vertical angles • Grade 7 • Geometry • 7.G.B.5

Two angles form a straight line. One measures 65°.

What does the other measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

Two lines intersect. One angle measures 40°.

What does the vertical angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

What is the complement of a 25° angle?

A. 65°

B. 75°

C. 155°

D. 25°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Supplementary angles add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has angles of 40°, 60°, and 80°.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

A triangle has a 90° angle.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

Which angle measures make an obtuse triangle?

A. 60°, 60°, 60°

B. 30°, 40°, 110°

C. 45°, 45°, 90°

D. 50°, 60°, 70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

True or false?

An acute triangle has three angles that each measure less than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 1

Specify a sequence of transformations that maps a figure onto another • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5

Triangle A(1, 1), B(3, 1), C(1, 4) is mapped onto A'(−1, −1), B'(−3, −1), C'(−1, −4).

Describe a transformation or sequence of transformations that does this.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

Which single transformation maps (2, 3) to (2, −3) and (5, 1) to (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

Which rotation about the origin maps (4, 0) to (0, 4)?

A. 270° counterclockwise

B. 180°

C. 90° clockwise

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across a line and then reflecting the image across the same line gives back the original figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

True or false?

A translation 3 units right followed by a translation 3 units left moves the figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 2

Prove theorems about parallelograms • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram. Prove that angle A and angle B are supplementary. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5

In parallelogram ABCD, angle A = 70°.

Find the other three angles and give a reason for each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5

Which statement about every parallelogram can be proved?

A. All of its angles are right angles.

B. Its diagonals are congruent.

C. Its diagonals bisect each other.

D. Its diagonals are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5

True or false?

Opposite sides of a parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5

True or false?

A quadrilateral with one pair of parallel sides must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5

SET 34 • 8.G.A.1.c • GEOMETRY

Verify that rigid motions take parallel lines to parallel lines

MAIN	Grade 8 main question	**8.G.A.1.c**
BACKWARD 1	Identify parallel lines and segments	**4.G.A.1**
BACKWARD 2	Draw polygons in the coordinate plane and find side lengths	**6.G.A.3**
FORWARD 1	Prove theorems about angles formed by parallel lines	**HSG-CO.C.9**
FORWARD 2	Use coordinates to prove simple geometric theorems	**HSG-GPE.B.4**

SET 34 • 8.G.A.1.c

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.c

Lines p and q are parallel. Both are rotated 90° about the same point.

Are the images parallel? Explain.

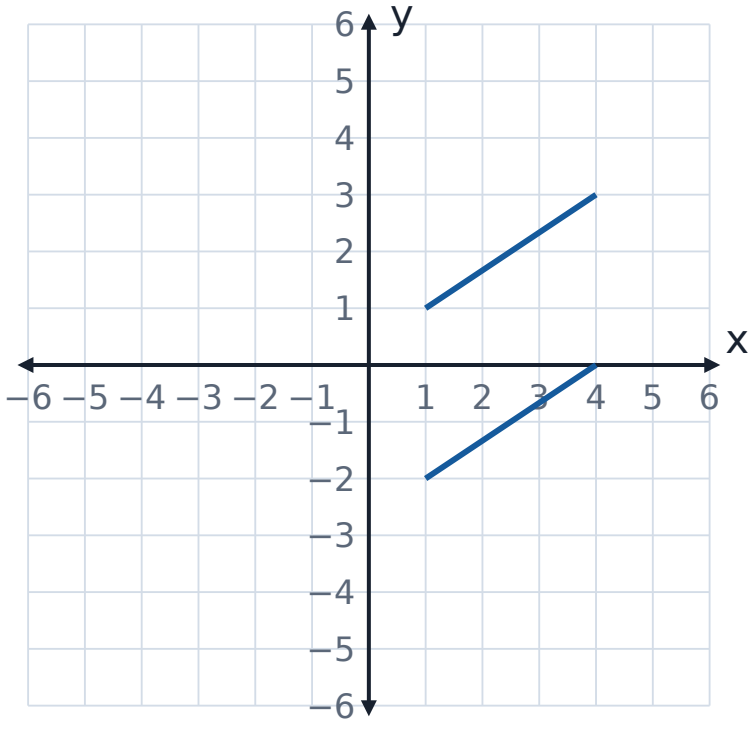
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

The two segments are parallel. Translate both segments 4 units to the left and draw the images.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

A parallelogram is reflected across a line.

What is its image?

A. A rectangle, in every case

B. A figure with no parallel sides

C. A parallelogram congruent to the original

D. A trapezoid with only one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

The line $y = 2x + 1$ is translated 3 units up.

What is the equation of the image? Is the image parallel to the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

True or false?

A rigid motion can map two parallel lines onto two lines that intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 1

Identify parallel lines and segments • Grade 4 • Geometry • 4.G.A.1

Name a quadrilateral that has two pairs of parallel sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

How many pairs of parallel sides does a trapezoid that is not a parallelogram have?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

Which capital letter contains a pair of parallel segments?

A. V

B. H

C. T

D. L

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

True or false?

Parallel lines never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

True or false?

Perpendicular lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

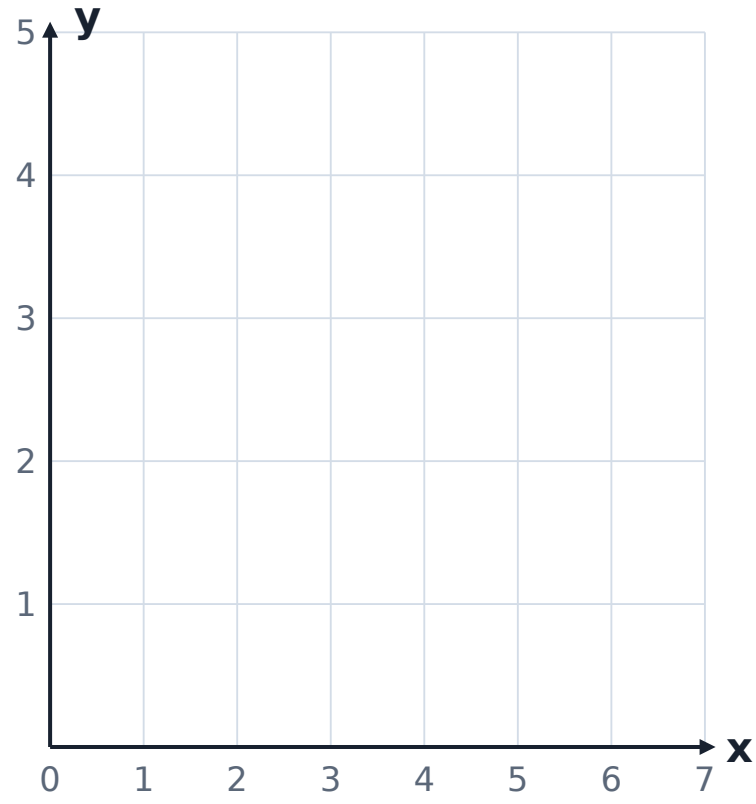
Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 2

Draw polygons in the coordinate plane and find side lengths • Grade 6 • Geometry • 6.G.A.3

Draw the quadrilateral with vertices (1, 1), (5, 1), (6, 4), and (2, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

Which side is parallel to the side from (1, 1) to (5, 1)?

A. From (1, 1) to (6, 4)

B. From (2, 4) to (6, 4)

C. From (1, 1) to (2, 4)

D. From (5, 1) to (6, 4)

True or false?

The side from (2, 4) to (6, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

True or false?

The segment from (1, 1) to (1, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

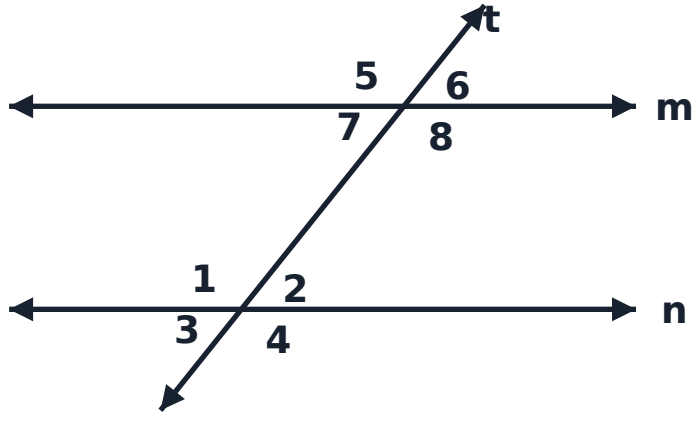
Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 8.G.A.1.c

FORWARD 1

Prove theorems about angles formed by parallel lines • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

In the figure, lines m and n are parallel. Prove that angle 2 = angle 7. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. One angle measures 72°.

What does the same-side interior angle measure? Give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

Which statement can be proved from "corresponding angles formed by parallel lines are congruent" and "vertical angles are congruent"?

A. Alternate interior angles are congruent.

B. The transversal is parallel to the lines.

C. The lines are perpendicular.

D. All the angles are right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

True or false?

If alternate interior angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

True or false?

Same-side interior angles formed by parallel lines are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

FORWARD 2

Use coordinates to prove simple geometric theorems • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Show that the quadrilateral with vertices (0, 0), (4, 0), (5, 3), and (1, 3) is a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

Is the triangle with vertices (0, 0), (4, 0), and (0, 3) a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

In order, (0, 0), (2, 0), (3, 2), and a fourth point form a parallelogram.

Which is the fourth point?

A. (0, 3)

B. (3, 0)

C. (1, 2)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

True or false?

A quadrilateral whose opposite sides have equal slopes is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

True or false?

The points (0, 0), (1, 1), and (2, 3) lie on one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 8.G.A.2 • GEOMETRY

Describe a sequence of rigid motions that shows two figures are congruent

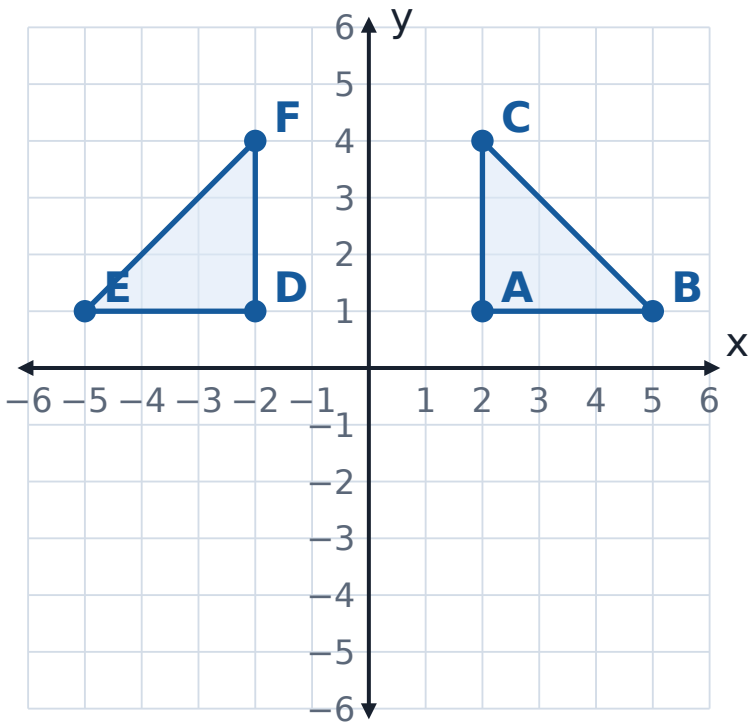
MAIN	Grade 8 main question	**8.G.A.2**
BACKWARD 1	Draw a polygon from its vertices	**6.G.A.3**
BACKWARD 2	Use signs to locate points across the axes	**6.NS.C.6.b**
FORWARD 1	Use corresponding parts to explain triangle congruence	**HSG-CO.B.7**
FORWARD 2	Use triangle congruence criteria	**HSG-CO.B.8**

SET 35 • 8.G.A.2

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.2

Describe a rigid motion that maps triangle ABC onto triangle DEF.



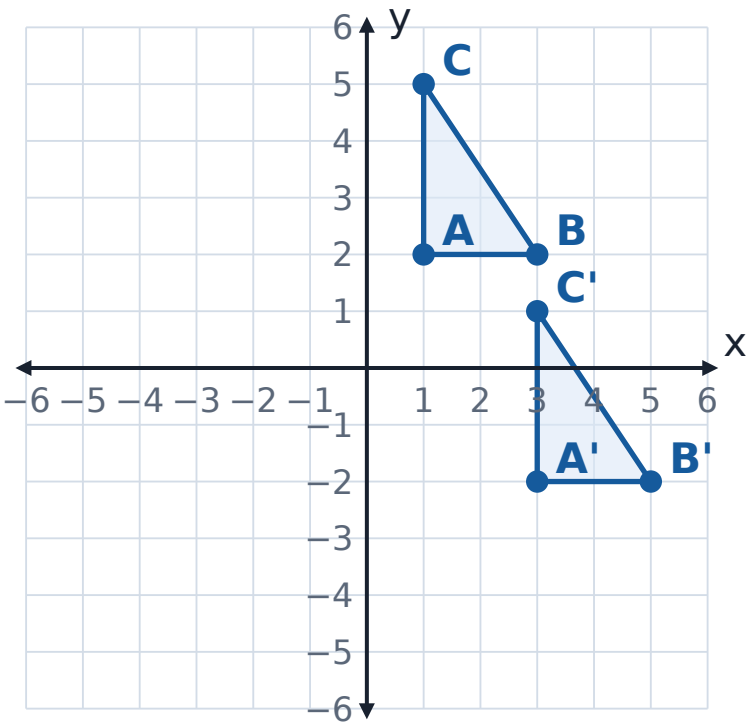
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

Describe a rigid motion that maps triangle ABC onto triangle A'B'C'.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

Figure P is mapped onto figure Q by a 90° rotation followed by a translation.

Which statement is true?

A. P and Q are congruent.

B. Q is larger than P.

C. P and Q are not congruent because they face different directions.

D. P and Q have different angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

Explain why a triangle and its image after a dilation with a scale factor of 2 are not congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

True or false?

Two figures are congruent if a sequence of rotations, reflections, and translations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

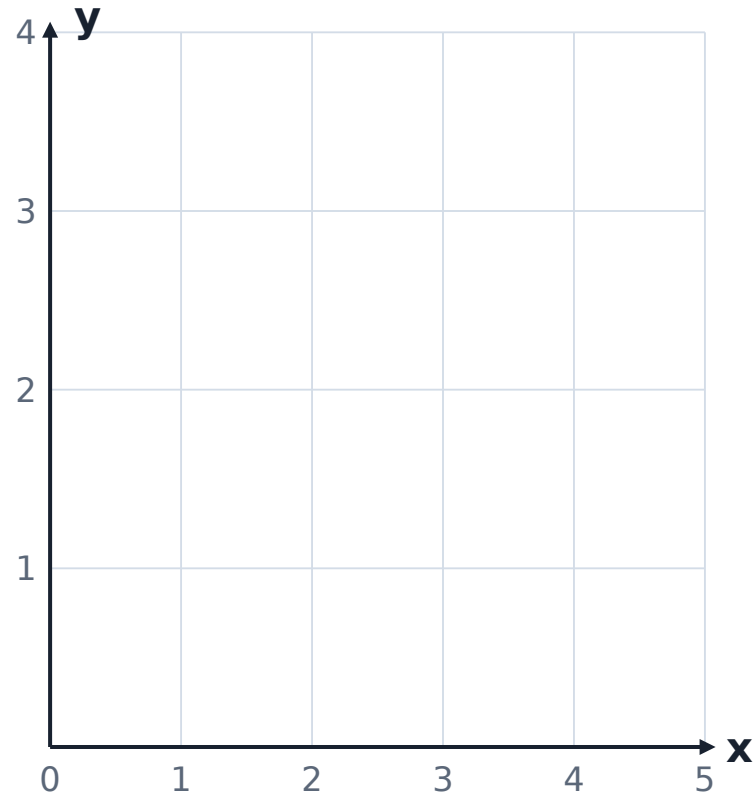
Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

SET 35 • 8.G.A.2

BACKWARD 1

Draw a polygon from its vertices • Grade 6 • Geometry • 6.G.A.3

Draw the triangle with vertices (1, 1), (4, 1), and (1, 3).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

A triangle has vertices (1, 1), (4, 1), and (1, 3).

How long is the side from (1, 1) to (4, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

A rectangle has vertices (0, 0), (5, 0), and (5, 2).

What is the fourth vertex?

A. (0, 2)

B. (2, 0)

C. (5, 5)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

True or false?

The side from (1, 1) to (1, 3) is 2 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The points (2, 1), (2, 5), and (6, 1) form a triangle with a side 3 units long.

A. True

B. False

SET 35 • 8.G.A.2

BACKWARD 2

Use signs to locate points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (5, 2) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

The point $(-1, -4)$ is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

Points (3, 2) and (3, −2) differ only in the sign of the y-coordinate.

How are they related?

A. They are reflections across the x-axis.

B. They are reflections across the y-axis.

C. They are the same point.

D. They are 3 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

True or false?

(−4, 1) and (4, 1) are reflections of each other across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

True or false?

(2, −6) and (−2, 6) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 8.G.A.2

FORWARD 1

Use corresponding parts to explain triangle congruence • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7

Triangle ABC is congruent to triangle DEF, with AB = 5 and angle B = 40°.

Which side and which angle of DEF must equal these?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

Two triangles have all corresponding sides and all corresponding angles congruent.

What can you conclude? Explain with rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

Triangle PQR is congruent to triangle XYZ.

Which statement must be true?

A. PQ = YZ

B. QR = YZ

C. Angle P = angle Z

D. PR = XY

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

True or false?

Corresponding parts of congruent triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

True or false?

If triangle ABC is congruent to triangle DEF, then AC = EF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

SET 35 • 8.G.A.2

FORWARD 2

Use triangle congruence criteria • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8

Two triangles have AB = DE, angle A = angle D, and AC = DF.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

Two triangles have angle A = angle D, AB = DE, and angle B = angle E.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

Which is NOT a valid congruence criterion for triangles?

A. AAA

B. ASA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

True or false?

SSS means that three pairs of equal sides show two triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

True or false?

Two sides and an angle that is not between them (SSA) always prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 8.G.A.3 • GEOMETRY

Translate figures on the coordinate plane

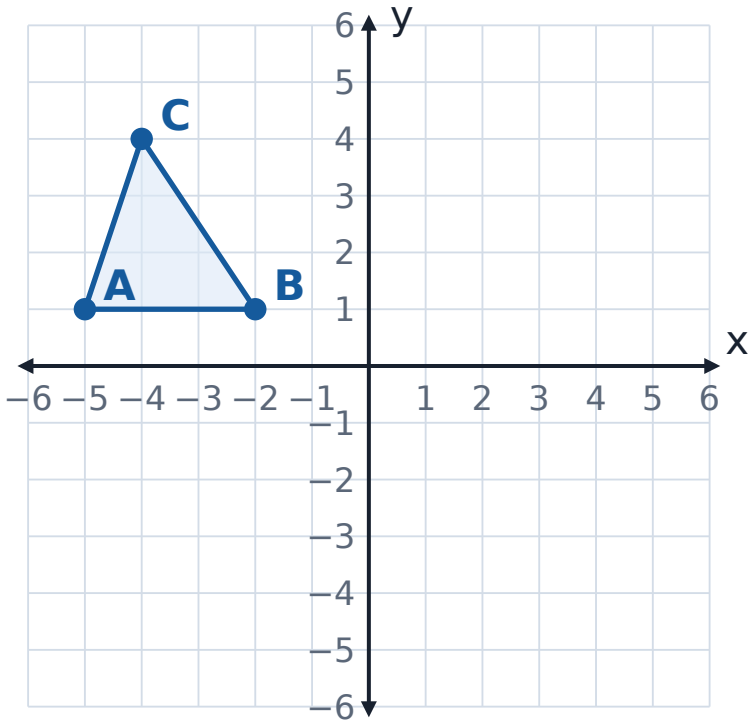
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Find and plot points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Add integers	**7.NS.A.1.b**
FORWARD 1	Describe transformations as functions and decide which preserve…	**HSG-CO.A.2**
FORWARD 2	Find the point that divides a segment in a given ratio	**HSG-GPE.B.6**

SET 36 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Translate triangle ABC 4 units right and 2 units down. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

The point P(3, −2) is translated 5 units left and 4 units up.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

Which rule describes a translation 6 units down?

A. $(x, y) \rightarrow (x, y - 6)$

B. $(x, y) \rightarrow (x, y + 6)$

C. $(x, y) \rightarrow (x + 6, y)$

D. $(x, y) \rightarrow (x - 6, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

A translation maps A(2, 5) to A'(−1, 7).

Where does it map B(4, 0)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

True or false?

The rule $(x, y) \to (x + 3, y - 1)$ moves every point 3 units right and 1 unit down.

A. True

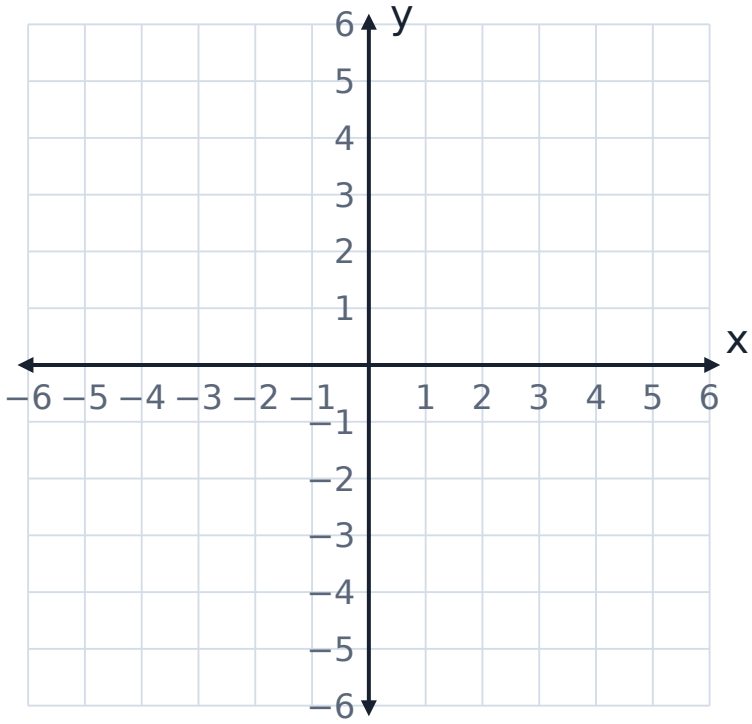
B. False

SET 36 • 8.G.A.3

BACKWARD 1

Find and plot points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot the points (−3, 2) and (4, −5).

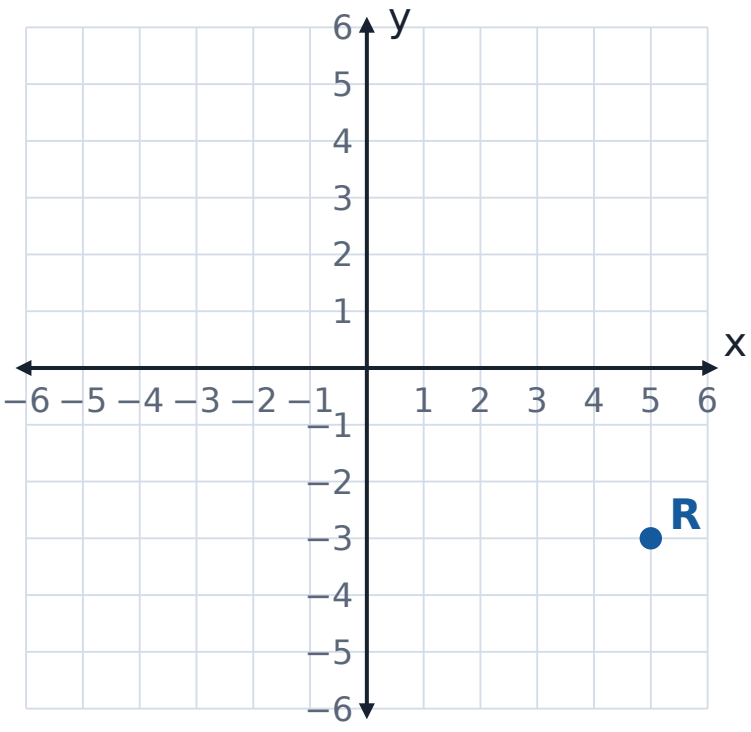


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

What are the coordinates of point R?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

Which point is 2 units to the left of (1, 4)?

A. (1, 6)

B. (−1, 4)

C. (1, 2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

Moving 3 units up from (2, −1) gives (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

True or false?

Moving 4 units left from (0, 0) gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 8.G.A.3

BACKWARD 2

Add integers • Grade 7 • The Number System • 7.NS.A.1.b

Add.

$-4 + 4$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

Add.

$3 + (-5)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

Add.

$-1 + 6$

A. 5

B. -5

C. 7

D. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

True or false?

$-2 + (-3) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

True or false?

$5 + (-8) = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 8.G.A.3

FORWARD 1

Describe transformations as functions and decide which preserve distance • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2

Write the rule for a translation 2 units left and 7 units up.

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

Does the rule (x, y) → (2x, y) keep distances the same? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

Which transformation keeps all distances and angles the same?

A. $(x, y) \to (x, 2y)$

B. $(x, y) \to (x^2, y)$

C. $(x, y) \to (3x, 3y)$

D. $(x, y) \to (x + 4, y - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

A translation is a function whose inputs and outputs are points of the plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

True or false?

The rule $(x, y) \rightarrow (x + 1, y + 1)$ changes the length of every segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 8.G.A.3

FORWARD 2

Find the point that divides a segment in a given ratio • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the midpoint of the segment from (2, 4) to (8, 10).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5

Find the point that is $\frac{1}{3}$ of the way from A(0, 0) to B(9, 6).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5

Point P is on segment AB from A(1, 1) to B(7, 4), with AP : PB = 2 : 1.

What are the coordinates of P?

A. (3, 2)

B. (4, 2.5)

C. (6, 3)

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5

True or false?

The midpoint of the segment from $(-4, 2)$ to $(6, -2)$ is $(1, 0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5

True or false?

The point halfway from (0, 0) to (10, 4) is (10, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5

SET 37 • 8.G.A.3 • GEOMETRY

Reflect figures across the x-axis and the y-axis

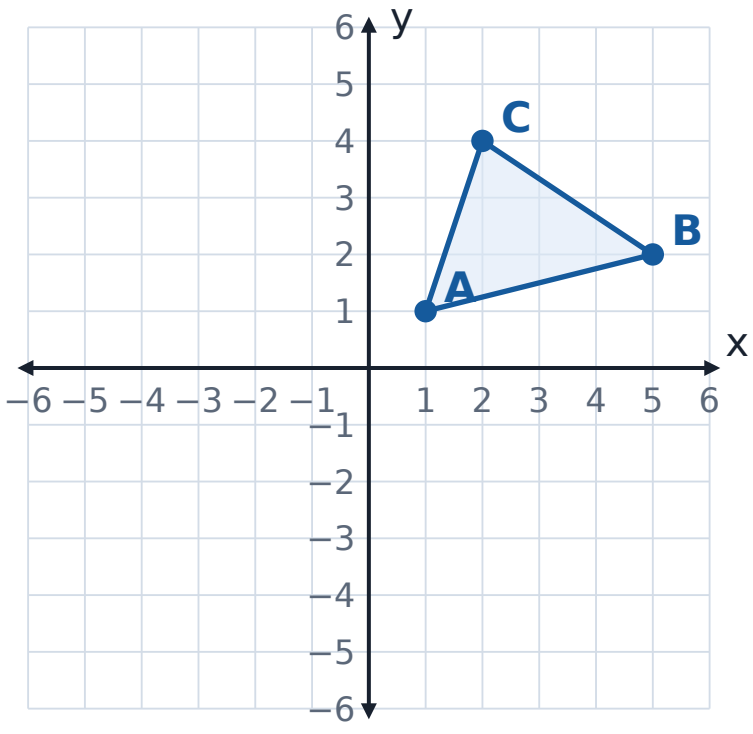
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Locate points that are reflections across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the opposite of a number	**6.NS.C.6.a**
FORWARD 1	Describe the rotations and reflections that carry a polygon onto it…	**HSG-CO.A.3**
FORWARD 2	Draw the image of a figure after a sequence of transformations	**HSG-CO.B.6**

SET 37 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Reflect triangle ABC across the x-axis. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

The point $(-3, 5)$ is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

Which rule describes a reflection across the x-axis?

A. $(x, y) \rightarrow (-x, y)$

B. $(x, y) \rightarrow (-x, -y)$

C. $(x, y) \rightarrow (y, x)$

D. $(x, y) \rightarrow (x, -y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

A segment has endpoints (2, 3) and (5, 1). It is reflected across the y-axis.

What are the endpoints of the image?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

True or false?

Reflecting (4, −6) across the x-axis gives (−4, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

SET 37 • 8.G.A.3

BACKWARD 1

Locate points that are reflections across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point $(7, -2)$ is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

The point $(-6, -3)$ is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

Reflecting a point across the y-axis changes...

A. the sign of the y-coordinate

B. neither sign

C. the sign of the x-coordinate

D. both signs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

True or false?

(0, 5) is its own reflection across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

True or false?

The reflection of $(-1, 8)$ across the y-axis is $(-1, -8)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 8.G.A.3

BACKWARD 2

Find the opposite of a number • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of -7?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

Which pair are opposites?

A. -4 and 4

B. 4 and $\frac{1}{4}$

C. 0 and 1

D. -4 and -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

The opposite of 0 is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 8.G.A.3

FORWARD 1

Describe the rotations and reflections that carry a polygon onto itself • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3

How many lines of symmetry does a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

Which rotations (between 0° and 360°) about its center carry a regular hexagon onto itself?

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

Which figure is carried onto itself by a 180° rotation but has no line of symmetry?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. An equilateral triangle

D. An isosceles trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A rectangle that is not a square has exactly 2 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

An isosceles trapezoid is carried onto itself by a 90° rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

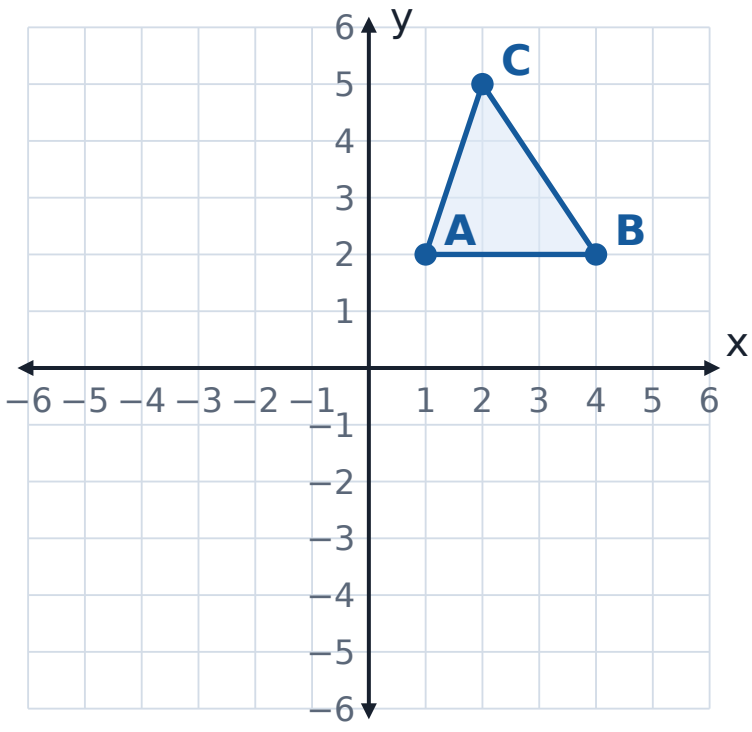
High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 8.G.A.3

FORWARD 2

Draw the image of a figure after a sequence of transformations • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Reflect triangle ABC across the x-axis, then reflect that image across the y-axis. Draw the final image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

What single transformation has the same effect as a reflection across the x-axis followed by a reflection across the y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

The point (3, 1) is reflected across the line y = x.

What is its image?

A. (3, −1)

B. (−1, −3)

C. (−3, 1)

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

True or false?

Reflecting a figure across one line and then across a parallel line has the same effect as a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

True or false?

Doing a translation and a reflection in the other order always gives the same image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 8.G.A.3 • GEOMETRY

Rotate figures 90° and 180° about the origin

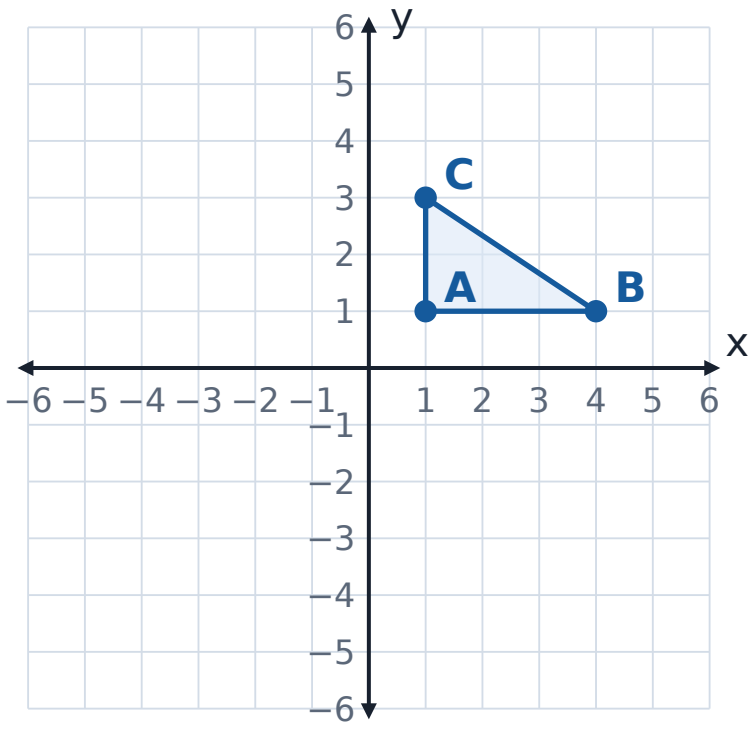
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Relate turns to angle measures	**4.MD.C.5.a**
BACKWARD 2	Locate a point reflected across both axes	**6.NS.C.6.b**
FORWARD 1	Describe a rotation in terms of distance and angle	**HSG-CO.A.4**
FORWARD 2	Use relationships among central angles, inscribed angles, and rad…	**HSG-C.A.2**

SET 38 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Rotate triangle ABC 90° counterclockwise about the origin. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

The point $(5, -2)$ is rotated $180°$ about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

Which rule describes a 90° clockwise rotation about the origin?

A. $(x, y) \to (y, x)$

B. $(x, y) \to (y, -x)$

C. $(x, y) \to (-y, x)$

D. $(x, y) \to (-x, -y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

The point (2, 3) is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

True or false?

Rotating $(-4, 1)$ by $180°$ about the origin gives $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

SET 38 • 8.G.A.3

BACKWARD 1

Relate turns to angle measures • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many degrees are in a quarter turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

How many degrees are in a half turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

How many degrees are in a three-quarter turn?

A. 360°

B. 180°

C. 270°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

True or false?

A full turn is 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

True or false?

A half turn is 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 8.G.A.3

BACKWARD 2

Locate a point reflected across both axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, 4) is reflected across the x-axis and then across the y-axis.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

The point $(-2, 5)$ is reflected across both axes.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

Reflecting a point across both axes changes...

A. the signs of both coordinates

B. only the y-coordinate

C. neither coordinate

D. only the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

Reflecting (1, −6) across both axes gives (−1, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

Reflecting (4, 4) across both axes gives (4, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 8.G.A.3

FORWARD 1

Describe a rotation in terms of distance and angle • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A rotation of 120° about point O maps A to A'.

Which lengths and which angle are determined by the rotation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

P(6, 0) is rotated 60° counterclockwise about the origin.

How far is the image from the origin?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

A rotation about point O maps every point P to a point P' so that...

A. P' is on the line of reflection

B. P' is twice as far from O as P

C. OP = OP' and angle POP' is the angle of rotation

D. PP' is parallel to one fixed line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

True or false?

In a rotation by a nonzero angle less than 360°, the center is the only point that does not move.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

A 90° rotation maps every horizontal segment to another horizontal segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 8.G.A.3

FORWARD 2

Use relationships among central angles, inscribed angles, and radii • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2

A central angle of a circle measures 80°.

What does an inscribed angle that intercepts the same arc measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5

An inscribed angle intercepts a semicircle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5

A radius is drawn to the point where a tangent line touches a circle.

What angle do the radius and the tangent line form?

A. 90°

B. 45°

C. 180°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5

True or false?

Rotating a circle about its center by any angle carries the circle onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5

True or false?

An inscribed angle measures twice its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5

SET 39 • 8.G.A.3 • GEOMETRY

Dilate figures on the coordinate plane with the origin as the center

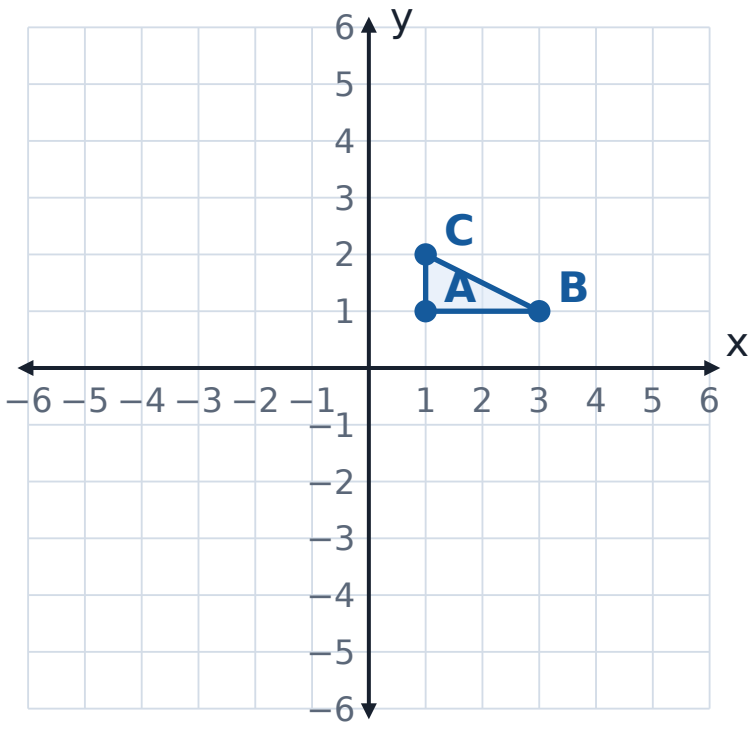
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Use a scale factor to find lengths in a scale drawing	**7.G.A.1**
BACKWARD 2	Multiply rational numbers	**7.NS.A.2.a**
FORWARD 1	Verify the properties of dilations	**HSG-SRT.A.1**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 39 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Dilate triangle ABC by a scale factor of 2 with the origin as the center. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

The point $(6, -4)$ is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

Which rule describes a dilation with a scale factor of 3 and the origin as the center?

A. $(x, y) \rightarrow (3x, y)$

B. $(x, y) \rightarrow (x + 3, y + 3)$

C. $(x, y) \rightarrow (\frac{x}{3}, \frac{y}{3})$

D. $(x, y) \rightarrow (3x, 3y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

A dilation with the origin as the center maps (4, 2) to (10, 5).

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

True or false?

A dilation with a scale factor of $\frac{1}{3}$ and the origin as the center maps (9, 6) to (3, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

SET 39 • 8.G.A.3

BACKWARD 1

Use a scale factor to find lengths in a scale drawing • Grade 7 • Geometry • 7.G.A.1

A 4 cm segment is enlarged by a scale factor of 3.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

A 10 cm segment is reduced by a scale factor of $\frac{1}{2}$.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

A rectangle 2 cm by 5 cm is enlarged by a scale factor of 4.

What are the new dimensions?

A. 8 cm by 20 cm

B. 8 cm by 5 cm

C. 6 cm by 9 cm

D. 2 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A scale factor greater than 1 makes a figure larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A scale factor of $\frac{1}{4}$ makes each length 4 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 8.G.A.3

BACKWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$2 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

Multiply.

$\frac{1}{2} \times (-6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

Multiply.

3.5×4

A. 12

B. 0.875

C. 14

D. 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

True or false?

$3 \times (-2) = -6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 9 = 27$

A. True

B. False

SET 39 • 8.G.A.3

FORWARD 1

Verify the properties of dilations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A dilation with the origin as the center and a scale factor of 3 is applied to the line $y = x + 1$.

What is the equation of the image?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

A segment 4 units long is dilated by a scale factor of 2.5.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

A dilation is applied to a line that passes through the center of dilation.

What is the image?

A. A parallel line

B. A single point

C. A perpendicular line

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

A dilation with a scale factor of 2 doubles every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 8.G.A.3

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-foot person casts a 4-foot shadow. At the same time, a tree casts a 22-foot shadow.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

Triangle ABC is similar to triangle DEF, with AB = 8, DE = 12, and BC = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

A segment parallel to one side of a triangle splits a second side into parts of 3 and 6. It splits the third side into a first part of 4 and a second part.

How long is the second part?

A. 2

B. 12

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

True or false?

In similar triangles, the ratio of the perimeters equals the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

True or false?

Triangles with sides 3, 4, 5 and 6, 8, 12 are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 8.G.A.4 • GEOMETRY

Describe a sequence of transformations that shows two figures are similar

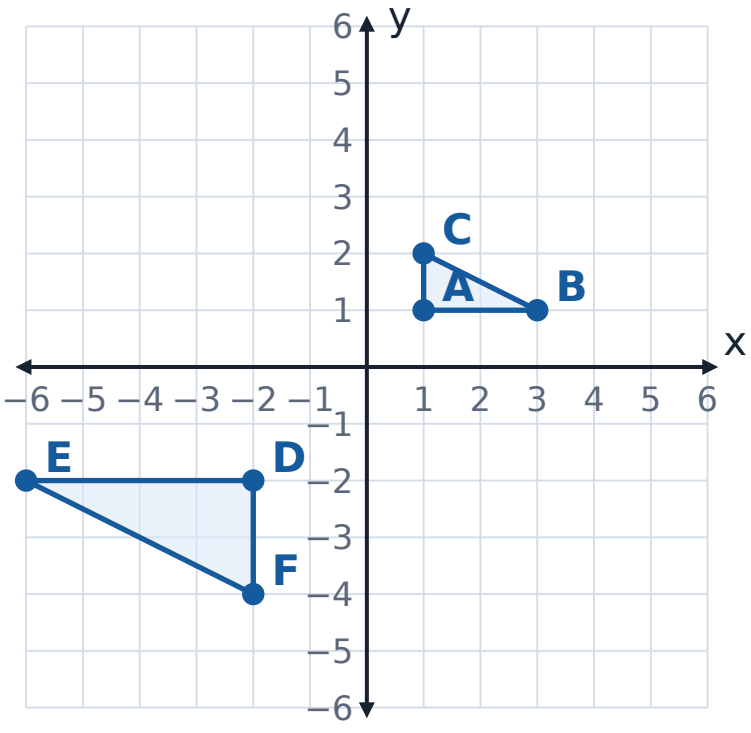
MAIN	Grade 8 main question	**8.G.A.4**
BACKWARD 1	Find a scale factor between scale copies	**7.G.A.1**
BACKWARD 2	Decide whether two ratios are equivalent	**7.RP.A.2.a**
FORWARD 1	Use similarity transformations to decide whether triangles are sim…	**HSG-SRT.A.2**
FORWARD 2	Use a translation and a dilation to show that any two circles are s…	**HSG-C.A.1**

SET 40 • 8.G.A.4

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.4

Describe a sequence of transformations that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

Explain why a 2-by-3 rectangle and a 4-by-6 rectangle are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

Which sequence of transformations maps a figure onto a figure that is similar but not congruent?

A. A rotation, then a translation

B. A translation only

C. Two reflections

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

Are a 3-4-5 triangle and a 5-12-13 triangle similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

True or false?

Two figures are similar if a sequence of rotations, reflections, translations, and dilations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

SET 40 • 8.G.A.4

BACKWARD 1

Find a scale factor between scale copies • Grade 7 • Geometry • 7.G.A.1

A 3 cm side becomes 12 cm in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

A 15 m side becomes 5 m in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

A triangle's sides are 2, 3, and 4. Which side lengths are a scale copy?

A. 6, 9, 12

B. 4, 5, 6

C. 2, 6, 8

D. 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A scale copy with a scale factor of 1 is the same size as the original.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A rectangle 2 by 3 and a rectangle 3 by 4 are scale copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 8.G.A.4

BACKWARD 2

Decide whether two ratios are equivalent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Are 4 : 6 and 10 : 15 equivalent ratios?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

Are 3 : 5 and 9 : 12 equivalent ratios?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 5 : 8?

A. 25 : 48

B. 15 : 24

C. 8 : 5

D. 10 : 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

True or false?

2 : 7 and 6 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

True or false?

4 : 5 and 5 : 6 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 8.G.A.4

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 5, 7, and 9. Triangle DEF has sides 10, 14, and 18.

Are they similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

Two triangles are similar. Explain what is true about their corresponding angles and sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

Triangle PQR is similar to triangle STU with a scale factor of 3. PQ = 4.

What is ST?

A. 12

B. 4

C. $\frac{4}{3}$

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

True or false?

If a dilation followed by a rotation maps one triangle onto another, the triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

True or false?

Any two rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 8.G.A.4

FORWARD 2

Use a translation and a dilation to show that any two circles are similar • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1

Circle A has center (0, 0) and radius 2. Circle B has center (5, 1) and radius 6.

Describe a translation followed by a dilation that maps circle A onto circle B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

What scale factor of a dilation about the center maps a circle with a radius of 4 onto a circle with the same center and a radius of 10?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

Which statement explains why any two circles are similar?

A. A rotation alone maps any circle onto any other circle.

B. All circles have the same radius.

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

D. All circles have the same area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

True or false?

Any two circles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

A circle with a radius of 3 can be mapped onto a circle with a radius of 6 by a translation alone.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 8.G.A.5 • GEOMETRY

Use the angle sum and exterior angles of a triangle

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for supplementary and complementary…	**7.G.B.5**
BACKWARD 2	Find an unknown angle in a sum of angles	**4.MD.C.7**
FORWARD 1	Prove theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Define trigonometric ratios from similar right triangles	**HSG-SRT.C.6**

SET 41 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle measure 52° and 71°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

The angles of a triangle measure $x°$, $(x + 10)°$, and $(2x - 30)°$.

Write and solve an equation to find x.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

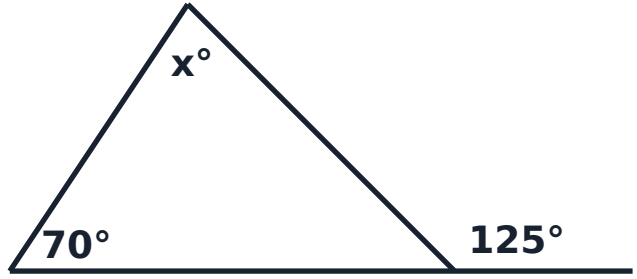
In the figure, what is the value of x?

A. 55

B. 70

C. 125

D. 35



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

SET 41 • 8.G.A.5

BACKWARD 1

Write and solve equations for supplementary and complementary angles • Grade 7 • Geometry • 7.G.B.5

Two supplementary angles measure x° and 130°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

Two complementary angles measure 2x° and 40°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

Angles of x° and (x + 30)° are supplementary.

What is x?

A. 75

B. 150

C. 105

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Angles of 47° and 43° are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

True or false?

Angles of 100° and 90° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 8.G.A.5

BACKWARD 2

Find an unknown angle in a sum of angles • Grade 4 • Measurement & Data • 4.MD.C.7

A 90° angle is split into two angles. One is 35°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

A straight angle (180°) is split into two angles. One is 110°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

An angle of 120° is split into angles of 75° and x°.

What is x?

A. 195

B. 55

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

Angles of 30° and 60° together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

Angles of 80° and 80° together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 8.G.A.5

FORWARD 1

Prove theorems about triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Use a line through C that is parallel to AB. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

The angles of a triangle are 2x°, 3x°, and 4x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

An isosceles triangle has base angles of 40°.

What does its third angle measure?

A. 100°

B. 40°

C. 80°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

True or false?

The base angles of an isosceles triangle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 8.G.A.5

FORWARD 2

Define trigonometric ratios from similar right triangles • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is sin A?

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is tan A?

tan A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is cos A?

A. $\frac{3}{5}$

B. $\frac{5}{4}$

C. $\frac{4}{5}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5

True or false?

In every right triangle with a 30° angle, the side opposite the 30° angle is half the hypotenuse, so $\sin 30° = \frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5

True or false?

sin A changes when a right triangle is enlarged but its angles stay the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5

SET 42 • 8.G.A.5 • GEOMETRY

Write and solve equations for angles formed by parallel lines cut by a transversal

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for vertical and adjacent angles	**7.G.B.5**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Prove that alternate interior angles are congruent	**HSG-CO.C.9**
FORWARD 2	Use slopes to show that lines are parallel or perpendicular	**HSG-GPE.B.5**

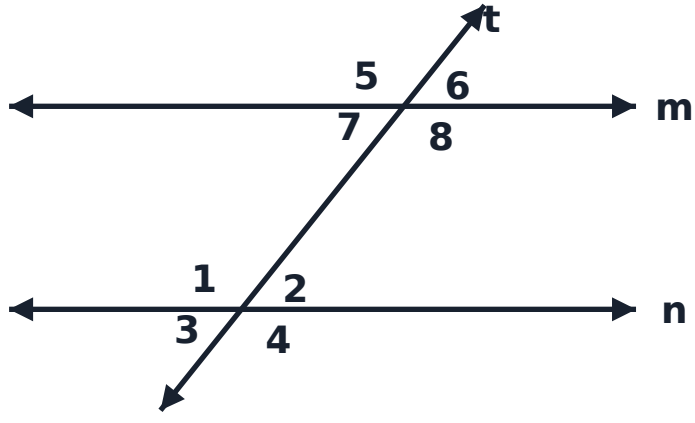
SET 42 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

In the figure, lines m and n are parallel and angle 2 = 52°.

What does angle 6 measure? Give the reason.



Answer: _____

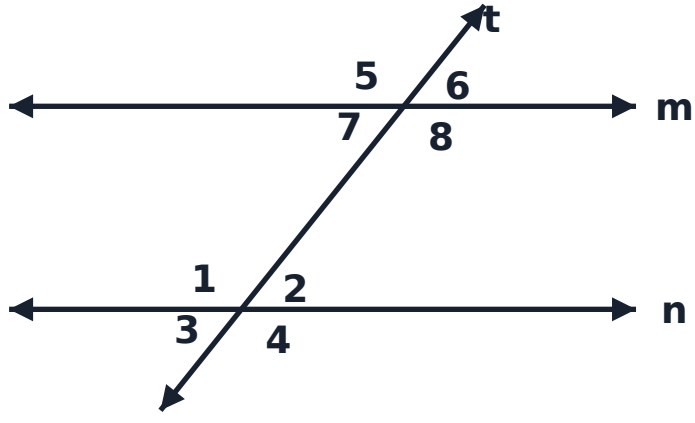
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

In the figure, lines m and n are parallel. Angle 1 = (3x + 10)° and angle 8 = 115°.

Write and solve an equation for x.



Equation and answer: _____

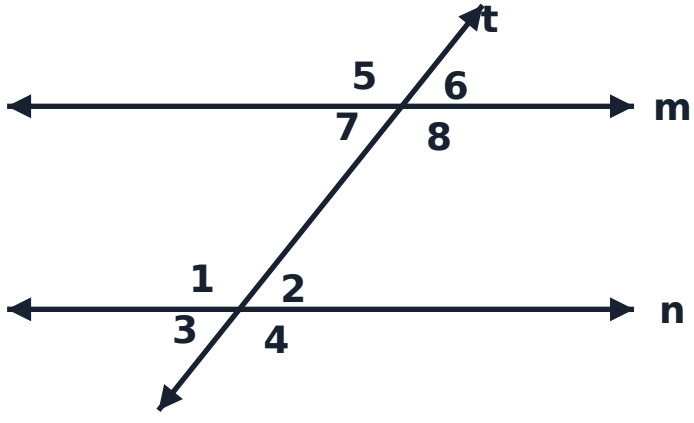
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

In the figure, lines m and n are parallel. Angle 2 = (2x + 4)° and angle 8 = (4x − 10)°.

What is x?



A. 93

B. 31

C. 43

D. 7

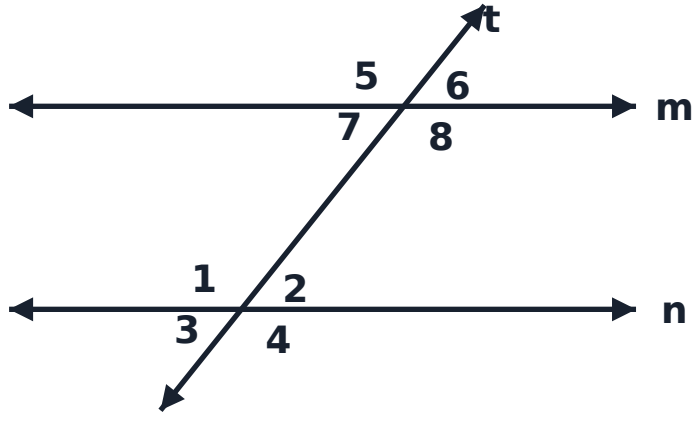
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

In the figure, lines m and n are parallel and angle 4 = 128°.

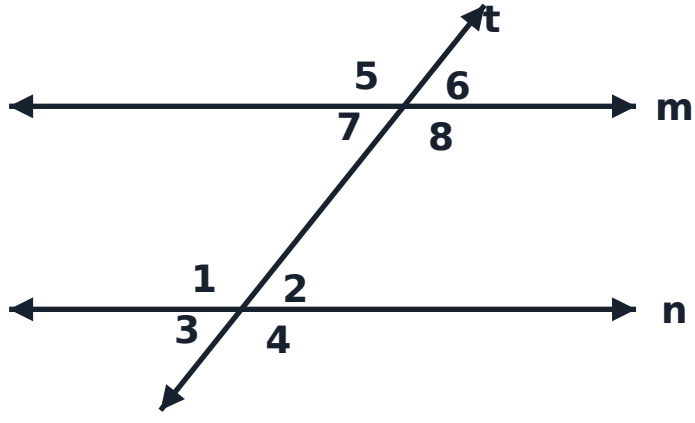
What does angle 5 measure? Give the reasons.



Answer: _____

True or false?

In the figure, angles 3 and 6 are alternate exterior angles, so they are congruent when m and n are parallel.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

SET 42 • 8.G.A.5

BACKWARD 1

Write and solve equations for vertical and adjacent angles • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. Vertical angles measure (2x + 20)° and 80°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

Two lines intersect. Adjacent angles measure x° and 3x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

Two lines intersect. Adjacent angles measure (x + 40)° and 2x°.

What is x?

A. $\frac{140}{3}$

B. 50

C. 40

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

True or false?

Vertical angles of 5x° and 75° give x = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

True or false?

Adjacent angles that form a straight line measure $(x + 10)°$ and $x°$. Then $x = 90$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 8.G.A.5

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$3x + 10 = 115$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

Solve for x.

$2x - 15 = 75$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

Solve for x.

$6x - 6 = 180$

A. $x = 174$

B. $x = 29$

C. $x = 31$

D. $x = 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

True or false?

$x = 20$ is the solution of $4x + 20 = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

True or false?

$x = 25$ is the solution of $5x - 5 = 115$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

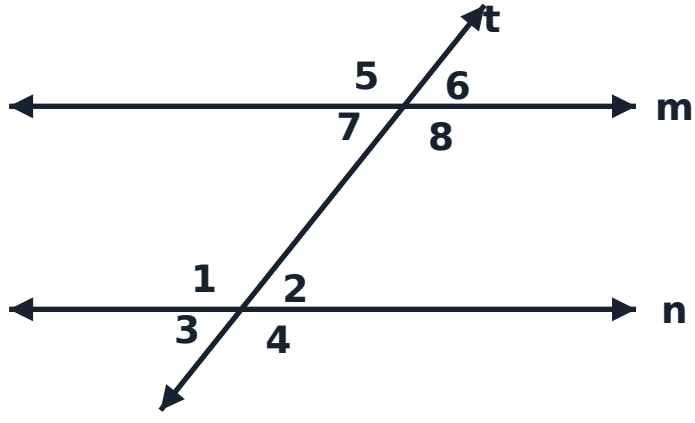
Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 8.G.A.5

FORWARD 1

Prove that alternate interior angles are congruent • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

In the figure, lines m and n are parallel. Prove that angle 1 = angle 8. Give a reason for each step.



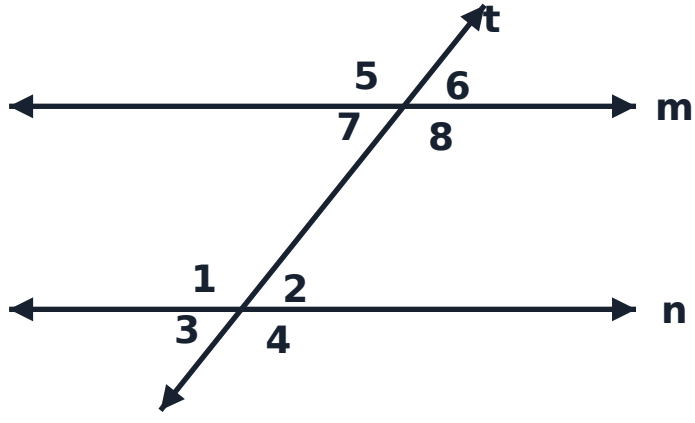
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

In the figure, lines m and n are parallel. Prove that angle 2 + angle 8 = 180°. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

A proof says: angle 1 = angle 5 (corresponding angles). Angle 5 = angle 8 (?). So angle 1 = angle 8.

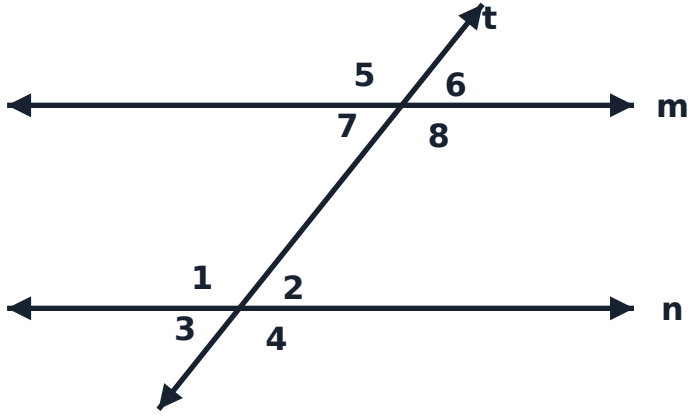
What is the missing reason?

A. Linear pairs are supplementary.

B. Vertical angles are congruent.

C. Alternate interior angles are congruent.

D. The triangle angle sum is 180°.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

If corresponding angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

True or false?

"The angles look equal in the figure" is a valid reason in a proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 8.G.A.5

FORWARD 2

Use slopes to show that lines are parallel or perpendicular • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

A transversal crosses the lines $y = 2x + 1$ and $y = 2x - 3$.

Are the two lines parallel? Explain with slopes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

Write an equation of the line through (1, 4) that is perpendicular to $y = -\frac{1}{2}x + 3$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

Which pair of lines is perpendicular?

A. $y = x$ and $y = x + 2$

B. $y = 3x$ and $y = -3x$

C. $y = 3x + 1$ and $y = 3x - 1$

D. $y = 3x + 1$ and $y = -\frac{1}{3}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

True or false?

Two non-vertical lines are parallel exactly when their slopes are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

The lines $y = \frac{2}{5}x$ and $y = \frac{5}{2}x$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 43 • 8.G.A.5 • GEOMETRY

Use the angle-angle criterion to show that triangles are similar

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Find a missing length in a scale copy	**7.G.A.1**
BACKWARD 2	Use the triangle angle sum to find a missing angle	**7.G.B.5**
FORWARD 1	Establish the angle-angle criterion with similarity transformations	**HSG-SRT.A.3**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 43 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Triangle ABC has angles of 35° and 85°. Triangle DEF has angles of 35° and 60°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

A 5-foot post casts a 3-foot shadow. At the same time, a tree casts a 12-foot shadow.

Explain why the two triangles are similar, and find the height of the tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

Which information shows that two triangles are similar by the angle-angle criterion?

A. One pair of sides is proportional.

B. All the sides are equal.

C. Two pairs of corresponding angles are equal.

D. One pair of corresponding angles is equal.

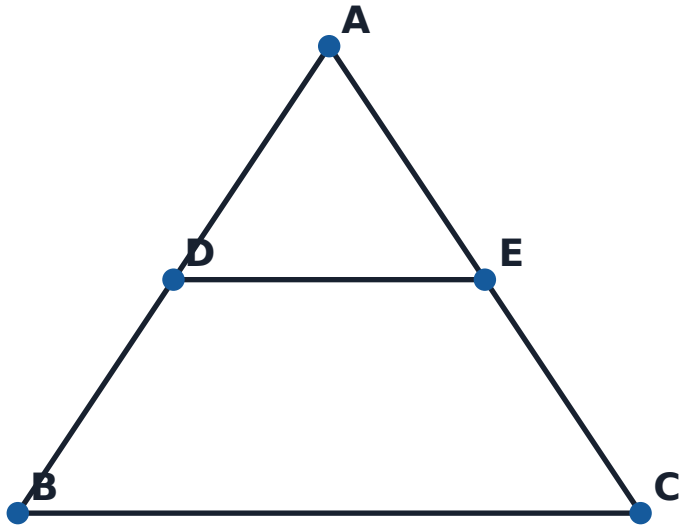
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

In the figure, segment DE is parallel to BC.

Explain why triangle ADE is similar to triangle ABC.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

True or false?

Two triangles that each have a 90° angle and a 40° angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

SET 43 • 8.G.A.5

BACKWARD 1

Find a missing length in a scale copy • Grade 7 • Geometry • 7.G.A.1

A triangle has sides 4, 6, and 8. In a scale copy, the side of 4 becomes 10.

What does the side of 6 become?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A drawing with a scale factor of 3 shows a 2 cm side.

How long is that side in the scale copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A 5-by-8 rectangle is scaled so the 5 becomes 15.

What does the 8 become?

A. 40

B. 24

C. 13

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

In a scale copy, every length is multiplied by the same scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

In a scale copy with a scale factor of 2, every angle is doubled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 8.G.A.5

BACKWARD 2

Use the triangle angle sum to find a missing angle • Grade 7 • Geometry • 7.G.B.5

A triangle has angles of 35° and 60°.

What is the third angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q1/5

A right triangle has a 40° angle.

What is the third angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q2/5

A triangle has angles of 90° and 25°.

What is the third angle?

A. 75°

B. 65°

C. 115°

D. 55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q3/5

True or false?

A triangle with angles of 50° and 60° has a third angle of 70°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q4/5

True or false?

A triangle with angles of 45° and 45° has a third angle of 45°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 8.G.A.5

FORWARD 1

Establish the angle-angle criterion with similarity transformations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Triangles ABC and DEF have angle A = angle D and angle B = angle E.

Which transformations show that the triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

Why is it enough to know two pairs of equal angles to show triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

Triangles with angles 30°, 70°, 80° and 70°, 30°, 80° are...

A. congruent in every case

B. similar

C. not similar

D. impossible to compare

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

True or false?

Two equilateral triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

True or false?

Two isosceles triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 8.G.A.5

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 1.5 m stick casts a 2 m shadow. A building casts a 36 m shadow at the same time.

How tall is the building?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

Triangle PQR is similar to triangle XYZ with PQ = 6, XY = 9, and QR = 8.

Find YZ.

YZ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

Two similar triangles have a scale factor of 3. The smaller one has a perimeter of 14.

What is the larger one's perimeter?

A. 17

B. 14

C. 42

D. 126

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

True or false?

A mirror can be used with similar triangles to find the height of a tall object.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

True or false?

If two triangles are similar with a scale factor of 2, their areas are also in the ratio 2 : 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • 8.G.B.6 • GEOMETRY

Explain a proof of the Pythagorean theorem and its converse

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the area of a right triangle	**6.G.A.1**
BACKWARD 2	Find the area of a square	**3.MD.C.7.b**
FORWARD 1	Prove the Pythagorean theorem with similar triangles	**HSG-SRT.B.4**
FORWARD 2	Derive the equation of a circle from the Pythagorean theorem	**HSG-GPE.A.1**

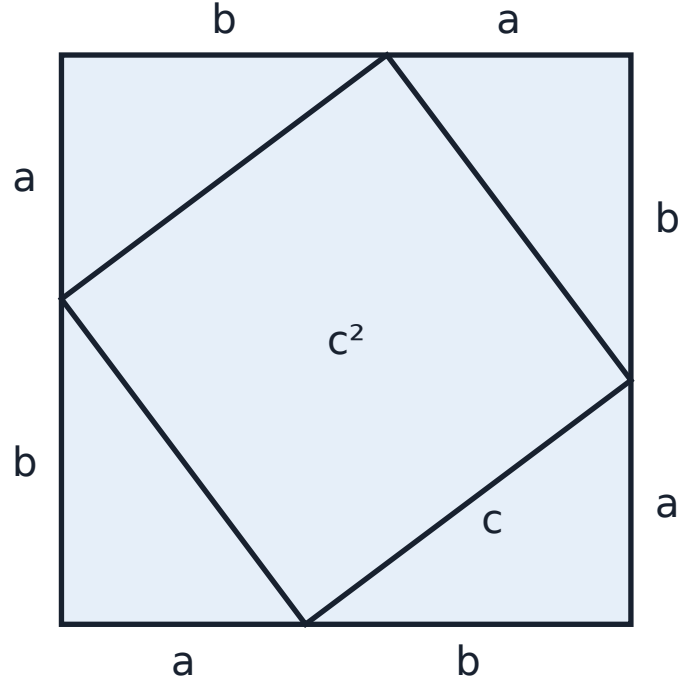
SET 44 • 8.G.B.6

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

The figure shows a square with sides a + b, made of four right triangles and a tilted square with side c.

Explain how the figure proves that $a^2 + b^2 = c^2$.



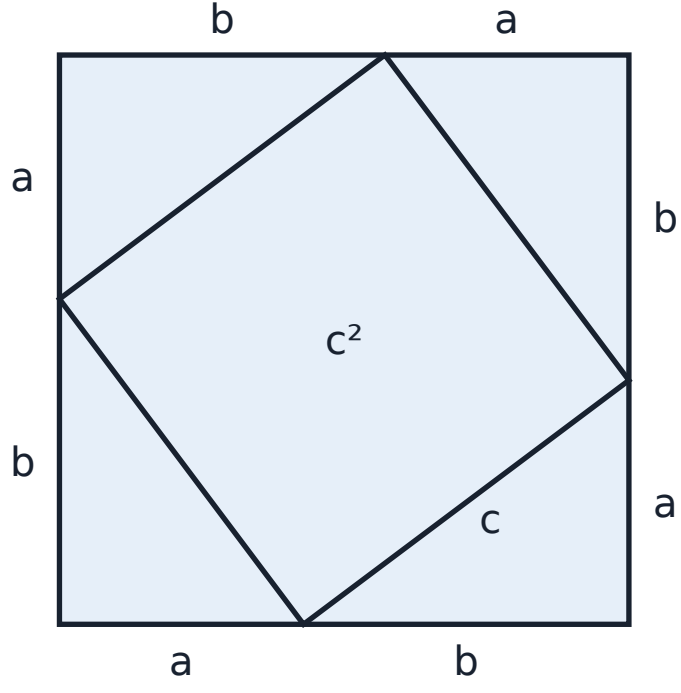
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

In the figure, write the area of the large square as the sum of the areas of the four triangles and the inner square.



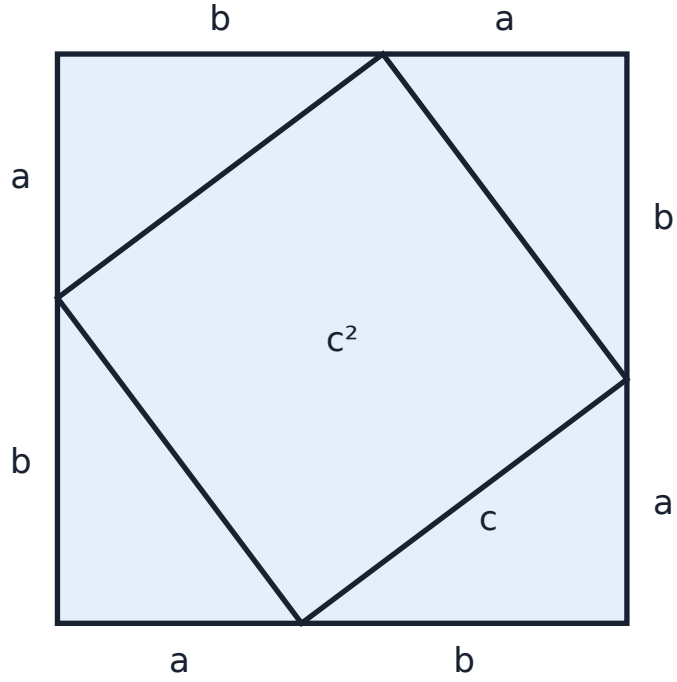
Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

In the proof with the figure, which step comes after writing $(a + b)^2 = 2ab + c^2$?



A. Divide both sides by c

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

C. Add 2ab to both sides

D. Take the square root of each term

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

A triangle has sides a, b, and c with $a^2 + b^2 = c^2$. Explain why it must be a right triangle.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

True or false?

The Pythagorean theorem $a^2 + b^2 = c^2$ is true for every triangle.

A. True

B. False

SET 44 • 8.G.B.6

BACKWARD 1

Find the area of a right triangle • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs of 3 and 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

A right triangle has legs of 5 and 12.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

What is the total area of four right triangles that each have legs of 2 and 6?

A. 24

B. 12

C. 48

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

A right triangle with legs of 8 and 3 has an area of 12 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

True or false?

A right triangle with legs of 6 and 6 has an area of 36 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 8.G.B.6

BACKWARD 2

Find the area of a square • Grade 3 • Measurement & Data • 3.MD.C.7.b

A square has sides of 7 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

A square has sides of 9 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

A square has sides of 5 units.

What is its area?

A. 15 square units

B. 25 square units

C. 10 square units

D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

True or false?

A square with 4-unit sides has an area of 16 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

A square with 6-unit sides has an area of 24 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 8.G.B.6

FORWARD 1

Prove the Pythagorean theorem with similar triangles • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In right triangle ABC with the right angle at C, altitude CD is drawn to hypotenuse AB.

Why is $AC^2 = AD \cdot AB$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

In the same triangle, $BC^2 = BD \cdot AB$. Add this to $AC^2 = AD \cdot AB$. What do you get?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

Which criterion shows that triangle ACD is similar to triangle ABC?

A. ASA

B. AA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

The altitude to the hypotenuse divides a right triangle into two triangles similar to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

The altitude to the hypotenuse always cuts the hypotenuse into two equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 8.G.B.6

FORWARD 2

Derive the equation of a circle from the Pythagorean theorem • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center (0, 0) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

Write the equation of the circle with center $(2, -1)$ and radius 3.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

Which point is on the circle $x^2 + y^2 = 100$?

A. (5, 5)

B. (10, 10)

C. (6, 8)

D. (6, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

True or false?

The equation $(x - h)^2 + (y - k)^2 = r^2$ comes from the Pythagorean theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

True or false?

The circle $x^2 + y^2 = 16$ has a radius of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 8.G.B.6 • GEOMETRY

Use the converse of the Pythagorean theorem to decide whether a triangle is a right triangle

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the squares of whole numbers	**6.EE.A.1**
BACKWARD 2	Evaluate an expression with exponents	**6.EE.A.2.c**
FORWARD 1	Classify a triangle as acute, right, or obtuse from its side lengths	**HSG-CO.C.10**
FORWARD 2	Use coordinates to show that a triangle is a right triangle	**HSG-GPE.B.4**

SET 45 • 8.G.B.6

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides 9, 12, and 15 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

Is a triangle with sides 7, 8, and 11 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

Which side lengths form a right triangle?

A. 6, 8, 12

B. 20, 21, 29

C. 5, 7, 9

D. 10, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

A carpenter checks a corner. Points 3 feet and 4 feet from the corner along the two edges are 5 feet apart.

Is the corner a right angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

True or false?

A triangle with sides 2, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

SET 45 • 8.G.B.6

BACKWARD 1

Find the squares of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

Find the value.

11^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

Find the value.

13^2

A. 26

B. 169

C. 139

D. 196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

True or false?

$20^2 = 400$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

$9^2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 8.G.B.6

BACKWARD 2

Evaluate an expression with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $a^2 + b^2$ when $a = 6$ and $b = 8$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

Evaluate $a^2 + b^2$ when $a = 5$ and $b = 12$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

Evaluate $c^2 - a^2$ when $c = 10$ and $a = 6$.

A. 16

B. 136

C. 64

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

True or false?

When $a = 3$ and $b = 4$, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

True or false?

When $a = 2$ and $b = 3$, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 8.G.B.6

FORWARD 1

Classify a triangle as acute, right, or obtuse from its side lengths • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

Is a triangle with sides 5, 6, and 8 acute, right, or obtuse? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

Is a triangle with sides 6, 7, and 8 acute, right, or obtuse? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

A triangle has sides 8, 15, and 17.

What kind of triangle is it?

A. Obtuse

B. Acute

C. Right

D. Not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

True or false?

If $a^2 + b^2 < c^2$ for the longest side c, the triangle is obtuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

True or false?

A triangle with sides 4, 4, and 7 is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 8.G.B.6

FORWARD 2

Use coordinates to show that a triangle is a right triangle • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Is the triangle with vertices (1, 1), (4, 1), and (4, 5) a right triangle? Use side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

Use slopes to decide whether the triangle with vertices (0, 0), (2, 4), and (6, 2) has a right angle at (2, 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

Which triangle is a right triangle?

A. (0, 0), (2, 2), (4, 1)

B. (0, 0), (3, 1), (1, 3)

C. (1, 1), (2, 3), (3, 1)

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

True or false?

The triangle with vertices (0, 0), (5, 0), and (0, 5) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

True or false?

The triangle with vertices (0, 0), (4, 0), and (2, 1) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in two dimensions

MAIN	Grade 8 main question	**8.G.B.7**
BACKWARD 1	Find the squares of decimals and whole numbers	**6.EE.A.1**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve right triangles with the Pythagorean theorem and trigonom...	**HSG-SRT.C.8**
FORWARD 2	Apply geometric methods to solve design problems	**HSG-MG.A.3**

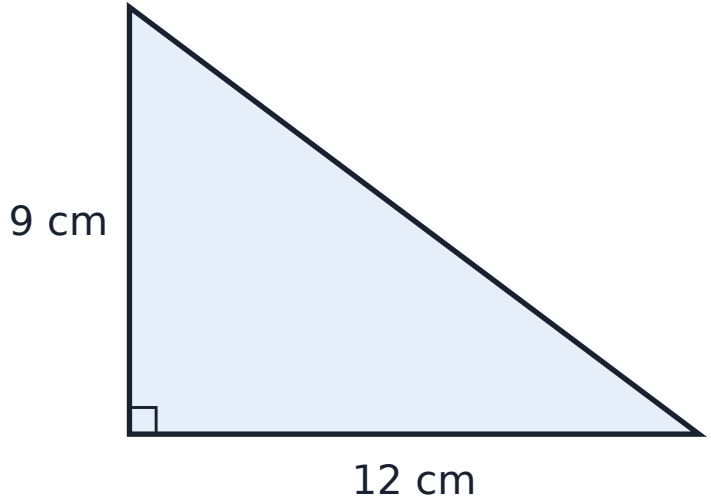
SET 46 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

Find the length of the hypotenuse.

Hypotenuse: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

A 13-foot ladder leans against a wall. Its base is 5 feet from the wall.

How high up the wall does it reach?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

A rectangular screen is 48 inches wide and 36 inches tall.

How long is its diagonal?

A. 60 inches

B. 84 inches

C. 72 inches

D. 42 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

A right triangle has a hypotenuse of 17 and a leg of 8.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

True or false?

A square with 10 cm sides has a diagonal of about 14.1 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

SET 46 • 8.G.B.7

BACKWARD 1

Find the squares of decimals and whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

14^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

Find the value.

2.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

Find the value.

17^2

A. 189

B. 279

C. 289

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

$1.5^2 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 8.G.B.7

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$x + 64 = 289$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

Solve for x.

$25 + x = 169$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

Solve for x.

$81 + x = 100$

A. $x = 19$

B. $x = 181$

C. $x = 9$

D. $x = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

True or false?

$x = 15$ is the solution of $x + 49 = 64$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

True or false?

$x = 46$ is the solution of $64 + x = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 8.G.B.7

FORWARD 1

Solve right triangles with the Pythagorean theorem and trigonometric ratios • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has a hypotenuse of 10 and a 30° angle. $\sin 30° = 0.5$.

How long is the side opposite the 30° angle?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

A ramp rises 3 feet over a horizontal distance of 4 feet.

How long is the ramp?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

A kite string is 50 m long and makes a right triangle with the ground. The kite is 30 m above the ground.

How far is the kite horizontally from the person?

A. 58.3 m

B. 40 m

C. 20 m

D. 80 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

True or false?

A right triangle with two 45° angles and legs of 7 has a hypotenuse of $7\sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

True or false?

$\tan 45° = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 8.G.B.7

FORWARD 2

Apply geometric methods to solve design problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular field is 120 m by 50 m.

How much shorter is walking along the diagonal than walking along two sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

A TV is sold as a 50-inch TV (its diagonal). It is 40 inches wide.

How tall is it?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

A cable runs from the top of a 12 m pole to a stake 5 m from the base of the pole.

How long is the cable?

A. 13 m

B. 17 m

C. $\sqrt{119}$ m

D. 7 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

True or false?

A 10-foot ladder whose base is 6 feet from a wall reaches 8 feet up the wall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

True or false?

The diagonal of a 9-by-12 rectangle is 21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in three dimensions

MAIN	Grade 8 main question	8.G.B.7
BACKWARD 1	Identify the faces, edges, and vertices of a prism	6.G.A.4
BACKWARD 2	Describe cross sections of right prisms	7.G.A.3
FORWARD 1	Use the Pythagorean theorem to solve problems in three dimensio	HSG-SRT.C.8
FORWARD 2	Identify solids made by rotating two-dimensional figures	HSG-GMD.B.4

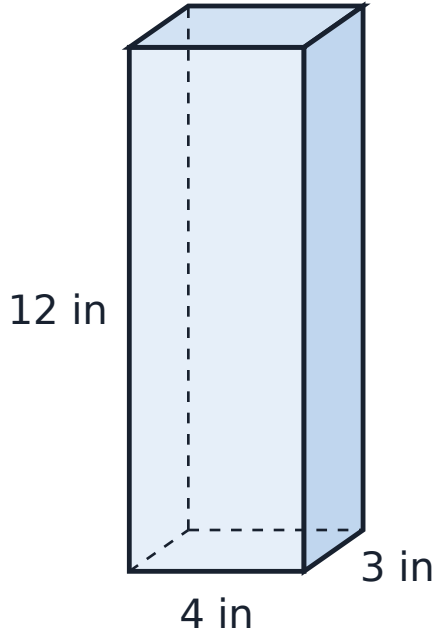
SET 47 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

A box is 4 inches long, 3 inches wide, and 12 inches tall.

How long is the diagonal from a bottom corner to the opposite top corner?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

A cube has 6 cm edges.

Find the diagonal of one face, and then the diagonal from one corner of the cube to the opposite corner. Give exact values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

Will a 26-inch stick fit inside a box that is 24 by 6 by 8 inches, from one corner to the opposite corner?

A. No; the box is too short.

B. Yes; the face diagonal is 30 inches.

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

D. No; the diagonal is only 24 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

A right square pyramid has its top directly above the center of its base. Its base edges are 10 m and its height is 12 m.

What is the slant height (from the top to the midpoint of a base edge)?

Slant height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

True or false?

The corner-to-corner diagonal of a 2 by 3 by 6 box is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

SET 47 • 8.G.B.7

BACKWARD 1

Identify the faces, edges, and vertices of a prism • Grade 6 • Geometry • 6.G.A.4

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

How many edges does a rectangular prism have?

Edges: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

How many vertices does a cube have?

A. 6

B. 4

C. 12

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

Each face of a rectangular prism is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A triangular prism has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 8.G.B.7

BACKWARD 2

Describe cross sections of right prisms • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane through two opposite vertical edges.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

A cube is sliced by a plane parallel to one face, through its interior.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

A right rectangular prism is sliced parallel to its base, through its interior.

What is the cross section?

A. A circle

B. A triangle

C. A pentagon

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

True or false?

A diagonal slice through a box can make a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A slice of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 8.G.B.7

FORWARD 1

Use the Pythagorean theorem to solve problems in three dimensions • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A room is 12 ft long, 9 ft wide, and 8 ft high.

How long is the diagonal of the floor?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

A room is 12 ft long, 9 ft wide, and 8 ft high.

How long is the diagonal from a floor corner to the opposite ceiling corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

A box is 1 by 2 by 2.

How long is its corner-to-corner diagonal?

A. 3

B. 9

C. $\sqrt{5}$

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

True or false?

The corner-to-corner diagonal of a cube with edge s is $s\sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

True or false?

A box 3 by 4 by 5 has a corner-to-corner diagonal of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 8.G.B.7

FORWARD 2

Identify solids made by rotating two-dimensional figures • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A right triangle with legs 3 and 4 is rotated around the leg of length 4.

What solid does it make, and what is its slant height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5

A 2-by-5 rectangle is rotated around one of its 5-unit sides.

What solid does it make?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5

A sphere is sliced by a plane through its center.

What is the cross section?

A. A circle with the same radius as the sphere

B. An oval that is not a circle

C. A square

D. A smaller circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5

True or false?

Rotating a semicircle about its diameter makes a sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5

True or false?

A cube can never have a triangular cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5

SET 48 • 8.G.B.8 • GEOMETRY

Find the distance between two points on the coordinate plane

MAIN	Grade 8 main question	**8.G.B.8**
BACKWARD 1	Find distances along grid lines	**6.NS.C.8**
BACKWARD 2	Find a distance as the absolute value of a difference	**7.NS.A.1.c**
FORWARD 1	Use the Pythagorean theorem with coordinates to solve problems	**HSG-SRT.C.8**
FORWARD 2	Use coordinates to find perimeters and areas of polygons	**HSG-GPE.B.7**

SET 48 • 8.G.B.8

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (7, 10).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

Find the distance between $(-3, 4)$ and $(2, -8)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

What is the distance between (0, 0) and (4, 4), to the nearest tenth?

A. 16.0 units

B. 4.0 units

C. 5.7 units

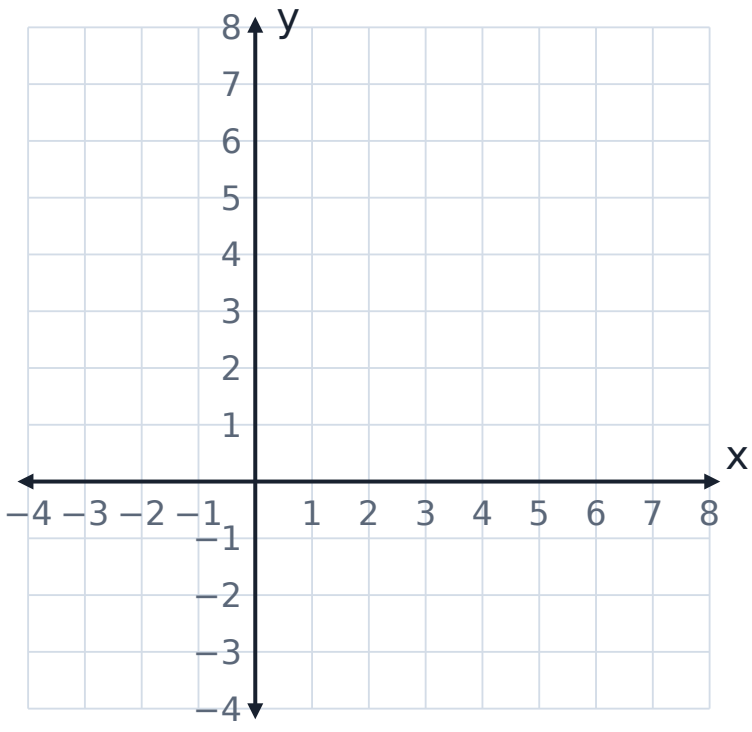
D. 8.0 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

Plot A(−2, −1) and B(4, 7). Draw a right triangle with AB as its hypotenuse, and find AB.



AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

True or false?

The distance between $(-1, -1)$ and $(2, 3)$ is 5 units.

A. True

B. False

SET 48 • 8.G.B.8

BACKWARD 1

Find distances along grid lines • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

Which two points are 8 units apart?

A. (1, 1) and (1, 8)

B. (−3, 0) and (3, 0)

C. (−3, 0) and (5, 0)

D. (0, −4) and (0, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

True or false?

(2, −5) and (2, 5) are 10 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

True or false?

$(-6, 3)$ and $(-1, 3)$ are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 8.G.B.8

BACKWARD 2

Find a distance as the absolute value of a difference • Grade 7 • The Number System • 7.NS.A.1.c

Find $|7 - (-2)|$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

How far apart are −4 and 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

Which expression gives the distance between -8 and -3?

A. $|-8 - (-3)|$

B. $-8 - 3$

C. $|-8 + 3| - 1$

D. $-8 + (-3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

True or false?

The distance between -5 and 1 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

True or false?

The distance between -7 and -2 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 8.G.B.8

FORWARD 1

Use the Pythagorean theorem with coordinates to solve problems • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

On a map grid, a park is at (2, 3) and a school is at (10, 9). Each unit is 1 km.

How far apart are they in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

A drone flies from (0, 0) to (12, 5) on a grid in meters.

How far does it fly in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

Which point is exactly 5 units from (1, 1)?

A. (6, 1)

B. (5, 5)

C. (1, 7)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

True or false?

Every point that is 5 units from the origin lies on a circle with radius 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

True or false?

The points (0, 0) and (3, 3) are 6 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 8.G.B.8

FORWARD 2

Use coordinates to find perimeters and areas of polygons • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the perimeter of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

Find the area of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

A rhombus has vertices (0, 0), (3, 4), (8, 4), and (5, 0).

What is its perimeter?

A. 20 units

B. 18 units

C. 16 units

D. 25 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

True or false?

A square with vertices (0, 0), (3, 1), (2, 4), and (−1, 3) has an area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

True or false?

The rectangle with vertices (0, 0), (4, 0), (4, 3), and (0, 3) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

SET 49 • 8.G.C.9 • GEOMETRY

Find the volume of a cylinder

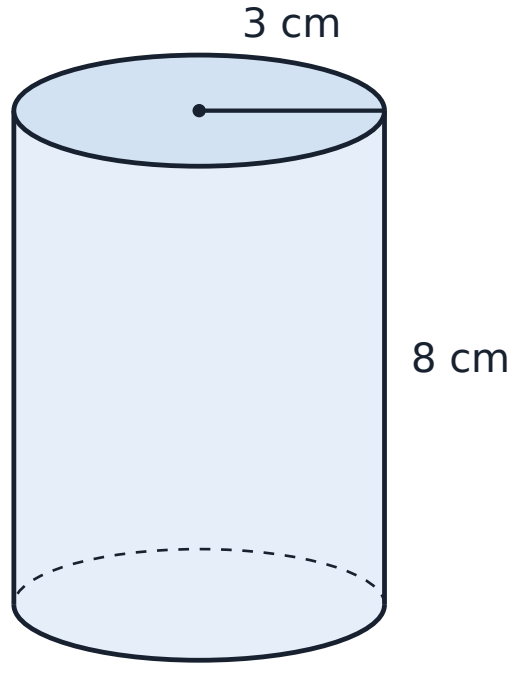
MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle	**7.G.B.4**
BACKWARD 2	Find the volume of a right prism as base area times height	**7.G.B.6**
FORWARD 1	Give an informal argument for volume formulas	**HSG-GMD.A.1**
FORWARD 2	Use density with volume in a model	**HSG-MG.A.2**

SET 49 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cylinder in terms of π.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

A cylinder has a diameter of 10 inches and a height of 6 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

A can holds 128π cubic centimeters and has a radius of 4 cm.

What is its height?

A. 8 cm

B. 2 cm

C. 16 cm

D. 32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

A cylinder-shaped tank has a radius of 2 m and a height of 5 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

True or false?

Doubling the radius of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

SET 49 • 8.G.C.9

BACKWARD 1

Find the area of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 3.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

A circle has a diameter of 10 cm.

What is its area? Use 3.14 for π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

A circle has a radius of 4 m.

What is its area in terms of π?

A. 64π square meters

B. 4π square meters

C. 16π square meters

D. 8π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 1 has an area of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

True or false?

A circle with a diameter of 6 has an area of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism as base area times height • Grade 7 • Geometry • 7.G.B.6

A right prism has a base area of 20 square cm and a height of 7 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

A right triangular prism has a triangle base with area 6 square inches and a length of 10 inches.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

A right prism has a base area of 15 square m and a height of 4 m.

What is its volume?

A. 19 cubic meters

B. 30 cubic meters

C. 225 cubic meters

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

True or false?

The volume of any right prism is the area of its base times its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

True or false?

A prism with a base area of 9 and a height of 2 has a volume of 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 8.G.C.9

FORWARD 1

Give an informal argument for volume formulas • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of a cylinder is the area of its base times its height.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

A straight stack of coins and a slanted stack have the same coins and the same height.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

An oblique cylinder and a right cylinder have the same radius and height.

How do their volumes compare?

A. The oblique one is larger.

B. It cannot be known.

C. The right one is larger.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

True or false?

Cavalieri's principle compares the areas of cross sections at every height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

True or false?

Leaning a stack of paper to one side increases its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 8.G.C.9

FORWARD 2

Use density with volume in a model • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

An aluminum cylinder has a radius of 2 cm and a height of 10 cm. Aluminum has a density of 2.7 grams per cubic centimeter.

What is its mass? Use 3.14 for π.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

A town has 12,000 people and covers 30 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

A 500 cubic-centimeter block has a mass of 4,000 grams.

What is its density?

A. 8 grams per cubic centimeter

B. 2 grams per cubic centimeter

C. 0.125 gram per cubic centimeter

D. 4,500 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

True or false?

If two objects have the same volume, the one with the greater density has the greater mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

True or false?

Density is volume divided by mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 8.G.C.9 • GEOMETRY

Find the volume of a cone

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle from its radius or diameter	**7.G.B.4**
BACKWARD 2	Multiply a fraction by a whole number	**5.NF.B.4.a**
FORWARD 1	Explain why a cone's volume is one-third of a cylinder's	**HSG-GMD.A.1**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

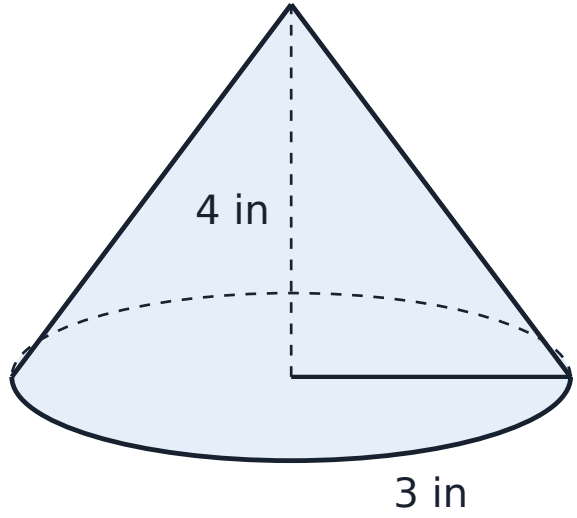
SET 50 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cone in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

A cone has a diameter of 12 feet and a height of 10 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

A cone and a cylinder have the same base and the same height. The cylinder holds 90 cubic cm.

How much does the cone hold?

A. 30 cubic cm

B. 90 cubic cm

C. 270 cubic cm

D. 45 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

A cone has a volume of 50π cubic units and a radius of 5 units.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

True or false?

A cone with a radius of 6 and a height of 2 has a volume of 24π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

SET 50 • 8.G.C.9

BACKWARD 1

Find the area of a circle from its radius or diameter • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 6.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

A circle has a diameter of 4 m.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

A circle has a radius of 10 cm.

What is its area? Use 3.14 for π.

A. 314 square cm

B. 31.4 square cm

C. 62.8 square cm

D. 628 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

True or false?

A circle with a radius of 5 has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 8.G.C.9

BACKWARD 2

Multiply a fraction by a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{1}{3} \times 36$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

Multiply.

$\frac{1}{3} \times 90$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

Multiply.

$\frac{1}{3} \times 48$

A. 144

B. 12

C. 16

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{3} \times 27 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 15 = 45$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 8.G.C.9

FORWARD 1

Explain why a cone's volume is one-third of a cylinder's • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cone and a cylinder have the same base and height. Water from 3 full cones exactly fills the cylinder.

What formula does this suggest for the cone's volume?

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

Explain why an oblique cone and a right cone with the same base and height have the same volume.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

A pyramid and a prism have the same base and height.

How do their volumes compare?

A. The pyramid's is $\frac{1}{3}$ of the prism's.

B. The prism's is $\frac{1}{3}$ of the pyramid's.

C. The pyramid's is $\frac{1}{2}$ of the prism's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

True or false?

A cone's volume formula has the same form as a pyramid's: $\frac{1}{3}$ × base area × height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

True or false?

A cone has half the volume of a cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A paper cone cup has a radius of 4 cm and a height of 9 cm.

How many full cups fill a cylinder-shaped jug with a radius of 8 cm and a height of 18 cm?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

A cone-shaped pile of sand has a radius of 3 m and holds 30π cubic meters.

How tall is the pile?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

A cone has a radius of 6 and a height of 8.

What is its slant height?

A. 14

B. 10

C. 48

D. $\sqrt{28}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

True or false?

Doubling the height of a cone doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

True or false?

Doubling the radius of a cone doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 8.G.C.9 • GEOMETRY

Find the volume of a sphere

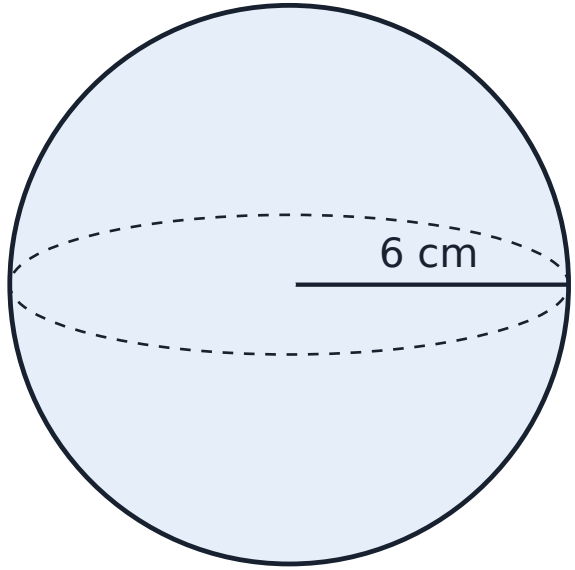
MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a cube	**6.G.A.2**
BACKWARD 2	Evaluate a formula with a fraction and an exponent	**6.EE.A.2.c**
FORWARD 1	Model objects with spheres, cylinders, and cones	**HSG-MG.A.1**
FORWARD 2	Use volume formulas to solve problems with spheres	**HSG-GMD.A.3**

SET 51 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the sphere in terms of π.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

A ball has a diameter of 10 cm.

Find its volume to the nearest tenth. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

A hemisphere (half of a sphere) has a radius of 3.

What is its volume in terms of π?

A. 36π

B. 18π

C. 9π

D. 27π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

A sphere has a volume of 36π cubic units.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

True or false?

A sphere with a radius of 1 has a volume of $\frac{4}{3}\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

SET 51 • 8.G.C.9

BACKWARD 1

Find the volume of a cube • Grade 6 • Geometry • 6.G.A.2

A cube has edges of 3 units.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

A cube has edges of $\frac{1}{2}$ foot.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

A cube has edges of 6 cm.

What is its volume?

A. 18 cubic cm

B. 216 cubic cm

C. 72 cubic cm

D. 36 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

True or false?

A cube with 5-meter edges has a volume of 15 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 8.G.C.9

BACKWARD 2

Evaluate a formula with a fraction and an exponent • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $V = \frac{4}{3}r^3$. Find V when r = 3.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

Use $V = \frac{4}{3}r^3$. Find V when r = 6.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

Use $V = \frac{1}{2}r^3$. Find V when r = 2.

A. 6

B. 4

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

True or false?

When $r = 1$, $\frac{4}{3}r^3 = \frac{4}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

True or false?

When r = 2, $\frac{4}{3}r^3 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 8.G.C.9

FORWARD 1

Model objects with spheres, cylinders, and cones • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

An orange is modeled as a sphere with a diameter of 8 cm.

Estimate its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

A tennis ball can holds 3 balls in a cylinder whose radius equals the ball's radius r and whose height is 6r.

What fraction of the can is filled by the balls?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

Which shape is the best model for a scoop of ice cream?

A. A cylinder

B. A sphere

C. A cone

D. A cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

True or false?

Modeling Earth as a sphere is a useful approximation even though Earth is slightly flattened.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

True or false?

A soccer ball is best modeled by a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems with spheres • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A spherical tank has a radius of 3 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r.

What is the ratio of the sphere's volume to the cylinder's volume?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

Doubling a sphere's radius multiplies its volume by...

A. 6

B. 2

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

True or false?

A sphere with volume 288π has a radius of 6.

A. True

B. False

True or false?

A hemisphere with radius 6 has a volume of 288π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • 8.G.C.9 • GEOMETRY

Solve real-world problems with the volumes of cylinders, cones, and spheres

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a solid made of two prisms	**5.MD.C.5.c**
BACKWARD 2	Find the volume of a right prism	**7.G.B.6**
FORWARD 1	Model real objects with combinations of solids	**HSG-MG.A.1**
FORWARD 2	Apply volume to design and capacity problems	**HSG-MG.A.3**

SET 52 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

A grain silo is a cylinder with a radius of 6 m and a height of 10 m, topped with half of a sphere with a radius of 6 m.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

An ice cream cone (radius 3 cm, height 10 cm) is filled and topped with half of a sphere of ice cream with a radius of 3 cm.

Find the total volume of ice cream in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

A cone and a cylinder each have a radius of 2 and a height of 6.

How many full cones of water fill the cylinder?

A. 6

B. 3

C. 1

D. 2

A cylinder with a radius of 5 cm holds water. A ball with a radius of 3 cm is dropped in and sinks completely.

How many centimeters does the water level rise?

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

True or false?

A sphere with radius r fits exactly in a cylinder with radius r and height 2r. The sphere's volume is $\frac{2}{3}$ of the cylinder's.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

SET 52 • 8.G.C.9

BACKWARD 1

Find the volume of a solid made of two prisms • Grade 5 • Measurement & Data • 5.MD.C.5.c

A 4 by 3 by 2 prism and a 2 by 3 by 2 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

Two cubes with 2-unit edges are joined face to face.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

A 5 by 2 by 1 prism and a 1 by 2 by 1 prism are joined with no overlap.

What is the total volume?

A. 12 cubic units

B. 7 cubic units

C. 20 cubic units

D. 10 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

Volumes of parts that do not overlap can be added to find a total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

A 3 by 3 by 3 cube and a 1 by 1 by 1 cube joined with no overlap have a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism • Grade 7 • Geometry • 7.G.B.6

A right rectangular prism is 6 cm by 4 cm by 5 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q1/5

A right prism has a base area of 12 square feet and a height of 3 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q2/5

A right triangular prism has a base area of 10 square inches and a length of 8 inches.

What is its volume?

A. 40 cubic inches

B. 80 cubic inches

C. 18 cubic inches

D. 800 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A cube with edges of 10 m has a volume of 1,000 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A 2 by 5 by 10 box has a volume of 17 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 8.G.C.9

FORWARD 1

Model real objects with combinations of solids • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

A capsule-shaped vitamin is a cylinder 1 cm long with a radius of 0.3 cm, with half of a sphere of radius 0.3 cm on each end.

What solids make up the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q1/5

A pencil is modeled as a cylinder with a cone at one end.

Which measurements are needed to find its volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q2/5

Which solids best model a traffic cone sitting on a square base?

A. A cone on a square prism

B. A sphere on a cylinder

C. Two cylinders

D. A cube on a pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q3/5

True or false?

A model can use simpler shapes even when the real object is not exactly that shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q4/5

True or false?

A water bottle is best modeled as one sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q5/5

SET 52 • 8.G.C.9

FORWARD 2

Apply volume to design and capacity problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a cylinder-shaped can that holds 500π cubic cm with a radius of 5 cm.

How tall must the can be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q1/5

A cone-shaped funnel has a radius of 6 cm and a height of 10 cm. Water drains at 6π cubic cm per second.

How long does a full funnel take to drain?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q2/5

Each can is a cylinder holding the same volume.

Which design uses the most metal for the curved side?

A. Radius 4 cm, height 6.25 cm

B. Radius 2.5 cm, height 16 cm

C. Radius 5 cm, height 4 cm

D. Radius 2 cm, height 25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q3/5

True or false?

Two cans with the same volume can have different surface areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q4/5

True or false?

A taller can always holds more than a shorter can.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 8.SP.A.1 • STATISTICS & PROBABILITY

Construct a scatter plot for bivariate measurement data

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Graph points that represent a real-world situation	**5.G.A.2**
BACKWARD 2	Display a numerical data set in a dot plot	**6.SP.B.4**
FORWARD 1	Represent two-variable data on a scatter plot and describe how t…	**HSS-ID.B.6**
FORWARD 2	Fit an exponential or quadratic function to data and use it to make	**HSS-ID.B.6.a**

SET 53 • 8.SP.A.1

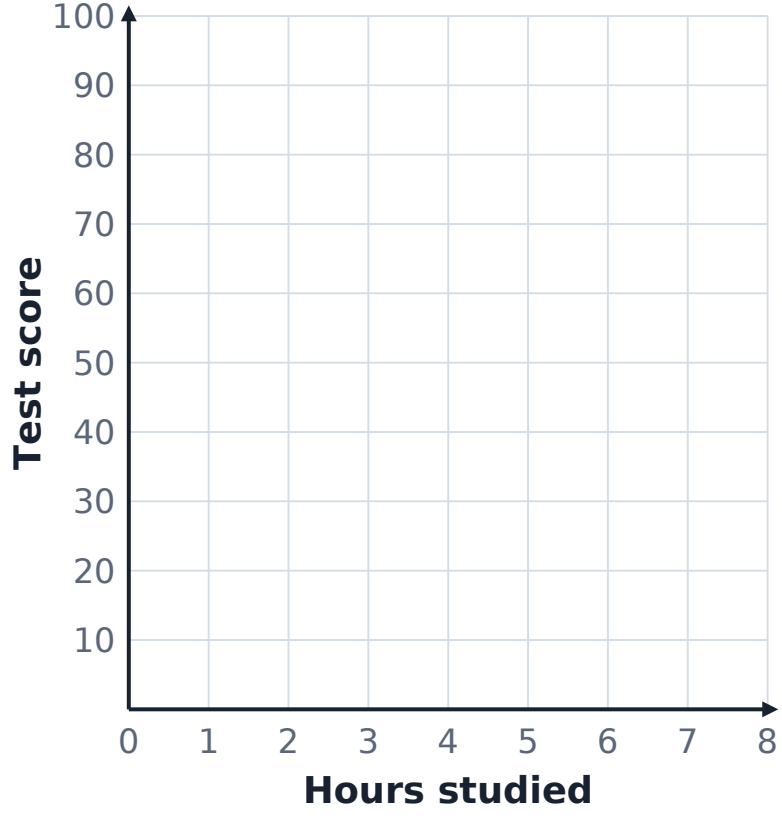
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Seven students recorded (hours studied, test score):

(1, 50), (2, 60), (3, 60), (4, 70), (5, 80), (6, 80), (7, 90)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

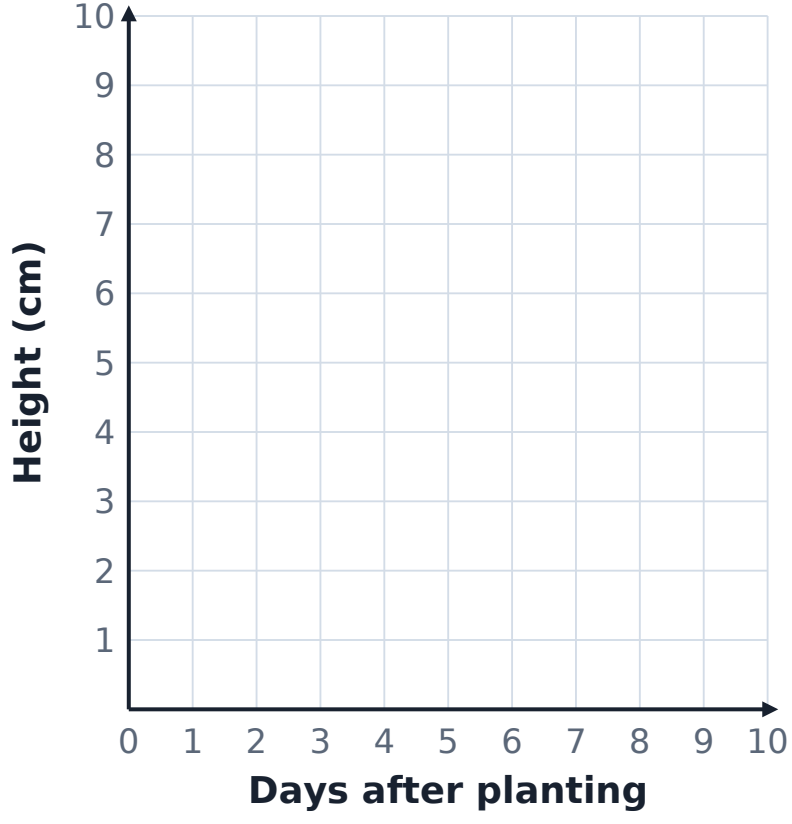
ID S53-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q1/5

A bean plant was measured on different days (days after planting, height in cm):

(2, 1), (4, 3), (5, 4), (6, 4), (8, 6), (9, 7), (10, 8)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

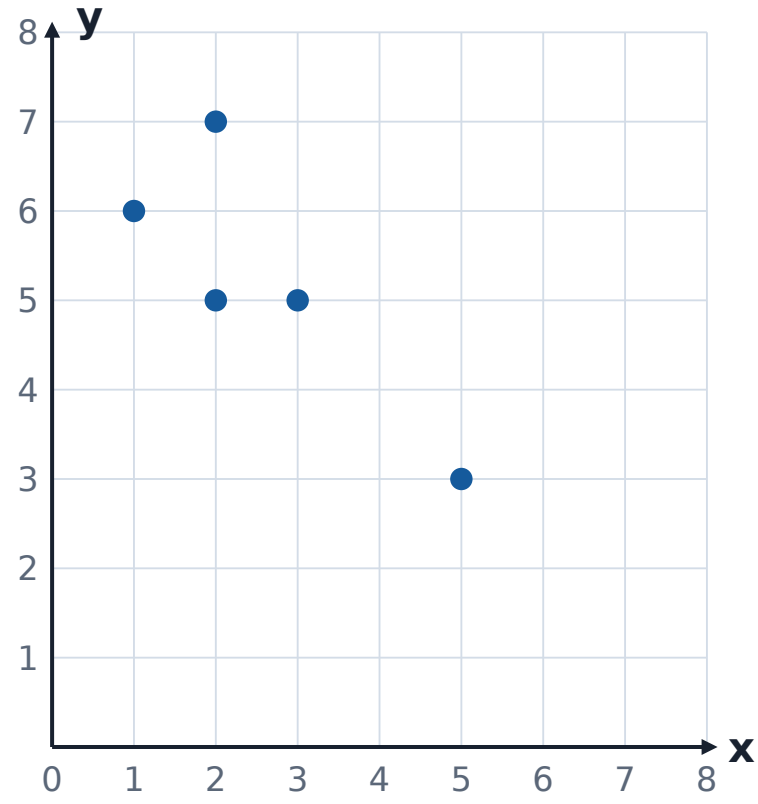
ID S53-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q2/5

The scatter plot should show the data (1, 6), (2, 5), (3, 5), (5, 3), and (7, 2), but one point is plotted wrong.

Which point is wrong, and where should it be?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q3/5

Ten students recorded their shoe sizes and heights to make a scatter plot.

What does each point on the scatter plot show?

A. How many students have each shoe size

B. One student's shoe size and height

C. The mean shoe size of the class

D. One student's shoe size only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q4/5

True or false?

A scatter plot of hours slept and test scores puts two points on the grid for each student, one for each variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q5/5

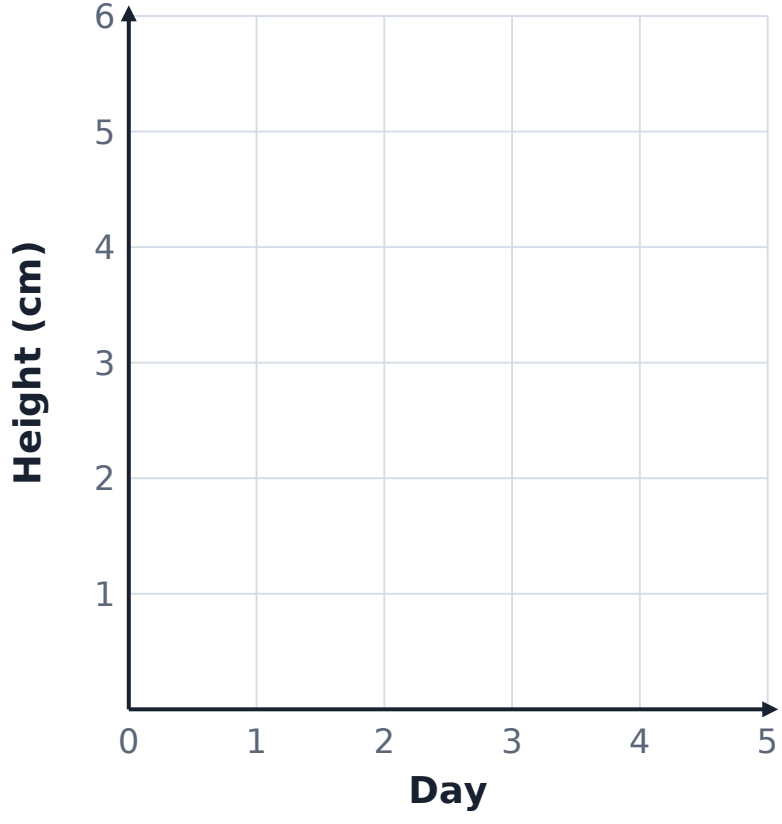
SET 53 • 8.SP.A.1

BACKWARD 1

Graph points that represent a real-world situation • Grade 5 • Geometry • 5.G.A.2

A plant was 2 cm tall on day 1, 4 cm tall on day 2, and 5 cm tall on day 3.

Plot the points (day, height) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q1/5

A graph shows points (day, height in cm).

What does the point (4, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q2/5

Points on a graph are (hours, miles).

Which point shows 5 hours and 2 miles?

A. (5, 5)

B. (2, 5)

C. (5, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q3/5

True or false?

To plot (3, 6), start at the origin, move 3 units right, then 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The point (0, 4) is on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 8.SP.A.1

BACKWARD 2

Display a numerical data set in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The numbers of pets of 10 students are 0, 1, 1, 2, 2, 2, 3, 3, 4, and 1.

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

How many students are shown in the dot plot?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

Which display shows each value of a data set as a dot above a number line?

A. A box plot

B. A bar graph

C. A histogram

D. A dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

In a dot plot, a value that appears 3 times has 3 dots stacked above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A dot plot shows two different measurements for each student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 1

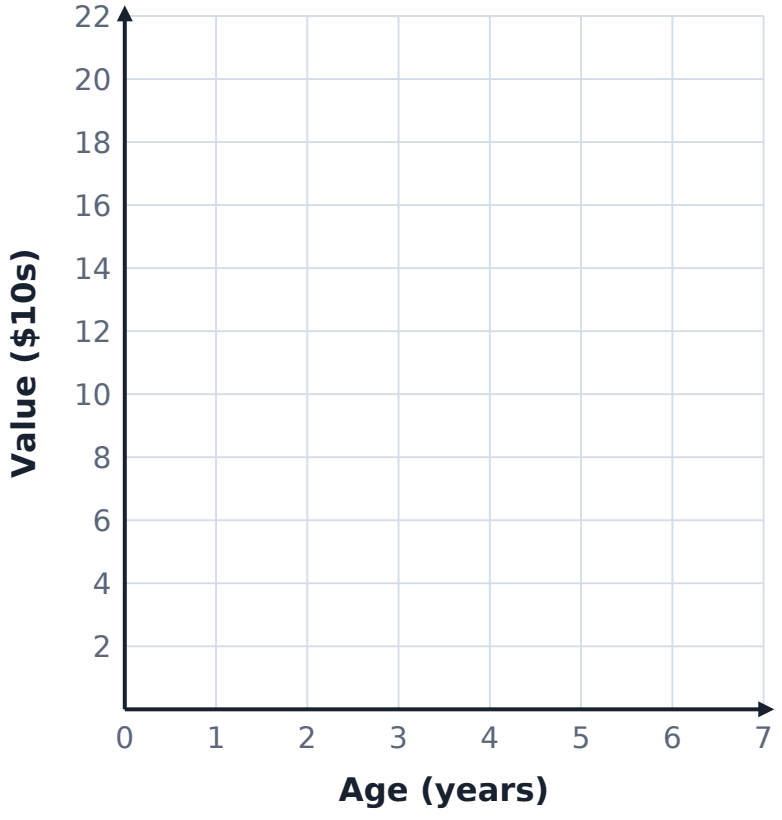
Represent two-variable data on a scatter plot and describe how the variables are related • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

Six used bikes have these (age in years, value in tens of dollars):

(1, 20), (2, 17), (3, 15), (4, 12), (5, 10), (6, 8)

Make a scatter plot of the data. Then describe how age and value are related.

Relationship: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q1/5

A scatter plot of hours of sleep and reaction time shows points falling from left to right.

Describe how the variables are related.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q2/5

Which pair of variables would most likely show a positive association on a scatter plot?

A. Height and arm span of adults

B. Outside temperature and heating bill

C. Shoe size and test score

D. Age of a car and its value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q3/5

True or false?

A scatter plot whose points rise from left to right shows a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q4/5

True or false?

If two variables have no association, their scatter plot points lie on a horizontal line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 2

Fit an exponential or quadratic function to data and use it to make predictions • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A bacteria count is 100 at hour 0, 200 at hour 1, 400 at hour 2, and 800 at hour 3.

Write a function that fits the data.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q1/5

A bacteria count fits y = 100 · 2^x after x hours.

Predict the count at hour 6.

Count: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q2/5

Which function fits the data exactly?

x	0	1	2	3	4
y	0	1	4	9	16

A. $y = x^2$

B. $y = 4x$

C. $y = x + 4$

D. $y = 2^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q3/5

True or false?

The data (0, 5), (1, 10), (2, 20), and (3, 40) fit $y = 5 \cdot 2^x$ exactly.

A. True

B. False

True or false?

Data whose y-values go up by the same amount at each step are best fit by an exponential function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q5/5

SET 54 • 8.SP.A.1 • STATISTICS & PROBABILITY

Describe clustering, outliers, and positive, negative, linear, or nonlinear association in a scatter plot

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Describe a data distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Describe the overall pattern of data and any striking deviations	**6.SP.B.5.c**
FORWARD 1	Distinguish between correlation and causation	**HSS-ID.C.9**
FORWARD 2	Interpret the correlation coefficient of a linear fit	**HSS-ID.C.8**

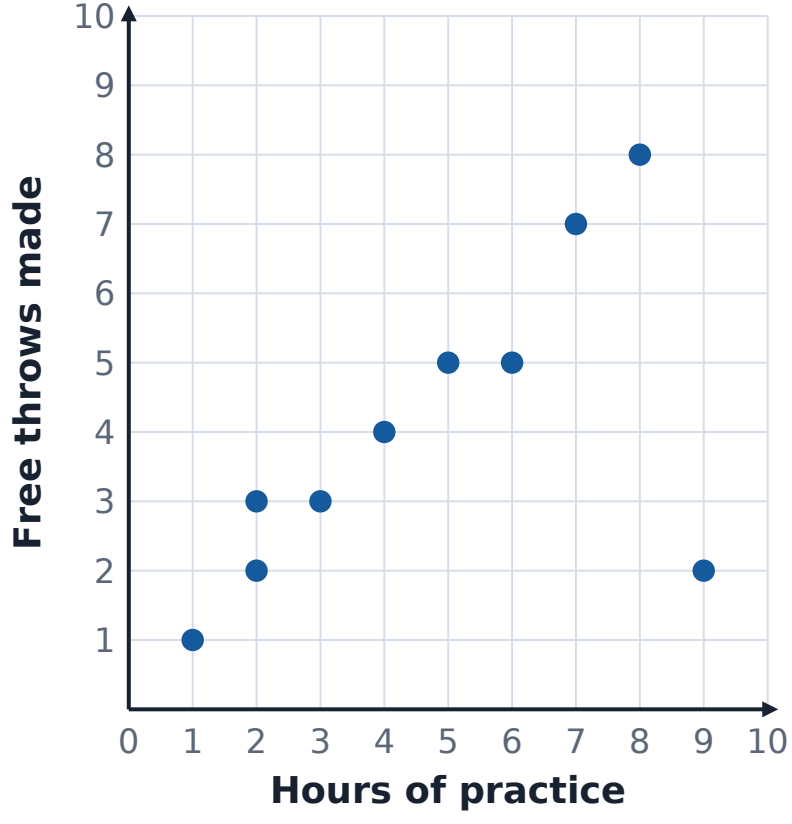
SET 54 • 8.SP.A.1

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Describe the association in the scatter plot (positive, negative, or none; linear or nonlinear), and name any outlier.

Answer: _____



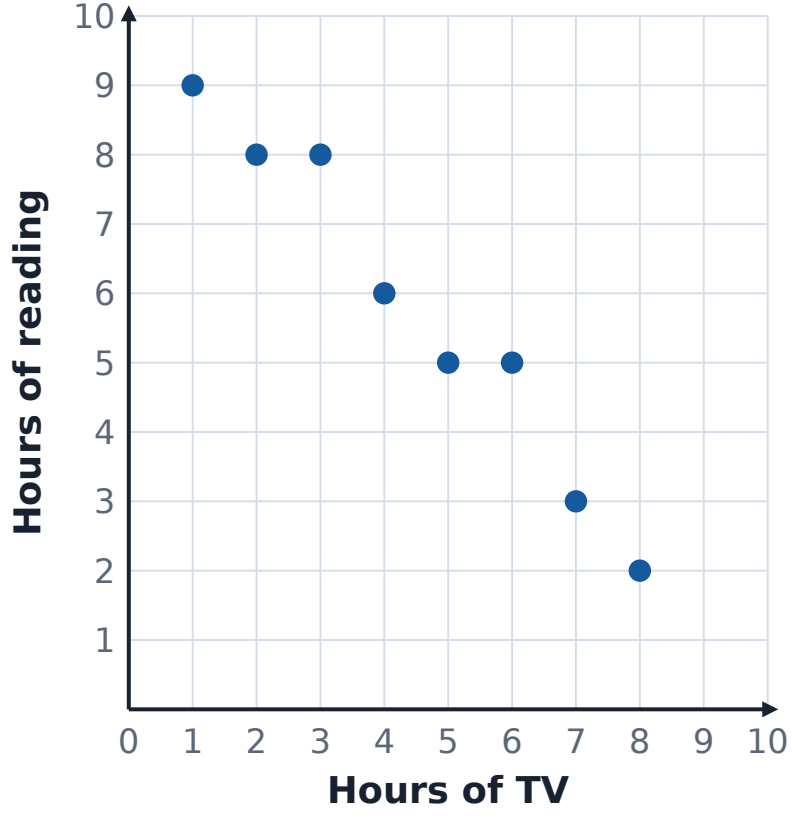
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q1/5

Describe the association in the scatter plot.

Answer: _____

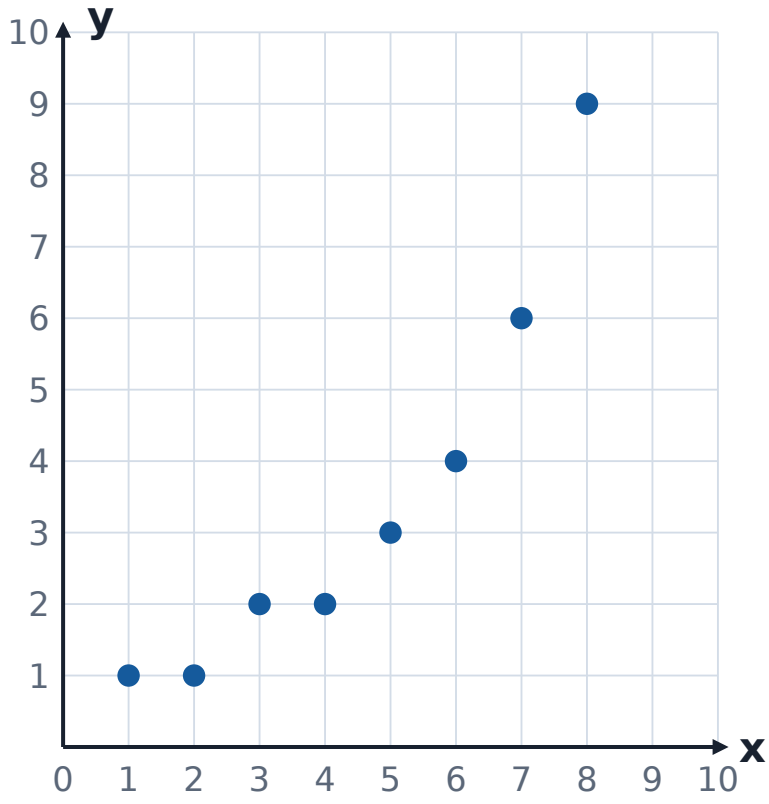


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q2/5

Which best describes the association in the scatter plot?



A. Positive and linear

B. No association

C. Negative and nonlinear

D. Positive and nonlinear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

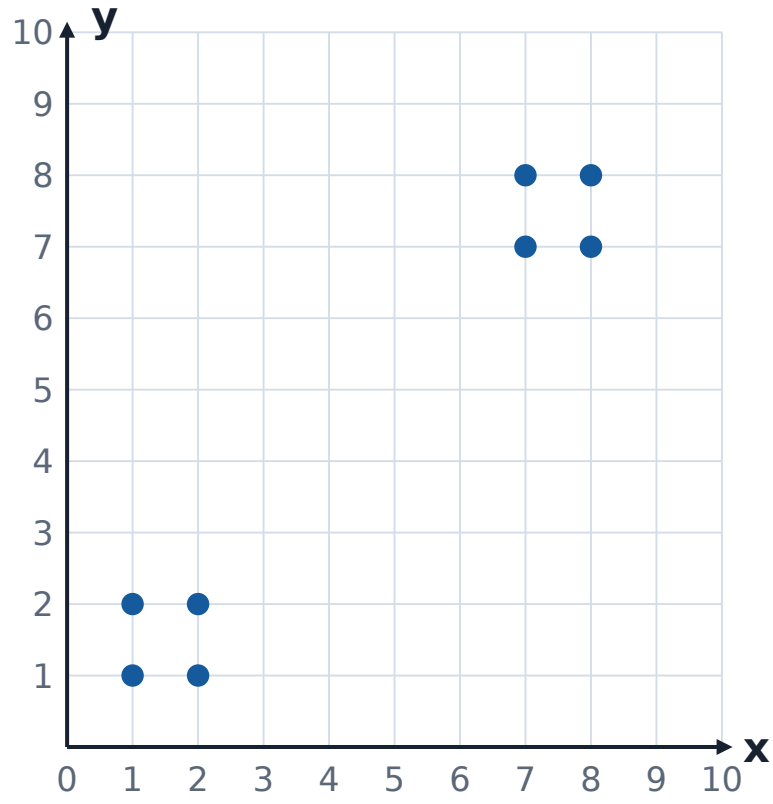
ID S54-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q3/5

The points in the scatter plot form two groups.

Describe the two clusters.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

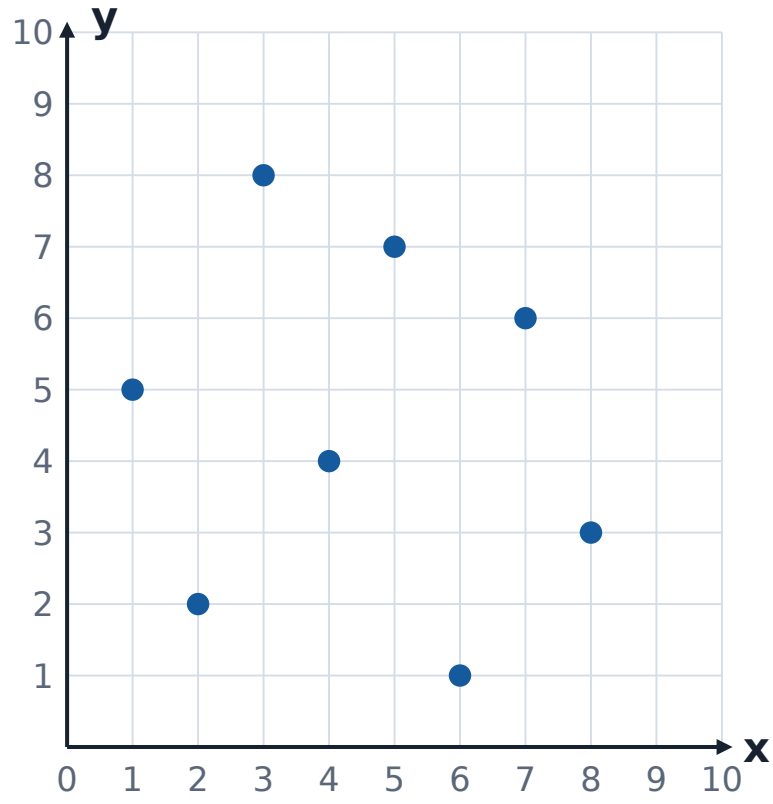
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q4/5

True or false?

The scatter plot shows a negative association.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

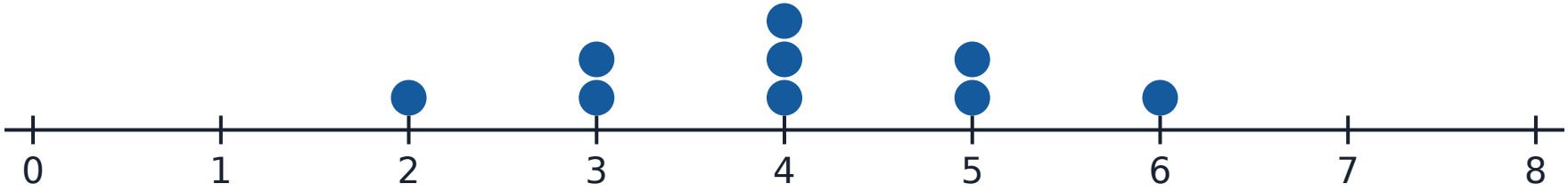
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 1

Describe a data distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

Is the data in the dot plot symmetric or skewed?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q1/5

What is the range of the data in the dot plot?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which value is far from the rest of the data?

3, 4, 4, 5, 5, 6, 18

A. 18

B. 6

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

The data set 2, 3, 3, 4, 4, 4, 5, 5, 6 is symmetric about 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

The range of a data set measures its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 2

Describe the overall pattern of data and any striking deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

4, 6, 7, 7, 8, 9, 30

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q1/5

Which value is a striking deviation from the pattern?

4, 6, 7, 7, 8, 9, 30

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q2/5

The data are 10, 12, 12, 13, 14, and 40.

Which measure of center is changed more by the value 40?

A. The mean

B. The median

C. Both are changed equally

D. Neither is changed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q3/5

True or false?

Removing a value that is much larger than the others lowers the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q4/5

True or false?

An outlier always changes the median by a large amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 1

Distinguish between correlation and causation • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more ice-cream sales also have more sunburns.

Does buying ice cream cause sunburns? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q1/5

A study finds that students who sleep more tend to have higher grades.

Can you conclude that more sleep causes higher grades? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q2/5

Which kind of study can show that one variable causes a change in another?

A. An observational study

B. A randomized experiment

C. A survey of volunteers

D. A scatter plot of survey data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q3/5

True or false?

Two variables can be strongly associated without one causing the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q4/5

True or false?

A strong correlation shows that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 2

Interpret the correlation coefficient of a linear fit • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

The correlation coefficient of a data set is $r = -0.92$.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q1/5

Which shows the stronger linear association, $r = 0.45$ or $r = -0.81$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q2/5

Points lie almost exactly on a line that rises from left to right.

Which value of r fits?

A. -0.98

B. 0.98

C. 0

D. 0.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q3/5

True or false?

The correlation coefficient r is always between −1 and 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q4/5

True or false?

$r = -0.9$ shows a weaker linear association than $r = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 8.SP.A.2 • STATISTICS & PROBABILITY

Informally fit a straight line to a scatter plot and judge how closely it fits

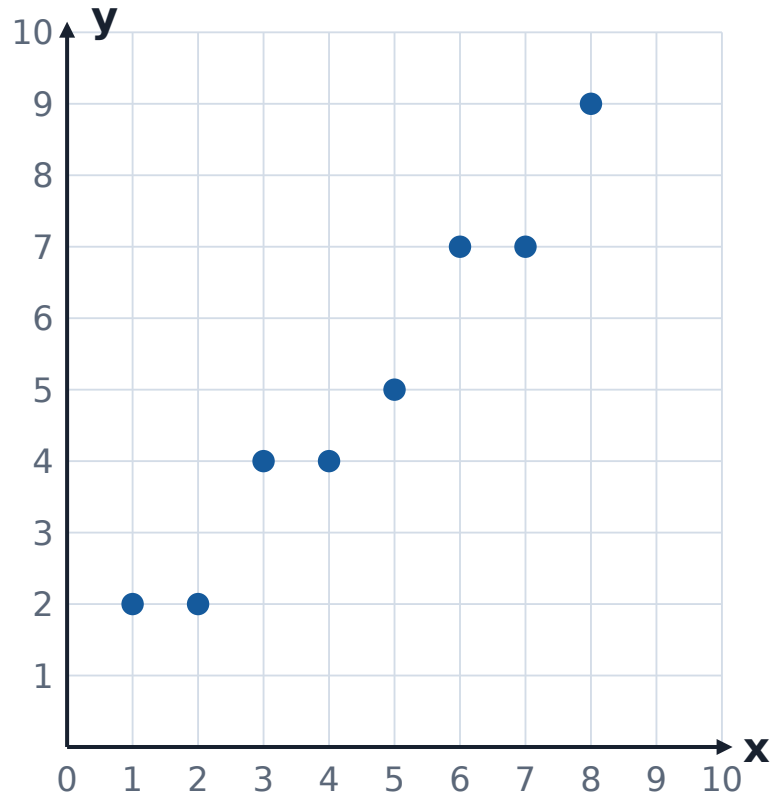
MAIN	Grade 8 main question	**8.SP.A.2**
BACKWARD 1	Decide from a table or graph whether a relationship is proportion…	**7.RP.A.2.a**
BACKWARD 2	Write an equation for a relationship shown in a table or graph	**6.EE.C.9**
FORWARD 1	Fit a linear function to data that suggest a linear association	**HSS-ID.B.6.c**
FORWARD 2	Judge the fit of a line by finding and plotting residuals	**HSS-ID.B.6.b**

SET 55 • 8.SP.A.2

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.2

Draw a straight line that fits the data. Then write an equation of your line.



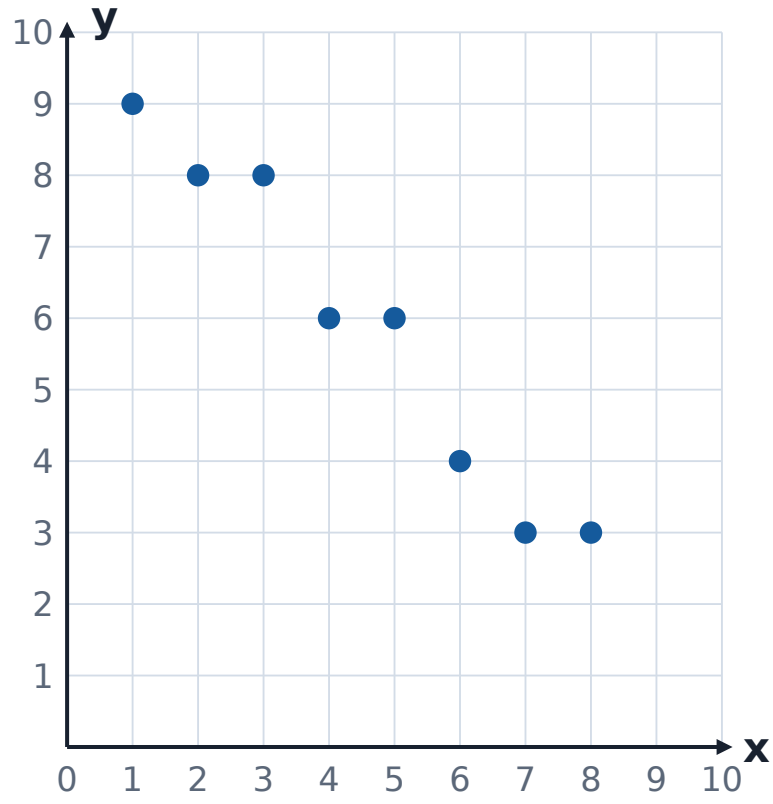
Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q1/5

Draw a straight line that fits the data. Use your line to estimate y when x = 9.



Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q2/5

Two students drew lines for the same scatter plot. Line A has every point above it. Line B has about half the points above it and half below, all close to it.

Which line fits better? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q3/5

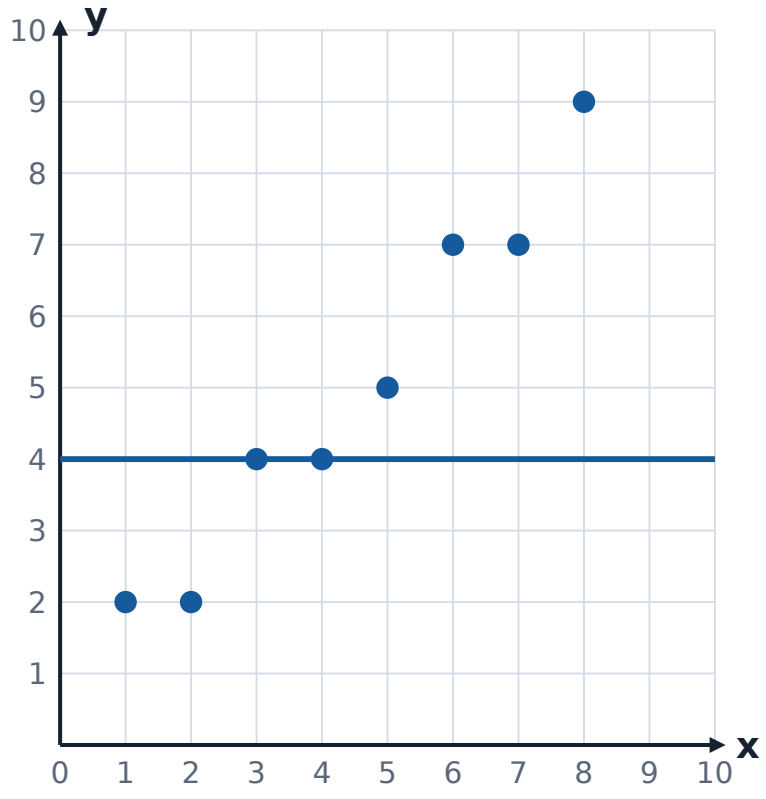
Why is the line on the scatter plot a poor fit?

A. It is not long enough.

B. It does not follow the upward trend of the points.

C. It starts at the y-axis.

D. It passes through too many points.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q4/5

True or false?

A line that fits a scatter plot well must pass through every point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q5/5

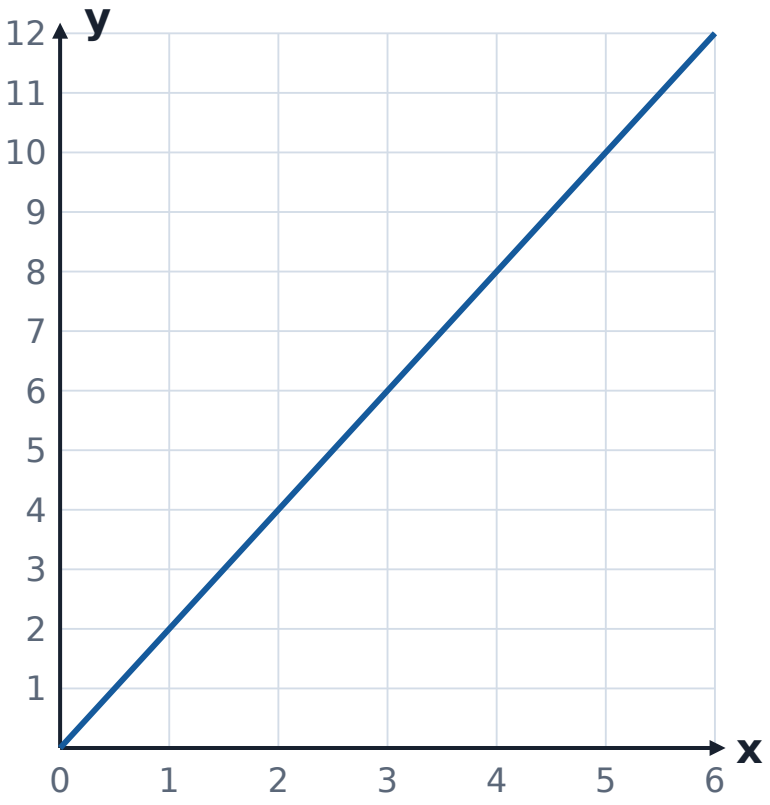
SET 55 • 8.SP.A.2

BACKWARD 1

Decide from a table or graph whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Is the relationship shown in the graph proportional? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	4
y	3	6	13

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q2/5

Which graph shows a proportional relationship?

A. A curve through (0, 0) and (2, 4)

B. A straight line through (0, 2) and (2, 6)

C. A horizontal line through (0, 3)

D. A straight line through (0, 0) and (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q3/5

True or false?

A straight line through (0, 0) and (3, 9) shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q4/5

True or false?

Every straight-line graph shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 8.SP.A.2

BACKWARD 2

Write an equation for a relationship shown in a table or graph • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	0	1	2	3
y	2	3	4	5

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q2/5

A line passes through (0, 0), (1, 5), and (2, 10).

Which equation does it show?

A. y = 5x

B. y = 10x

C. x = 5y

D. y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q3/5

True or false?

The points (1, 3), (2, 6), and (3, 9) fit y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q4/5

True or false?

The points (0, 1), (1, 2), and (2, 3) fit y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 8.SP.A.2

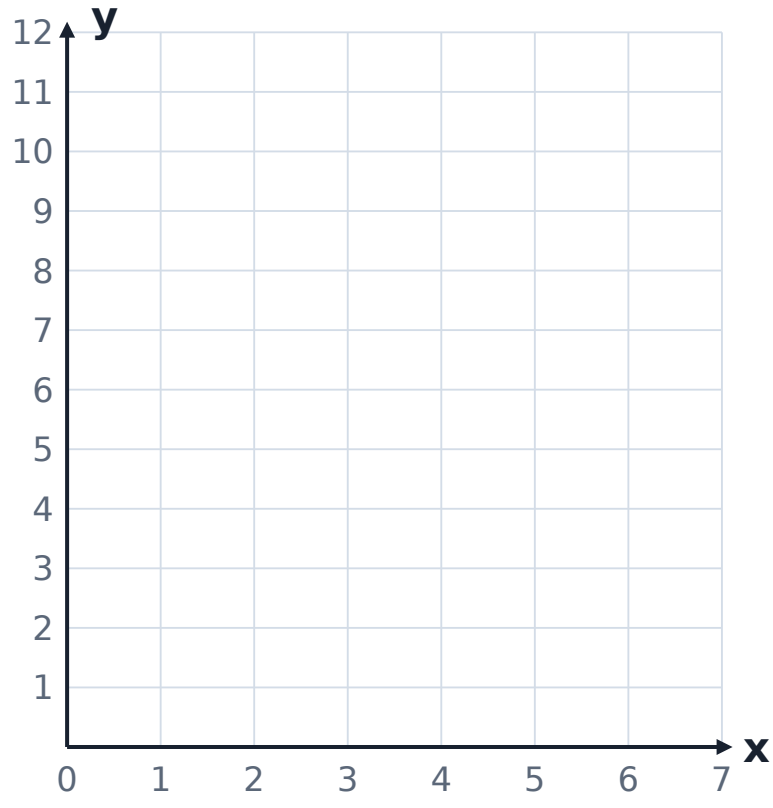
FORWARD 1

Fit a linear function to data that suggest a linear association • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

The data are (1, 3), (2, 5), (3, 6), (4, 8), (5, 10), and (6, 11).

Make a scatter plot, draw a line that fits the data, and write an equation of your line.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q1/5

A line of fit passes through (2, 5) and (6, 13).

Write its equation.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q2/5

A line of fit for (hours, dollars) data is $y = 12x + 8$.

What does it predict for 5 hours?

A. $100

B. $68

C. $25

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q3/5

True or false?

A line of fit through (0, 4) and (4, 12) has a slope of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q4/5

True or false?

A line of fit should always be drawn through the first and last data points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 2

Judge the fit of a line by finding and plotting residuals • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

A line of fit is $y = 2x + 1$. A data point is $(3, 8)$.

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q1/5

A line of fit is $y = 2x + 1$. A data point is $(5, 9)$.

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q2/5

A residual plot shows points scattered above and below 0 with no pattern.

What does this suggest?

A. A curve must be used.

B. Every prediction is exact.

C. The line is a poor fit.

D. The line is a good fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q3/5

True or false?

A residual of 0 means the data point lies on the line of fit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q4/5

True or false?

A residual plot with a clear U shape shows that a linear model fits well.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q5/5

SET 56 • 8.SP.A.3 • STATISTICS & PROBABILITY

Use the equation of a linear model to interpret the slope and intercept and to make predictions

MAIN	Grade 8 main question	**8.SP.A.3**
BACKWARD 1	Explain what a point on the graph of a proportional relationship m…	**7.RP.A.2.d**
BACKWARD 2	Evaluate a formula for given values of its variable	**6.EE.A.2.c**
FORWARD 1	Interpret the slope and intercept of a linear model in context	**HSS-ID.C.7**
FORWARD 2	Evaluate reports and predictions based on data	**HSS-IC.B.6**

SET 56 • 8.SP.A.3

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.3

A line of fit for the height h (inches) of a plant after w weeks is h = 1.5w + 4.

What do the slope and the intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q1/5

A line of fit for the gas g (gallons) left in a car's tank after m miles is $g = 14 - 0.04m$.

Predict the gas left after 200 miles.

Gas left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q2/5

A model for the water level w (feet) of a pond t days after the rain stops is $w = 12 - 0.3t$.

What does -0.3 mean?

A. The level rises about 0.3 foot each day.

B. The pond is empty after 0.3 day.

C. The level starts at 0.3 foot.

D. The level drops about 0.3 foot each day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q3/5

A model for the score s after h hours of practice is $s = 6h + 58$.

Predict the score after 5 hours of practice.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q4/5

True or false?

In the model $y = 3.2x + 15$ for the cost of a taxi ride of x miles, 15 is the cost per mile.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 1

Explain what a point on the graph of a proportional relationship means • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph of cost (dollars) against pounds is proportional and passes through (1, 3).

What does (1, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q1/5

A proportional graph of miles against hours passes through (0, 0).

What does (0, 0) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q2/5

A proportional graph of (gallons, cost) passes through (1, 4).

What is the unit rate?

A. $5 per gallon

B. $4 per gallon

C. $0.25 per gallon

D. $1 per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q3/5

True or false?

On a proportional graph of (hours, pages), the point (1, 20) means 20 pages are read per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q4/5

True or false?

On a proportional graph of (tickets, cost), the point (2, 18) shows the unit rate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 2

Evaluate a formula for given values of its variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 1.5w + 4 when w = 6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q1/5

Evaluate $300 - 40t$ when $t = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q2/5

Evaluate 6h + 58 when h = 2.

A. 62

B. 128

C. 66

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q3/5

True or false?

When $x = 4$, $3x + 5 = 35$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q4/5

True or false?

When $t = 2$, $10 - 3t = 14$.

A. True

B. False

SET 56 • 8.SP.A.3

FORWARD 1

Interpret the slope and intercept of a linear model in context • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for the number n of ice pops sold on a day with a high temperature of t °F is $n = 4t - 200$.

What does the slope mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q1/5

A model for ice-pop sales is $n = 4t - 200$, where t is the temperature in °F.

Does the intercept -200 make sense in this situation? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q2/5

A model for a phone's battery level b (percent) after h hours of use is $b = 100 - 12.5h$.

What does 100 mean?

A. The battery lasts 100 hours.

B. The battery charges 100% each hour.

C. The battery loses 100% each hour.

D. The battery starts at 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q3/5

True or false?

In the model $y = 0.8x + 2$, y increases by about 0.8 for each increase of 1 in x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q4/5

True or false?

In a linear model, the intercept always has a sensible meaning in the situation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q5/5

SET 56 • 8.SP.A.3

FORWARD 2

Evaluate reports and predictions based on data • High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A model of children's heights for ages 2 to 10 is $h = 2.5a + 31$ inches at age a.

A report uses it to predict the height of a 40-year-old. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q1/5

A line of fit was made from data collected between 60 °F and 90 °F.

A report uses it to predict sales at 70 °F. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q2/5

A report says: "People who own more books live longer, so buying books makes you live longer."

What is the main flaw?

A. It counts too few books.

B. It reports a positive association.

C. It treats an association as a cause.

D. It uses a line of fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q3/5

True or false?

Predictions far outside the range of the data are less reliable than predictions inside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q4/5

True or false?

A survey of volunteers always represents the whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 8.SP.A.4 • STATISTICS & PROBABILITY

Construct a two-way table summarizing data on two categorical variables

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Report the number of observations in a data set	**6.SP.B.5.a**
BACKWARD 2	Solve "how many more" and "how many in all" problems with a sc…	**3.MD.B.3**
FORWARD 1	Find joint, marginal, and conditional relative frequencies in a two…	**HSS-ID.B.5**
FORWARD 2	Use a two-way table to find conditional probabilities and decide…	**HSS-CP.A.4**

SET 57 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

60 students were asked whether they have a curfew and whether they have chores. 36 have a curfew. 24 of the students with a curfew have chores. 8 of the students without a curfew have chores.

Complete the two-way table.

	Chores	No chores	Total
Curfew			
No curfew			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q1/5

50 students were asked whether they prefer summer or winter. 22 are in Grade 7 and 28 are in Grade 8. 15 of the Grade 7 students and 18 of the Grade 8 students prefer summer.

Complete the two-way table.

	Summer	Winter	Total
Grade 7			
Grade 8			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q2/5

90 people were asked whether they exercise daily and whether they eat breakfast. 40 exercise daily. 30 of the people who exercise daily eat breakfast. 50 of the 90 people eat breakfast.

Complete the two-way table.

	Breakfast	No breakfast	Total
Exercises			
Does not			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q3/5

Ten students answered two questions: "Do you have a pet?" and "Do you have a sibling?" Their answers (pet, sibling) are:

yes–yes, yes–no, no–no, yes–yes, no–yes, yes–yes, no–no, no–yes, yes–no, no–no

Complete the two-way table.

	Sibling	No sibling	Total
Pet			
No pet			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q4/5

True or false?

A completed two-way table for 60 students can have row totals of 36 and 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 1

Report the number of observations in a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.a

In a survey, 12 students chose soccer, 9 chose basketball, and 7 chose tennis.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q1/5

How many observations are shown in the dot plot?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q2/5

Each of 25 students named a favorite fruit. 5 different fruits were named.

How many observations are in the data set?

A. 5

B. 125

C. 25

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

If 14 students answer yes and 11 answer no, there are 25 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

The data set 3, 3, 5, 7 has 3 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

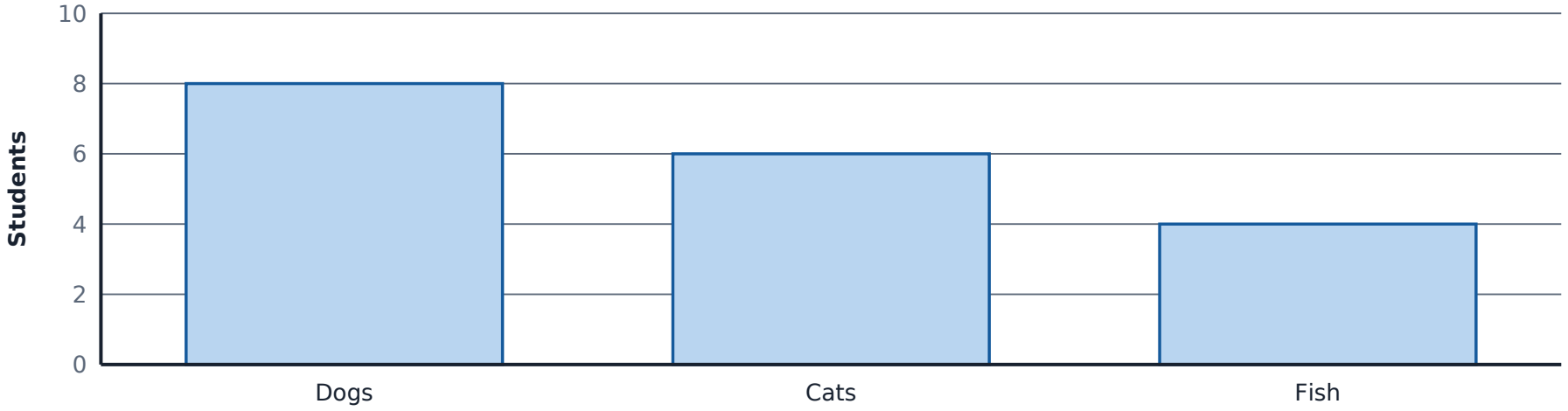
Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 2

Solve "how many more" and "how many in all" problems with a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many more students chose dogs than fish?



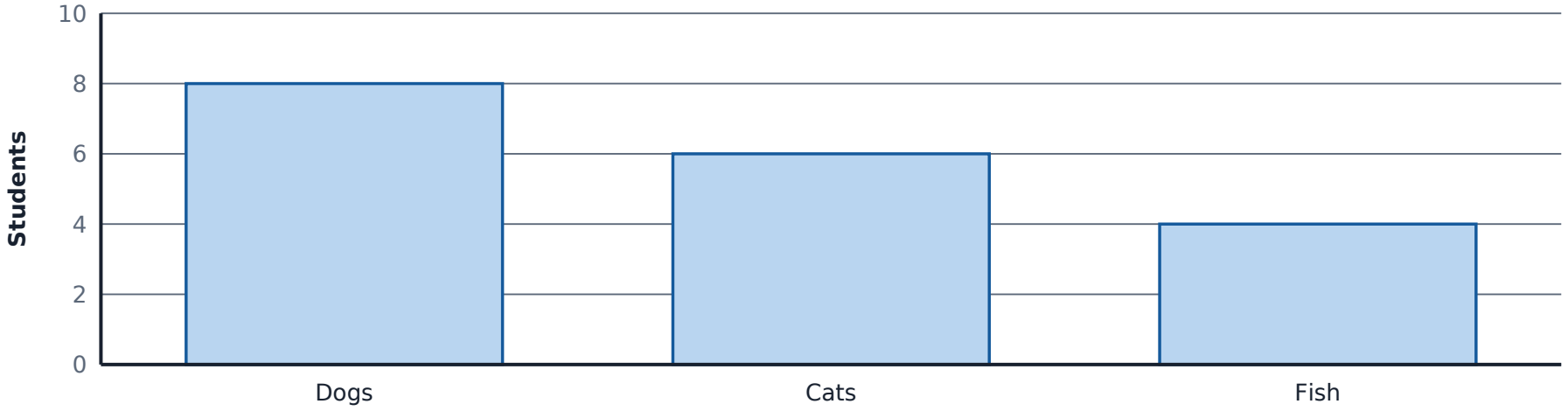
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q1/5

How many students chose a pet in all?



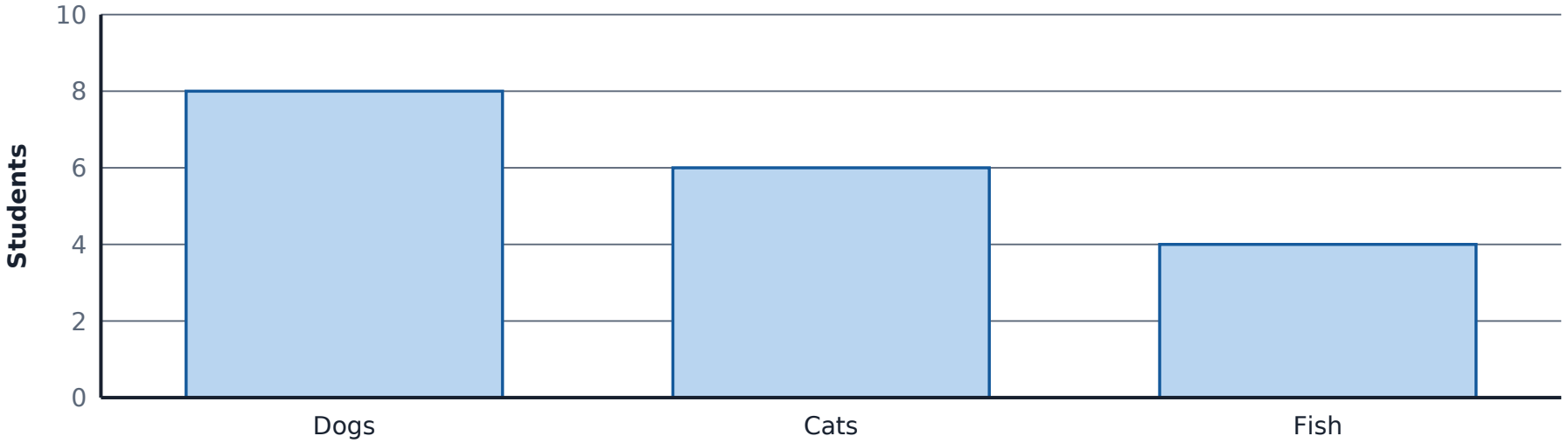
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q2/5

How many fewer students chose fish than cats?



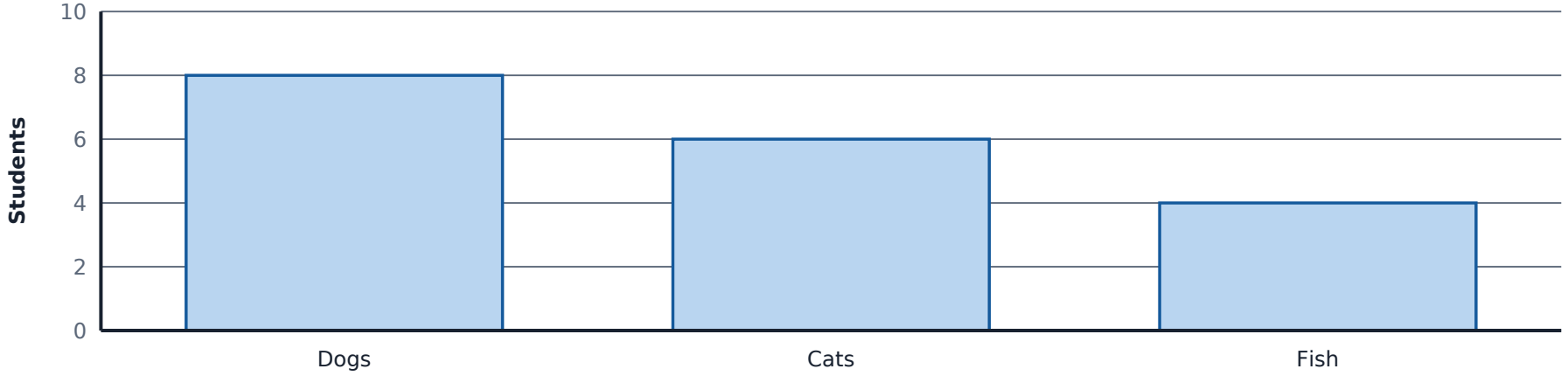
A. 4 **B.** 6 **C.** 10 **D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-B2-Q3**

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q3/5

True or false?

Fewer students chose cats than fish.



A. True

B. False

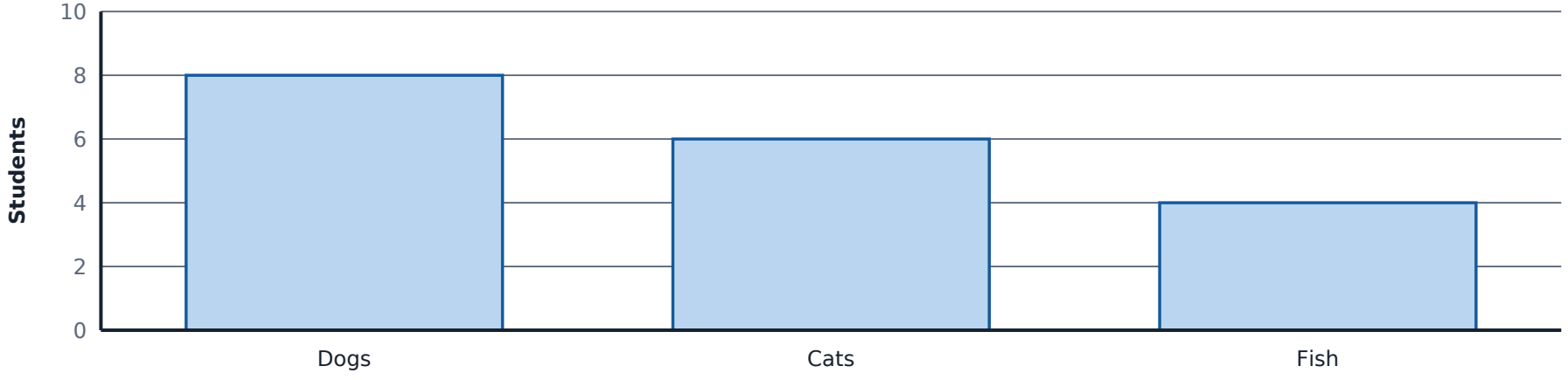
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The same number of students chose dogs and cats.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 1

Find joint, marginal, and conditional relative frequencies in a two-way table • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

What is the joint relative frequency of students who have a curfew and have chores?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q1/5

What is the marginal relative frequency of students who have chores? Round to the nearest hundredth.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q2/5

What is the conditional relative frequency of having chores among students with no curfew?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. $\frac{2}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{3}$ **D.** $\frac{2}{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q3/5

True or false?

The marginal relative frequency of students with a curfew is 0.36.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q4/5

True or false?

The joint relative frequencies in the four inner cells add to 2.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 2

Use a two-way table to find conditional probabilities and decide whether events are independent • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student from the table is chosen at random.

Find P(in club | Grade 8).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q1/5

Are being in the club and being in Grade 7 independent? Explain with conditional probabilities.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q2/5

A student from the table is chosen at random.

Find P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. 0.6 **B.** 0.12 **C.** 0.4 **D.** 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q3/5

True or false?

For a student chosen at random from the table, P(not in club) = 0.7.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q4/5

True or false?

For the table, P(in club | Grade 7) equals P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q5**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 8.SP.A.4 • STATISTICS & PROBABILITY

Use relative frequencies in a two-way table to describe a possible association

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Find a percent of a quantity and the percent one number is of ano…	**6.RP.A.3.c**
BACKWARD 2	Use ratio language to describe two quantities	**6.RP.A.1**
FORWARD 1	Recognize conditional probability and independence in everyday…	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as the fraction of B's outcomes that…	**HSS-CP.B.6**

SET 58 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

For each row, what percent of the people use a phone at night?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q1/5

Is there evidence of an association between sleep and phone use at night? Explain using the row relative frequencies.

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q2/5

Which conclusion do the row relative frequencies support?

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. There is a strong association because 45 is greater than 30.

B. Grade 7 students are more likely to eat breakfast.

C. Grade 8 students are more likely to eat breakfast.

D. There is no association: 60% of each grade eat breakfast.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q3/5

What percent of the people who use a phone at night sleep under 8 hours?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q4/5

True or false?

Grade 8 students are more likely to eat breakfast than Grade 7 students, because 45 is greater than 30.

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 1

Find a percent of a quantity and the percent one number is of another • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What percent of 40 is 12?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q1/5

Find 60% of 75.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q2/5

36 out of 48 is what percent?

A. 75%

B. 48%

C. 12%

D. 36%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q3/5

True or false?

15 out of 60 is 15%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q4/5

True or false?

20% of 50 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 2

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

In a group, 12 people use a phone at night and 28 do not.

Write the ratio of users to non-users in simplest form.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q1/5

A class has 18 students who walk to school and 12 who ride.

For every 3 students who walk, how many ride?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q2/5

In a survey, 30 people eat breakfast and 20 do not.

Which statement describes the ratio?

A. For every 3 who eat breakfast, 2 do not.

B. For every 1 who eats breakfast, 30 do not.

C. For every 3 who eat breakfast, 5 do not.

D. For every 2 who eat breakfast, 3 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q3/5

True or false?

The ratio 24 : 16 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The ratio 10 : 15 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 1

Recognize conditional probability and independence in everyday situations • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Explain the difference between P(sleeps under 8 hours | uses a phone at night) and P(uses a phone at night | sleeps under 8 hours).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q1/5

A fair coin is flipped twice, and the flips are independent.

Is P(second flip is heads | first flip is heads) equal to P(second flip is heads)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q2/5

Which pair of events is most likely independent?

A. Smoking and getting lung disease

B. Rolling a 6 on a fair number cube and flipping heads on a fair coin

C. Rain today and carrying an umbrella today

D. Studying more and getting a higher grade

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

If two events are independent, knowing that one happened does not change the probability of the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 2

Find a conditional probability as the fraction of B's outcomes that are also in A • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is even, what is the probability that it is greater than 3?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q1/5

Two fair coins are flipped, and the flips are independent.

Given that at least one coin shows heads, what is the probability that both show heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q2/5

Of 100 people, 48 use a phone at night, and 36 of those sleep under 8 hours.

A night phone user is chosen at random. What is P(sleeps under 8 hours | uses a phone at night)?

A. 0.75

B. 0.36

C. 0.48

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q3/5

True or false?

A card is drawn at random from cards numbered 1 to 10.

Given that the number is odd, the probability that it is 5 or less is $\frac{3}{5}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q4/5

True or false?

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is less than 3, the probability that it is 1 is $\frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q5/5